EarthCraft data sources

One master table of every map and data source gathered for EarthCraft: what it is, where it is, how to access it, under which licence, how hard it is to integrate and how long it will last.

207	14	57	70	47	18
sources	categories	easy to integrate	moderate	need an offline pre-bake	excluded (licence, paywall)

Column	How to read it
ID	Stable code: category letters + number. Sources were sorted by integration effort, then global before regional, then name; rows added later are appended at the end of their category so existing IDs never change.
Region / Format	Where the data applies, and how it is delivered: XYZ/WMTS tiles, WMS, vector tiles / PMTiles, REST or JSON API, feed, file download, cloud files (S3/COG/Zarr), GeoParquet, WebSocket, viewer only, catalog.
Auth	None, free key, free account, paid, or other access conditions.
Licence / Comm. use	Short licence label and whether commercial use is allowed: Yes, Yes · SA (share-alike), Check (unverified or mixed), No.
Integration	Easy: fits EarthCraft's existing tile or fetch code. Moderate: needs a new decoder, client or crate. Pre-bake: large data, offline step. Excluded: licence or paywall. n/a: lead, reference or catalog. P1 / P2 / P3 under the label is the suggested build order.
Longevity	● Inst. = institutional (agency or statute, effectively permanent) · ● Org = well-funded project · ● Comm. = depends on one community or company's goodwill · ● Fragile = single maintainer, no SLA or revocable.
Verification	Researched by web search on 2026-10-18 and 2026-10-19. Rows marked † are from memory and unverified. Endpoints other than GitHub files and Rust crates could not be live-tested from the research sandbox; Appendix I lists smoke tests to run.

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Category overview

Category	Code	Sources	Easy	Moderate	Pre-bake	Excluded	n/a
Terrain & Bathymetry	TER	36	5	15	11	2	3
Imagery & Basemaps	IMG	28	13	11	2	2	0
Weather, Ocean & Climate	WTH	17	7	5	4	0	1
Wind & Solar Resource	RES	6	2	2	2	0	0
Energy Infrastructure & Emissions	INF	9	2	5	0	1	1
Land Cover, Farmland, Soil & Forest	LND	26	1	4	15	1	5
Buildings & Population	BLD	5	0	1	4	0	0
Boundaries, Places & Reference	REF	23	12	2	4	5	0
Live Tracking & Transport	TRK	8	2	4	1	1	0
Hazards & Space Weather	HAZ	12	5	5	1	1	0
Marine & Maritime	MAR	5	1	2	0	1	1
Heritage, Archaeology & Nature	HER	25	7	10	3	4	1
Other Worlds	OTH	3	0	3	0	0	0
Catalogs & Discovery	CAT	4	0	1	0	0	3
All categories		207	57	70	47	18	15

Terrain & Bathymetry (TER) — 36 sources

ID	Source	Region	Format	Endpoint / access	Coverage / resolution	Auth	Licence	Comm. use	Integration	Longevity	Notes, terms and caveats
TER-01	**GEBCO WMS — zero-effort ocean-floor tiles**	Global	WMS	WMS live-verified: https://wms.gebco.net/mapserv — layers GEBCO_LATEST (GEBCO_2026) and GEBCO_LATEST_SUB_ICE_TOPO ; GetMap over the Atlantic = 200 image/png. Styled, shaded seafloor rendering, works in EPSG:3857	Global land + ocean floor (no sub-ice separate unless using the sub-ice layer; TID grid not in WMS)	None	Public domain	**Yes**	**Easy** P1	● Inst.	**Terms:** Same GEBCO public-domain policy as TER-22 **How:** Tile path overlay — the seafloor basemap many globe apps render at ocean zoom (GEBCO was the Google-Ocean source class of choice); cite GEBCO grid source + DOI request **Assessment:** Styled seafloor basemap; GetMap verified 200 PNG **Needs:** WMS adapter (A) **Longevity:** GEBCO (IHO+IOC) — public domain; WMS served by BODC-hosted mapserver
TER-02	**NASA GEDI L3 + ICESat-2 ATL08 (spaceborne, combined)**	Global	File download	LP DAAC Data Pool HTTPS; GEDI L3 1 km DTM also in OT's globaldem ; ATL08 via Earthdata/NSIDC	Global 1 km grid (GEDI) and track photon spots (ATL08)	Free account	NASA open / CC BY 4.0	**Yes**	**Easy** P3	● Inst.	**Terms:** NASA EOSDIS: "full and open sharing of all data..." / GEDI CC BY 4.0 in DCAT **How:** Geo fetch as background ground-elevation sanity check only — neither reveals earthworks or tells **Longevity:** NASA archive, decades-scale
TER-03	**OpenZenith DEM (Hugging Face** `aliasfox/openzenith-dem`**)**	Global	Tiles (XYZ/WMTS)	Terrarium PNG pyramid z0–8: `https://openzenith.cyopsys.com/api/dem-tile/{z}/{x}/{y}` (no key); point API `/api/elevation?lat=&lon=`	Global: Copernicus GLO-30 on land + GEBCO 2025 for the ocean	None	MIT tiles (+ source licences)	**Yes**	**Easy** P1	● Fragile	**Terms:** MIT for the tiles; sources keep their licences (GLO-30 ESA Open, GEBCO 2025 CC BY 4.0) **How:** Tile path with the Terrarium decode; the z0–8 pyramid can be vendored as an offline overview **Assessment:** Tile path; vendor z0–8 to avoid host risk **Needs:** Terrarium decode (B) **Longevity:** single hobby host — vendor the pyramid or fall back to TER-06
TER-04	**EMODnet Bathymetry**	Europe	WMS	WMS live-verified: https://ows.emodnet-bathymetry.eu/wms — layer emodnet:mean_2018 (also emodnet:mean_2016 , emodnet:hr_bathymetry_area coverage mask, emodnet:contours); GetMap = 200 image/png. WMTS host tiles.emodnet-bathymetry.eu exists but catalog path unverified †	European seas	None	Open (confirm terms)	**Check**	**Easy** P3	● Inst.	**Terms:** Open data. Confirm current terms **How:** Tile path (sea-only overlay); WMS bbox requests, not XYZ **Assessment:** One GetMap per tile bbox; sea-only overlay **Needs:** WMS adapter (A) **Longevity:** EU / CINEA programme funding
TER-05	**GSI Japan elevation tiles** 地理 院標高タイル **(ja)**	Japan	Tiles (XYZ/WMTS)	XYZ PNG: `https://cyberjapandata.gsi.go.jp/xyz/dem5a_png/{z}/{x}/{y}.png` (also dem1a_png). Rust decoder crate: demtile (MIT)	Japan. DEM5A / 5B / 5C / 10B / DEMGM, finest first. Sea pixels are RGB (128,0,0) = no data	None	GSI terms (check)	**Check**	**Easy** P1	● Inst.	**Terms:** GSI terms of use (利用規約). Confirm. Not Terrarium encoding. Use GSI's own formula (maps.gsi.go.jp/development/demtile.html) **How:** Tile path + per-source DEM decoder (demtile crate) — a second decode branch next to Terrarium **Assessment:** Plain XYZ + ~15-line decode (not Terrarium) **Needs:** GSI decoder (B) **Longevity:** GSI statutory free-use (with survey-law notice for derived sheets)
TER-06	**AWS Terrain Tiles (Terrarium)**	Global	Tiles (XYZ/WMTS)	XYZ PNG: `https://s3.amazonaws.com/elevation-tiles-prod/terrarium/{z}/{x}/{y}.png` . Also s3://elevation-tiles-prod	Global, z0–16, 256 px. Land blended with ocean bathymetry Europe content is older (mostly 30 m EU-DEM per the OSM community thread on Sonny's DTMs, 2026-10-19) — prefer TER-09 for Europe. EU replica bucket: `s3://elevation-tiles-prod-eu`.	None	Attribution (Tilezen)	**Yes**	**Moderate** P1	● Inst.	**Terms:** Attribution to Joerd/Tilezen. Decode: elev_m = (R*256 + G + B/256) - 32768 **How:** Tile path; terrarium PNG decode ((R*256+G+B/256)-32768 , formula in row) **Assessment:** Tile fetch is trivial; decode in the tile worker. Hillshade or elevation readout is easy; true 3D relief means ray-caster heightfield work **Needs:** elevation decoder (B) **Longevity:** AWS Registry of Open Data, hosted since 2016 — snapshot fallback would also be licence-clean
TER-07	**EMAG2 v3 magnetic anomalies**	Global	File download	GeoTIFF downloads www.ngdc.noaa.gov/geomag/emag2_download.html ; keyless getSamples on gis.ngdc.noaa.gov/arcgis/rest/services/EMAG2v3/ImageServer	Global 2 arc-min (nT)	None	Public domain	**Yes**	**Moderate** P3	● Inst.	**Terms:** Public domain — "not subject to copyright", no navigation use; cite doi 10.7289/V5H70CVX **How:** Bundle GeoTIFF; ImageServer is a handy verification tool **Longevity:** NOAA

ID	Source	Region	Format	Endpoint / access	Coverage / resolution	Auth	Licence	Comm. use	Integration	Longevity	Notes, terms and caveats
TER-08	**ETOPO 2022 / ETOPO1 relief**	Global	File download	`https://www.ngdc.noaa.gov/mgg/global/…` GeoTIFF/netCDF; ArcGIS ImageServer gis.ngdc.noaa.gov/arcgis/rest/services/etopo1/MapServer	1 arc-min (old) / 15 arc-sec 2022 blend of topography + bathymetry	None	Public domain	**Yes**	**Moderate** P3	● Inst.	**How:** Mid-res gauss grid is bundleable; the ImageServer can also be hit as tiles **Assessment:** Mid-res grid bundleable **Longevity:** NOAA NCEI — permanent
TER-09	**Mapterhorn**	Global	Vector tiles / PMTiles	PMTiles (WebP) at download.mapterhorn.com (planet.pmtiles). Pipeline: github.com/mapterhorn/mapterhorn	Global 30 m (Copernicus GLO-30) with national LiDAR down to 0.4 m over many European regions. Confirmed 2026-10-18 via attribution.json (151 sources): the Netherlands (AHN5), Spain (MDT02/MDT50, CC BY 4.0 SCNE), Switzerland (swissALTI3D), Japan (GSI DEM), all 16 German states' 1 m grids, Italy (regional DTMs: Trentino, Lombardia, Piemonte, Sardegna, Sicilia, Alto Adige CC0, plus TINITALY 10 m), Luxembourg (LiDAR 2019 MNT CC0), Portugal mainland + Madeira, Romania (LAKI II 0.5 m), Slovakia, Slovenia, Latvia, Estonia (1 m), Iceland (ÍslandsDEM), Faroes, Svalbard, Greenland, Taiwan (20 m), Israel (10 m), UK 1 m LiDAR (England / Scotland / Wales, OGL), US 3DEP 1 m and 1/3 arc-sec (PD), NZ (1 m), Australia (5 m + Tasmania 2 m), Canada, French overseas territories, Rwanda (10 m)	None	Mixed per country (CC BY, CC0, OGL…)	**Yes**	**Moderate** P2	● Comm.	**Terms:** Attribution required (mapterhorn.com/attribution , machine-readable download.mapterhorn.com/attribution.json). Per-source licences are mixed (CC BY 4.0, DL-DE/Zero, CC0, OGL, PD, Licence Ouverte) — the aggregated planet file is usable with the combined attribution list. Aerial imagery in development **How:** Tile path (PMTiles over HTTP range-requests) or pre-extract a region with pmtiles extract and ship it **Assessment:** Best data (LiDAR to 0.4 m); one PMTiles reader also unlocks five other rows **Needs:** `pmtiles` + `image-webp` + decoder (B), (C) **Longevity:** NLnet-backed mirror today, but every underlying source is national OGD — a dead mirror is replaceable
TER-10	**Open Topo Data and Open-Elevation (point APIs)**	Global	REST / JSON API	`https://api.opentopodata.org/v1/{dataset}?locations=lat,lng` (Google Elevation API-compatible) and `open-elevation.com`. Both self-hostable with Docker; Open-Elevation's global dataset is 253 GB zipped	Point elevation	None	Goodwill public APIs	**Check**	**Moderate** P3	● Comm.	**Terms:** Public hosted APIs are goodwill services; code is open **How:** Geo fetch for a cursor-elevation readout. Prefer decoding Terrarium locally **Assessment:** Cursor-elevation readout **Longevity:** volunteer-run
TER-11	**OpenTopography global DEM API — incl. ANADEM**	Global	REST / JSON API	REST portal.opentopography.org/API/globaldem?demtype=ANADEM… &outputFormat=GTiff bbox cut-out; catalogue via STAC (/stac/raster_catalog.json) and DCAT/CSW (both live-read)	Global 30 m set: ANADEM (South America DTM, vegetation removed from Copernicus GLO-30 — geoglyphs become visible), COP30/90, SRTM, NASADEM, GEDI L3 1 km; ≤450,000 km²/request	Free key	CC BY 4.0 (key limits)	**Check**	**Moderate** P3	● Inst.	**Terms:** DCAT: all listed CC BY 4.0; policy: 50 calls/24 h non-academic, keys "may not… be embedded within an application", commercial integration needs an Enterprise API key **How:** Geo fetch — user-supplied key from config (never shipped); server-proxy or pre-bake otherwise; STAC for catalogue verification **Assessment:** Keys may not be embedded in a distributed app **Needs:** user key (D) **Longevity:** NSF-funded facility, DOIs pinned
TER-12	**CD:NGI South Africa + eThekwini footprints**	South Africa	File download	ngigeoportal.dlrrd.gov.za (email-gated downloads) + eThekwini open-data portal	SA-wide orthos/topo/25 m DEM; nationwide 1–2 m DSM/DTM is commercial (Hexagon/GeoSpace); Durban: LiDAR-derived building footprints only	Email request	Free (hi-res is commercial)	**Check**	**Moderate** P3	● Inst.	**Terms:** Free use with attribution; hi-res elevation commercial † **How:** On-demand pulls only; do not bundle **Longevity:** long-lived custodian; portal restructures change URLs
TER-13	**City of Cape Town LiDAR DEM/DTM**	South Africa	REST / JSON API	ArcGIS open-data portal + MapServers Elevation_CCT_GLM2019_5m_DEM and Elevation_DTM_2m (both live-read)	~2,400 km² municipality; 2019 LiDAR → 5 m bare-earth DEM; 2025 LiDAR → 2 m ground-only DTM (2–3 pts/m², 0.1 m v-acc)	None	City open data terms	**Yes**	**Moderate** P3	● Org	**Terms:** City Open Data Terms of Use — free, attribution, reuse OK **How:** Pre-bake or live ArcGIS REST pull — the cleanest fully-open African municipal LiDAR **Longevity:** municipal program, 2019→2025 refresh cycle

ID	Source	Region	Format	Endpoint / access	Coverage / resolution	Auth	Licence	Comm. use	Integration	Longevity	Notes, terms and caveats
TER-14	**CNIG MDT05 / MDS05 (es)**	Spain	File download	Download centre centrodedescargas.cnig.es . WCS 2.0 https://servicios.idee.es/wcs-inspire/mdt (1000, 500, 200, 25, 5 m)	Spain, 5 m terrain (MDT05) and surface (MDS05) from PNOA-LiDAR	None	CC BY 4.0	Yes	Moderate P3	● Inst.	**Terms:** CC BY 4.0 (SCNE) . First LiDAR coverage dates 2008–2015 **How:** WCS client or pre-extract per region **Longevity:** CC BY 4.0 (SCNE)
TER-15	**GeoSampa — city of São Paulo LiDAR**	Brazil (São Paulo)	File download	Download page geosampa.prefeitura.sp.gov.br/PaginasPublicas/_SBC.aspx ; GeoNetwork record "Nuvem de pontos MDT 2020" (read directly)	~1,522 km² city; LAS/LAZ 10 pts/m² + ground-filtered MDT 0.50 m, 2017/2020 vintages	None	State open data (attribution)	Check	Moderate P3	● Inst.	**Terms:** Record states access "semRestricoes"; state open-data licence with attribution **How:** Pre-bake MDT excerpt — the model open Brazilian municipal LiDAR stack **Longevity:** PMSP institutional catalogue (ISO 19115)
TER-16	**IGC-SP — São Paulo state LiDAR 2023/24**	Brazil (São Paulo)	File download	igc.sp.gov.br downloads + state WMS geoportal (bot-blocked to curl from here)	State-wide slices, 10 pts/m², 2023/2024 flights (1:10,000 base)	None	Unverified †	Check	Moderate P3	● Inst.	**Terms:** State open-data assumed † — licence wording and per-tile download unverified **How:** Pre-bake via IGC download flow once reachable **Assessment:** Pre-bake via the IGC download flow once reachable; the WMS geoportal is bot-blocked **Longevity:** state cartographic institute † downloads unverified
TER-17	**IGN RGE ALTI + LiDAR HD — Réunion & Mayotte**	France (Réunion, Mayotte)	File download	Mapterhorn tarballs (frrgealti1reunion ≈ 3.1 GB; frhd2975 LiDAR HD 0.5 m ≈ 14 GB) + cartes.gouv.fr	Full-island 1 m/5 m MNT both islands; Réunion LiDAR HD 0.5 m	None	Licence Ouverte 2.0	Yes	Moderate P3	● Inst.	**Terms:** Licence Ouverte 2.0; IGN docs name "archéologie" as an intended application **How:** Bundle/mirror via Mapterhorn **Longevity:** French open data, Mapterhorn mirrored
TER-18	**INEGI MDE / historical LiDAR tiles (Mexico)**	Mexico	File download	inegi.org.mx/programas/mde/ ; Biblioteca (upc 702825796341); Continuo de Elevaciones Mexicano	National; LiDAR 2010–2017 vintages, MDT ~1:10k blocks + CEM 3.0 continuous	None	Libre Uso MX	Yes	Moderate P3	● Inst.	**Terms:** "Libre Uso MX" open-data terms with attribution **How:** Geo fetch of tile catalogue — the workflow pattern to copy (per-tile LAS/LAZ + raster MDT) **Longevity:** national statistical-agency portal
TER-19	**Rwanda 10 m DEM (via Mapterhorn)**	Rwanda	File download	download.mapterhorn.com/sources/rw.tar (~316 MB; origin github.com/WASAC/rwanda-dem)	Whole country, 10 m grid	None	CC BY 4.0	Yes	Moderate P3	● Inst.	**Terms:** CC BY 4.0 (NLA + WASAC/UNDP — attribution string read from attribution.json) **How:** Bundle; cross-ref the TER-09 pipeline **Longevity:** open gov source, Mapterhorn mirror
TER-20	**USGS 3DEP**	United States	REST / JSON API	National Map raster + point-query epqs.nationalmap.gov/v1 (also via OT mirror; 1 m there is academic-only)	USA only, 10 m seamless + 1 m tiles	None	Public domain	Yes	Moderate P3	● Inst.	**Terms:** Public domain (US federal); the reference model for EarthCraft's licence-verified catalog **How:** Tile path / point query; out of primary target area but free of every caveat **Longevity:** permanent USGS archive
TER-21	**Copernicus DEM (GLO-30 / GLO-90)**	Global	Cloud files (S3/COG/Zarr)	COGs: s3://copernicus-dem-30m , s3://copernicus-dem-90m	Global. GLO-30 Public has gaps for some countries	None	Copernicus licence	Yes	Pre-bake P3	● Inst.	**How:** Pre-bake (per-region COG reads via async-tiff); not a runtime source at whole-globe scale **Needs:** offline step **Longevity:** Copernicus permanent, free & open
TER-22	**GEBCO grid (GEBCO_2026, released April 2026)**	Global	File download	Global netCDF (~4 GB) / GeoTIFF / Esri ASCII via CEDA (dap.ceda.ac.uk/bodc/gebco/global/gebco_2026/... , also 8× 90°×90° tiles), user-area download app download.gebco.net , OPeNDAP	Global land + ocean, 15 arc-second. Includes ice-surface / sub-ice variants and a TID grid	None	Public domain	Yes	Pre-bake P3	● Inst.	**Terms:** Verified 2026-10-18: placed in the public domain, free including commercial use; GEBCO asks (not requires) the DOI citation **How:** Pre-bake, or a user-directed ~4 GB regional download at runtime **Needs:** offline step **Longevity:** GEBCO (IHO+IOC) — public domain by stated policy
TER-23	**JAXA ALOS World 3D (AW3D30) (ja / en)**	Global	File download	Tiled GeoTIFF from JAXA EORC; also OpenTopography	Global 30 m DSM (trees and buildings included), v4.1 April 2024	Free account	Free (JAXA)	Check	Pre-bake P3	● Comm.	**Terms:** Free. DSM, not bare earth **How:** Pre-extract after free registration **Needs:** account login; offline step **Longevity:** JAXA free with account; DSM caveat
TER-24	**AHN Actueel Hoogtebestand Nederland (nl)**	Netherlands	File download	Via PDOK (WCS, download)	Netherlands LiDAR terrain and surface. AHN5 confirmed in Mapterhorn (2026-10-18) as 5 m grid, plus 0.5 m for Bonaire, Saba and St. Eustatius	None	CC BY 4.0	Yes	Pre-bake P3	● Inst.	**Terms:** CC BY 4.0 per mapterhorn.com/attribution (the older OSM-thread "CC0" claim refers to earlier editions — use CC BY 4.0 with attribution to AHN.nl) **How:** Pre-extract (already inside Mapterhorn TER-09 — prefer that) **Needs:** offline step **Longevity:** CC BY 4.0
TER-25	**ArcticDEM (Polar Geospatial Center)**	Arctic	Cloud files (S3/COG/Zarr)	COG on S3, anonymous: `s3://pgc-opendata-dems/arcticdem/mosaics/` and `/strips/`	Arctic 2 m mosaics from 2007; strips added twice a year	None	CC BY 4.0	Yes	Pre-bake P3	● Inst.	**How:** Pre-bake or COG window reads (`async-tiff`); EPSG:3413 needs `proj4rs` **Needs:** offline step **Longevity:** NSF-funded PGC on AWS Open Data

ID	Source	Region	Format	Endpoint / access	Coverage / resolution	Auth	Licence	Comm. use	Integration	Longevity	Notes, terms and caveats
TER-26	**geoportal.gov.pl NMT / LiDAR point clouds (pl)**	Poland	File download	"Pobierz dane" module + open-data servers	Poland: NMT 1 m grid + raw LAS clouds, cycles 2011–2024	None	Open PZGIK	**Yes**	**Pre-bake** P3	● Inst.	**Terms:** Open PZGIK, free download (bulk fetching tolerated but official bulk API would need a key) **How:** Pre-extract LAS/GeoTIFF **Needs:** offline step **Longevity:** same as IMG-06
TER-27	**IGN LiDAR HD (fr)**	France	File download	GeoTIFF tiles (MNT 5 m, MNS 0.5 m, MNH) and point clouds via cartes.gouv.fr/telechargement and data.gouv.fr	France, at least 10 pulses/m². Rollout in progress (track at macarte.ign.fr). Restricted zones (ZICAD) withheld. MNS 50 cm tile ≈ 24 MB	None	Unconfirmed	**Check**	**Pre-bake** P3	● Comm.	**Terms:** Licence not confirmed. Check **How:** Pre-extract LiDAR tiles per region **Needs:** offline step **Longevity:** IGN licence to confirm; ZICAD zones withheld
TER-28	**Kruger NP river LiDAR 2012 (CEDA)**	South Africa	File download	CEDA catalogue uuid a2e82c7f... , direct web download + OPeNDAP	3× 50 km river reaches (Sabie/Olifants/Letaba); Optech M200 point cloud + contours + orthos, 12 GB	Free account	OGL v3	**Yes**	**Pre-bake** P3	● Inst.	**Terms:** OGL v3 — "Permitted Use: Any", citation required **How:** Link-only (point cloud) — the only verified open airborne point-cloud archive on the continent **Needs:** offline step **Longevity:** NERC-funded permanent archive
TER-29	**REMA (Reference Elevation Model of Antarctica)**	Antarctica	Cloud files (S3/COG/Zarr)	`s3://pgc-opendata-dems/rema/mosaics/` and `/strips/`	Antarctica, built from ~0.5 m imagery	None	PGC terms †	**Check**	**Pre-bake** P3	● Inst.	**How:** As TER-25 (EPSG:3031) **Needs:** offline step **Longevity:** NSF-funded PGC on AWS Open Data
TER-30	**Sonny's LiDAR Digital Terrain Models of Europe**	Europe	File download	Free downloads at `sonny.4lima.de`: `.hgt` at 1″ and 3″, GeoTIFF at 20 m and 50 m (Europe v24)	Europe: DTMs from national open LiDAR, filled with SRTM elsewhere in the 0.5″/1″/3″ sets	None	Free, credit required	**Check**	**Pre-bake** P3	● Comm.	**Terms:** Free; credit "Sonny" with a link. Underlying national licences vary — verify per country **How:** Mostly superseded by Mapterhorn (TER-09). Use for gaps, offline only **Needs:** offline step **Longevity:** one-person project; used by OpenTopoMap and OpenAndroMaps
TER-31	**swissALTI3D / swissSURFACE3D (de / fr)**	Switzerland	File download	Download via swisstopo (also read through OpenTopography)	Switzerland, 0.5 m terrain + surface from the 2017–2025 national LiDAR campaign	None	Open Government Data (CH)	**Yes**	**Pre-bake** P3	● Inst.	**Terms:** Open Government Data, commercial use allowed, attribution "Federal Office of Topography swisstopo" **How:** Pre-extract **Needs:** offline step **Longevity:** swisstopo OGD
TER-32	**FABDEM**	Global	File download	—	Global bare-earth model derived from Copernicus GLO-30	—	Purchase required	**No**	**Excluded**	—	**Terms:** Commercial use requires purchase (per a community listing) **How:** Not for an open app **Assessment:** Commercial use requires purchase **Needs:** licence
TER-33	**Viewfinder Panoramas (J. de Ferranti)**	Global	File download	Free `.hgt` downloads: 3″ global without serious voids, 1″ Europe	Global	None	Research / private use only	**No**	**Excluded**	● Comm.	**Terms:** Research and private use; written permission for other uses **How:** Not for an open app **Assessment:** Research/private use **Needs:** licence **Longevity:** individual
TER-34	**GMRT 4.0 (100 m multibeam) and SRTM15+ V2.1 (15″) †**	Global	—	Listed in a community DEM catalogue; not verified directly	Ocean bathymetry	—	—	—	n/a	—	**How:** Not assessed **Assessment:** Leads only **Status:** Lead only, not verified
TER-35	**HeightMap (`jaluebbe/HeightMap`)**	Global	Reference (not data)	Offline combination of elevation sources (SRTM `.hgt`, Sonny LiDAR, GEBCO_2023, regional sets) with a web UI	Reference	—	—	—	n/a	—	**How:** Reference for source-priority merging **Assessment:** Reference only
TER-36	**opendem.info link directory**	Global	Catalog / portal	Catalogue of free DEMs (SRTM public domain, MERIT non-commercial, EU-DEM, EuroDEM, Viewfinder)	Reference	—	Per dataset	—	n/a	—	**How:** Discovery only

Imagery & Basemaps (IMG) — 28 sources

ID	Source	Region	Format	Endpoint / access	Coverage / resolution	Auth	Licence	Comm. use	Integration	Longevity	Notes, terms and caveats
IMG-01	**EOX Sentinel-2 cloudless**	Global	Tiles (XYZ/WMTS)	WMTS (cloudless.eox.at)	Global 10 m, yearly mosaics 2016–2025	None	CC BY / BY-SA (sources conflict)	**Check**	**Easy** P3	● Comm.	**Terms:** Attribution required. Sources disagree on licence (CC BY vs BY-SA). Verify before shipping **How:** Tile path now; licence must be resolved before it can ship in a default layer **Assessment:** Plain WMTS; ship only after the licence check **Needs:** gate: licence **Longevity:** company goodwill (free for over a decade, no guaranteed right) — keep GIBS as fallback
IMG-02	**Esri World Imagery + Wayback (EarthCraft's current default imagery)**	Global	Tiles (XYZ/WMTS)	Unauthenticated XYZ/WMTS: server.arcgisonline.com/.../World_Imagery/MapServer/tile/ and wayback.maptiles.arcgis.com/.../tile/{release}/{z}/{y}/{x}	Global imagery + 500+ historical Wayback releases	None	Esri terms (non-commercial)	**No**	**Easy** P3	● Fragile	**Terms:** Verified 2026-10-18 against Esri's own ToU: free services are for "internal purposes or noncommercial external purposes" only; Esri may alter or discontinue them at its sole discretion; unauthenticated endpoint use is reserved for Esri-product users per Esri staff; no bulk tile export. Attribution string: "Imagery © Esri, Maxar, Earthstar Geographics, and the GIS User Community" **How:** Already wired — but should become an opt-in layer, never a default, for an open app **Assessment:** Make opt-in per Esri ToU (see watch-list) **Needs:** already wired **Longevity:** non-commercial-only and revocable; a free ArcGIS Developer account (2M tiles/mo) is the compliant variant; GIBS (IMG-04) and EOX (IMG-01) are the attribution-clean fallbacks
IMG-03	**GIBS Earth at Night (Black Marble city lights)**	Global	Tiles (XYZ/WMTS)	WMTS `https://gibs.earthdata.nasa.gov/wmts/epsg3857/best/{VIIRS_Black_Marble|VIIRS_CityLights_2012|VIIRS_Night_Lights}/default/{date}/{z}/{y}/{x}.png` — resolve the fixed date from GetCapabilities once and hardcode	Global static night composites (2012 / 2016)	None	NASA open data	**Yes**	**Easy** P1	● Inst.	**Terms:** NASA open **How:** Plain tile-path layer — this is the zero-effort night-lights basemap many browser globes use **Assessment:** GIBS night lights; hard-code the date **Needs:** resolve date once **Longevity:** NASA GIBS
IMG-04	**NASA GIBS**	Global	Tiles (XYZ/WMTS)	WMTS https://gibs.earthdata.nasa.gov/wmts/epsg3857/best/ . Read GetCapabilities for layers. Omit the time dimension for today Projections: `epsg4326`, `epsg3857`, `epsg3413` (Arctic), `epsg3031` (Antarctic); types `all`/`best`/`nrt`/`std`; load-spreading hosts `gibs-a`/`gibs-b`/`gibs-c.earthdata.nasa.gov`; pattern `https://gibs.earthdata.nasa.gov/{wmts|wms|twms}/epsg{4326|3857|3413|3031}/{all|best|nrt|std}`.	Global, dozens of daily layers (clouds, aerosols, sea ice, SST, fires)	None	NASA open data	**Yes**	**Easy** P1	● Inst.	**How:** Tile path; the only change is a time-dimension parameter in the URL template **Assessment:** Reuse the tile path; default epsg3857 **Needs:** time parameter; `proj4rs` for polar **Longevity:** NASA EOSDIS — institutionally guaranteed
IMG-05	**Dataforsyningen (Klimadatastyrelsen, da)**	Denmark	Tiles (XYZ/WMTS)	WMTS / WMS, anonymous access confirmed (dataforsyningen.dk); Datafordeler needs a key for extra products	Denmark: ortofoto_forår (~10–12.5 cm), topo kort, DHM terrain 0.4 m (the last also inside Mapterhorn TER-09)	None	Danish open geodata (attribution)	**Yes**	**Easy** P2	● Inst.	**Terms:** Frie offentlige geodata — attribution (kildeangivelse) mandatory; exact WMTS URL parameterisation still to be read from GetCapabilities **How:** Tile path (anonymous access confirmed) **Assessment:** Anonymous WMTS confirmed **Needs:** read GetCapabilities **Longevity:** Danish free-geodata statute
IMG-06	**geoportal.gov.pl Ortofotomapa WMTS (pl)**	Poland	Tiles (XYZ/WMTS)	Known-good WMTS: https://mapy.geoportal.gov.pl/wss/service/PZGIK/ORTO/WMTS/StandardResolution and .../HighResolution	Poland: ortho 5 cm urban / 10 cm / 25 cm pixel	None	Open PZGIK (CC BY †)	**Check**	**Easy** P2	● Inst.	**Terms:** Open PZGIK data (attribution-style open licence commonly cited as CC BY 4.0 †) **How:** Tile path (plain WMTS, already confirmed live) **Assessment:** Plain WMTS confirmed live **Longevity:** Polish geodetic law (PZGIK) — statutory open
IMG-07	**Kartverket topo4 WMTS + NiB + hoydedata (no)**	Norway	Tiles (XYZ/WMTS)	WMTS (topo4 open cache), ortho at norgeibilder.no , DEM via hoydedata.no	Norway: topo4 raster, NiB orthophotos, DTM 1 m (Terrengmodellar also inside Mapterhorn TER-09, CC BY 4.0)	None (bulk needs agreement)	CC BY 4.0	**Yes**	**Easy** P2	● Inst.	**Terms:** CC BY 4.0 for free products; NiB bulk access is a separate licence **How:** Tile path **Assessment:** CC BY 4.0 **Longevity:** Kartverket CC BY 4.0

ID	Source	Region	Format	Endpoint / access	Coverage / resolution	Auth	Licence	Comm. use	Integration	Longevity	Notes, terms and caveats
IMG-08	**Maanmittauslaitos avoimet aineistot (fi)**	Finland	Tiles (XYZ/WMTS)	WMTS (Karttakuvapalvelu , free API key) + file-table downloads	Finland: taustakartta / raster maps WMTS, ortokuva 50 cm, korkeusmalli 2 m (also in Mapterhorn TER-09)	Free key (tiles)	CC BY 4.0	**Yes**	**Easy** P2	● Inst.	**How:** Tile path with free API key; file downloads need none **Assessment:** Free key for the tile service **Needs:** key (D) **Longevity:** NLS Finland CC BY 4.0
IMG-09	**PDOK aerial photos Luchtfoto (nl)**	Netherlands	Tiles (XYZ/WMTS)	WMTS `https://service.pdok.nl/hwh/luchtfotorgb/wmts/v1_0/Actueel_orthoHR/{TileMatrixSet}/{TileMatrix}/{TileCol}/{TileRow}.jpeg`	Netherlands, 8 cm and 25 cm, newest year	None	CC BY	**Yes**	**Easy** P1	● Inst.	**Terms:** CC BY (capabilities keyword "open data cc-by"). Attribution to Kadaster / PDOK **How:** Tile path **Assessment:** WMTS REST, CC BY **Needs:** TileMatrixSet name **Longevity:** Kadaster/PDOK statutory open data
IMG-10	**PDOK BRT Achtergrondkaart (nl)**	Netherlands	Tiles (XYZ/WMTS)	WMTS https://service.pdok.nl/brt/achtergrondkaart/wmts/v2_0 . Also OGC API Vector Tiles	Netherlands topographic base map, EPSG:3857 available	None	CC BY 4.0	**Yes**	**Easy** P1	● Inst.	**How:** Tile path **Assessment:** WMTS REST, CC BY 4.0 **Longevity:** same as IMG-09
IMG-11	**Portale Geocartografico Trentino (it)**	Italy (Trentino)	Tiles (XYZ/WMTS)	WMS / WMTS + DTM downloads (www.territorio.provincia.tn.it)	Trentino: ortofoto 2021/2011, DTM 2 m, LiDAR	None	Attribution †	**Check**	**Easy** P3	● Comm.	**Terms:** Regional attribution terms † (verify per dataset) **How:** Tile path **Assessment:** Verify per dataset **Needs:** gate: licence † **Longevity:** same
IMG-12	**Regione Piemonte Geoportale (it)**	Italy (Piemonte)	Tiles (XYZ/WMTS)	WMTS + downloads (www.geoportale.piemonte.it)	Piemonte: ortofoto AGEA 2021/2024, BDTRE raster 1:10,000, DTM LiDAR 1 m	None	CC BY-type †	**Check**	**Easy** P3	● Comm.	**Terms:** Regional open data, licence reported CC BY-type † (verify per dataset) **How:** Tile path **Assessment:** Verify per dataset **Needs:** gate: licence † **Longevity:** regional OGD — verify per dataset
IMG-13	**swisstopo geo.admin.ch (de / fr)**	Switzerland	Tiles (XYZ/WMTS)	WMTS / WMS / vector tiles. Layers include swissIMAGE, swissALTI3D hillshade, swissBATHY3D hillshade	Switzerland	None	Open Government Data (CH)	**Yes**	**Easy** P1	● Inst.	**Terms:** Open Government Data since 2021-03-01. Free, commercial use allowed with attribution "©swisstopo" **How:** Tile path **Assessment:** Open Government Data; attribution **Longevity:** swisstopo OGD policy (since 2021)
IMG-14	**Protomaps OSM basemap**	Global	Vector tiles / PMTiles	Daily PMTiles builds (docs.protomaps.com/basemaps/downloads)	Planet about 120 GB, z0– 15. pmtiles extract for regions	None	ODbL (OSM)	**Yes · SA**	**Moderate** P2	● Org	**Terms:** ODbL, OSM attribution **How:** PMTiles path; or pre-extract a country and bundle **Needs:** `pmtiles` (C) **Longevity:** Protomaps Foundation; ODbL data is perpetual
IMG-15	**Geoportale Nazionale / PCN (it)**	Italy	Tiles (XYZ/WMTS)	WMS / WMTS view services (www.pcn.minambiente.it)	Italy: national orthophoto 0.5 m (several epochs 1988–2012), CTR vector map, DTM 10 m	None	CC BY-NC-ND 3.0 IT	**No**	**Moderate** P3	● Comm.	**Terms:** Metadata says CC-BY-NC-ND 3.0 IT and portal "termini di utilizzo" is stricter/murkier on redistribution — keep tiles client-side, attribution + same licence **How:** Tile path, display-only — respect the NC-ND clause (don't republish tiles outside the app) **Assessment:** Display-only; terms murky **Needs:** gate: CC BY-NC-ND **Longevity:** national metadata says CC BY-NC-ND and portal terms are stricter — region sources are safer
IMG-16	**IGN Géoplateforme (fr)**	France	Tiles (XYZ/WMTS)	WMTS / WMS / WFS (geoservices.ign.fr). Separate altimetry services	France	Check	Unconfirmed	**Check**	**Moderate** P3	● Inst.	**Terms:** Licence not confirmed in my sources. Check before use **How:** Tile/WMS path once licence confirmed **Assessment:** Confirm the licence first **Needs:** gate: licence unconfirmed **Longevity:** IGN — Licence Ouverte presumed, verify
IMG-17	**İBB Şehir Haritası + data.ibb.gov.tr (tr)**	Turkey (Istanbul)	Tiles (XYZ/WMTS)	Tile service (sehirharitasiapi.ibb.gov.tr) + open-data portal	Istanbul only: uydu (satellite) imagery, building-height layer	None	Municipal open data †	**Check**	**Moderate** P3	● Comm.	**Terms:** İBB open-data portal licence, attribution-style † (exact terms unverified) **How:** Tile path (Istanbul only) **Needs:** terms † **Longevity:** municipal OGD, exact terms †

ID	Source	Region	Format	Endpoint / access	Coverage / resolution	Auth	Licence	Comm. use	Integration	Longevity	Notes, terms and caveats
IMG-18	**Landmælingar Íslands gagnatorg (is)**	Iceland	Tiles (XYZ/WMTS)	WMS / WFS / WMTS at gagnatorg.lmi.is + DEM downloads (ÍslandsDEM 2 m / 10 m, also inside Mapterhorn TER-09, CC BY 4.0)	Iceland: IS50V topo, ÍslandsDEM v1.0	None	CC BY 4.0 †	**Check**	**Moderate** P3	● Comm.	**Terms:** CC BY 4.0 † (exact terms text to double-check) **How:** Tile path **Assessment:** Small agency; confirm terms **Needs:** licence text † **Longevity:** small agency, statutory open (CC BY 4.0, exact text †)
IMG-19	**Lantmäteriet öppna geodata (sv)**	Sweden	Tiles (XYZ/WMTS)	WMTS under opengeodata.lantmateriet.se (paths not fully verified) + LIDAR FFM download tables	Sweden: ortofoto 0.5 m, topografisk webbkarta, national DEM from LiDAR	None	CC0	**Yes**	**Moderate** P3	● Inst.	**Terms:** CC0 (confirmed on lantmateriet.se; attribution welcome, not required) **How:** Tile path **Assessment:** CC0; read capabilities first **Needs:** paths unverified **Longevity:** Lantmäteriet CC0 — the strongest licence class
IMG-20	**NGII background and satellite maps** 영상지도·배경지도 **(ko)**	South Korea	Tiles (XYZ/WMTS)	WMTS. Layers: general, multilingual, satellite, night, hybrid	South Korea	Free key	Korean public work (attribution)	**Check**	**Moderate** P3	● Inst.	**Terms:** Public work, attribution (공공저작 물_출처표시) **How:** Key client with attribution string in the map footer **Needs:** key (D) **Longevity:** Korean public-works statutes
IMG-21	**NLS Scotland historic maps + NL Topotijdreis**	UK & Netherlands	Tiles (XYZ/WMTS)	NLS georeferenced maps viewer/WMTS (maps.nls.uk) † · Topotijdreis WMTS via PDOK (service.pdok.nl/hwh/topotijdrei s) — capabilities not greppable, † verify	Scotland 1840s–1960s; Netherlands 1815–today, national historic raster stack	None	Per collection †	**Check**	**Moderate** P3	● Inst.	**Terms:** NLS: free to use with attribution for most, per-collection † · Topotijdreis: open (PDOK family) † **How:** Tile path with a year-parameter (time-travel mode for Europe's oldest mapped places) **Longevity:** NLS + Kadaster institutions
IMG-22	**Tianditu** 天地图 **(zh)**	China	Tiles (XYZ/WMTS)	`https://t{0-7}.tianditu.gov.cn/{vec|img|ter}_{c|w}/wmts?...&tk={key}` . Vector, imagery, terrain + labels	China, CGCS2000 and Web Mercator. A Chinese tutorial says its imagery has no coordinate offset, unlike commercial Chinese maps	Free key	ToS (check)	**Check**	**Moderate** P3	● Comm.	**Terms:** Terms of service apply. Confirm redistribution and caching rules **How:** Key client (registration + phone verification) **Needs:** key (D) **Longevity:** state ToS; redistribution and caching rules unclear — cautious
IMG-23	**VWorld** 브이월드 **(ko)**	South Korea	Tiles (XYZ/WMTS)	WMTS / WMS / WFS, 168 layers. WebGL 3D API	South Korea	Free key	Free (check terms)	**Check**	**Moderate** P3	● Inst.	**Terms:** Free. Traffic limits depend on the agency. Check terms **How:** Key client **Needs:** key (D); WMS adapter (A) **Longevity:** / traffic limits per agency
IMG-24	**VIIRS night lights (NASA Black Marble)**	Global	File download	HDF5 archive at NASA LAADS DAAC: VNP46A2 daily gap-filled ~15 arc-sec (~500 m), 2012–now, plus monthly VNP46A4 and JPSS-1 (VJ146) equivalents; also free in GEE (NASA/VIIRS/002/VNP46A2)	Global	Free account	NASA open data	**Yes**	**Pre-bake** P3	● Inst.	**Terms:** Free, released Aug 2020, citation DOI:10.5067/VIIRS/VNP46A2.002. EOG / Colorado School of Mines products are the paywalled ones — use NASA Black Marble. Big HDF5 files → offline preprocessing **How:** Pre-bake HDF5 per region **Needs:** account login; offline step **Longevity:** NASA (free since 2020, permanent URL)
IMG-25	**USGS historical topographic maps (topoView)**	United States	File download	Scan downloads per quad, keyless; also WMTS current-scales basemap.nationalmap.gov/arcgis/rest/services/USGSTopo/MapServer for the modern equivalent	USA, 1884–2006 topographic quads	None	Public domain	**Yes**	**Pre-bake** P3	● Inst.	**Terms:** Public domain (US federal) **How:** Pre-bake — georeferenced quad scans for "time-travel" imagery over US landmarks **Needs:** offline step **Longevity:** USGS permanent
IMG-26	**Bhuvan / NRSC Open EO Data Archive (NOEDA) (hi)**	India	File download	Web viewer + archive downloads (bhuvan.nrsc.gov.in , bhuvan-app3.nrsc.gov.in/data/)	India: LISS-III / LISS-IV, Cartosat DEM, ortho products	Free account	NRSC rules †	**Check**	**Excluded**	● Comm.	**Terms:** NRSC Open Data rules — free for registered users with non-commercial-style restrictions † (licence text unverified) **How:** Registration-gated download; viewer tiles are unofficial **Assessment:** Not a runtime source **Needs:** registration + restrictions **Longevity:** NRSC restrictions; registration needed
IMG-27	**map.is / Loftmyndir ehf (is)**	Iceland	Viewer only	Tile viewer (www.map.is)	Iceland: aerial photo mosaic < 0.5 m	None	Commercial, no redistribution	**No**	**Excluded**	● Fragile	**Terms:** Commercial — personal viewing, no open tile API, no redistribution **How:** Not usable — exclude **Assessment:** No open tile API **Needs:** commercial **Longevity:** commercial, no toggle endpoint

ID	Source	Region	Format	Endpoint / access	Coverage / resolution	Auth	Licence	Comm. use	Integration	Longevity	Notes, terms and caveats
IMG-28	**Earth Timelapse annual mosaics (Google, via Earth Engine)**	Global land + coast (82.6°S–83.69° N)	Earth Engine tiles (REST)	Earth Engine ImageCollection `projects/ee-earth-timelapse/assets/annual` (publisher catalog; contact earth-timelapse@google.com). Tiles over REST: POST `https://earthengine.googleapis.com/v1beta/projects/{project}/maps` returns a map ID (valid a few hours); then GET `https://earthengine.googleapis.com/v1/projects/{project}/maps/{id}/tiles/{z}/{x}/{y}` with an OAuth token (scope earthengine). Select one year with calendarRange.	One gap-filled, water-masked RGB mosaic per year, 1984–2022 (39 years). 30 m for 1984–2014 (Landsat); about 19.11 m from 2015 (Sentinel-2 fused with Landsat 8/9). 8-bit R, G, B bands plus a year property; no visualization parameters needed	Free account (Cloud project)	CC BY 4.0 (data) + Earth Engine terms (API)	Check	Moderate P2	● Comm.	**Terms:** Required attribution: “Google Earth Timelapse (Google, Landsat, Copernicus)”. Contains modified Copernicus Sentinel data [2015–present]. CC BY 4.0 covers the data (copying and redistribution allowed with attribution); using the Earth Engine API falls under Earth Engine’s own terms (free for noncommercial, research and education; commercial use is paid). **How:** Extends the time jump about 30 years before Wayback (which, from memory, starts in 2014); Wayback stays the choice for sub-metre detail. A fixed one-node map expression per year should be enough because no visualization parameters are needed (expression JSON not verified). The cache key must be (collection, year), not the URL, because the map ID expires. Years stop at 2022. Self-hosting estimate (my back-of-envelope: ~20–25 KB per JPEG tile, ~35% of tiles containing land): z0–6 about 1 GB for all years, z0–8 about 18 GB, native detail at z12 about 5–6 TB. Earth Engine export terms for a mirror were not checked. **Assessment:** Good fit for a year slider; not a replacement for Wayback detail (30 m / 19 m versus sub-metre). **Needs:** Earth Engine REST client, OAuth (desktop loopback), user-supplied Google Cloud project, token management (D) **Longevity:** Published by Google as a publisher-catalog dataset labelled v0 and not managed by Earth Engine itself. CC BY 4.0 would allow a mirror if hosting is ever needed.

Weather, Ocean & Climate (WTH) — 17 sources

ID	Source	Region	Format	Endpoint / access	Coverage / resolution	Auth	Licence	Comm. use	Integration	Longevity	Notes, terms and caveats
WTH-01	**NOAA AviationWeather**	Global	REST / JSON API	/api/data/metar , /api/data/airsigmet	Global METAR, AIRMET / SIGMET polygons	None	NOAA public data	Yes	Easy P2	● Inst.	**Terms:** NOAA public **How:** Geo fetch (METAR TAF text + SIGMET JSON on a 2-min cycle) **Assessment:** METAR / SIGMET JSON **Longevity:** NOAA
WTH-02	**NOAA NDBC buoys**	Global	Feed (JSON/CSV/text)	latest_obs.txt	Global ocean buoys: wind, waves, pressure	None	NOAA public data	Yes	Easy P2	● Inst.	**Terms:** NOAA public. No CORS, fine natively **How:** Geo fetch of text products on the 50-min cycle **Assessment:** `latest_obs.txt` **Needs:** text parser **Longevity:** NOAA
WTH-03	**Open-Meteo**	Global	REST / JSON API	JSON: https://api.open-meteo.com/v1/forecast , archive-api.open-meteo.com/v1/archive	Point queries, 80-year history, ERA5-based	None	Non-commercial (free tier)	No	Easy P3	● Comm.	**Terms:** Non-commercial only. About 10,000 calls/day. Data CC BY 4.0 **How:** Geo fetch client with per-place cache **Assessment:** Point JSON for a cursor weather probe; NC only **Needs:** gate: non-commercial **Longevity:** company goodwill with an explicit non-commercial clause — keep DWD as fallback
WTH-04	**OpenWeatherMap Maps 1.0**	Global	Tiles (XYZ/WMTS)	XYZ PNG: `https://tile.openweathermap.org/map/{layer}/{z}/{x}/{y}.png?appid={key}`	Global clouds, precipitation, pressure, temperature, wind	Free key	Free tier + attribution	Check	Easy P1	● Fragile	**Terms:** Maps 1.0 included in free plan. Attribution **How:** Key client (appid from env) **Assessment:** XYZ with appid from env **Needs:** key (D) **Longevity:** company free tier; ToS-dependent — optional plugin, never a default

ID	Source	Region	Format	Endpoint / access	Coverage / resolution	Auth	Licence	Comm. use	Integration	Longevity	Notes, terms and caveats
WTH-05	**RainViewer radar**	Global	Tiles (XYZ/WMTS)	Index https://api.rainviewer.com/public/weather-maps.json , then {host} {path}/{size}/{z}/{x}/{y}/{color}/{options}.png . Rust crate: rain_viewer	Global radar, past 2 h at 10 min	None	Personal / educational only	**No**	**Easy** P1	● Fragile	**Terms:** Free for personal, educational and small community use only. Attribution. No SLA **How:** Tile path via the two-step JSON index; rain_viewer crate fits **Assessment:** Check whether the existing radar layer already covers it **Needs:** two-step JSON index **Longevity:** no SLA, personal/small-community terms — non-default layer with a fallback
WTH-06	**DWD radar and weather WMS (de)**	Germany	WMS	WMS at maps.dwd.de (layer dwd:Radar_rv_product_1x1km_ger) and cdc.dwd.de/geoserver	Germany radar, 1 km, 5-minute steps, past plus 2 h forecast	None	CC BY 4.0	**Yes**	**Easy** P1	● Inst.	**Terms:** CC BY 4.0, commercial use allowed. Attribution "Deutscher Wetterdienst". WMS needs bbox requests, not XYZ **How:** WMS path (bbox, not XYZ) — new fetch branch; or WTH-12 for point data **Assessment:** CC BY 4.0, commercial use allowed **Needs:** WMS adapter (A) **Longevity:** DWD CC BY 4.0
WTH-07	**NOAA CO-OPS tides and currents**	United States	REST / JSON API	Live-tested `https://api.tidesandcurrents.noaa.gov/api/prod/datagetter?…` (products water_level , predictions , currents, wind, met) + Metadata API …/mdapi/prod/webapi/stations/<id>.json	US tide and current stations, observations + hi/lo predictions	None	Public domain	**Yes**	**Easy** P2	● Inst.	**Terms:** Public domain. No separate "v2" confirmed; documented set = Data API + Metadata API **How:** Geo fetch datagetter + station metadata cache **Assessment:** JSON per station; public domain **Longevity:** US federal public domain
WTH-08	**ECMWF Open Data (IFS + AIFS)**	Global	File download	GRIB2 from the ECMWF open-data portal and cloud mirrors; `ecmwf-opendata` client; rolling archive; `doi:10.21957/open-data`	Global real-time subset of IFS (medium-range, ensemble) and AIFS, including tropical-cyclone tracks. IFS cycle 50r1 since 13 May 2026 (06z/18z under `stream=oper`)	None	CC BY 4.0 + ECMWF terms	**Yes**	**Moderate** P3	● Inst.	**Terms:** CC BY 4.0 + ECMWF Terms of Use: redistribution and commercial use allowed with attribution **How:** Pre-bake first; prototype a runtime fetch of one parameter/step with the `grib` crate (TL-14) **Assessment:** Small GRIB2 subsets; otherwise pre-bake **Needs:** `grib` crate (prototype) **Longevity:** ECMWF
WTH-09	**NASA Earth Observations (NEO)**	Global	File download	JPEG / GeoTIFF downloads	50+ climate layers (aerosols, chlorophyll, SST)	None	NASA open data	**Yes**	**Moderate** P3	● Inst.	**How:** Vendored snapshot (static files) **Assessment:** Vendored equirectangular images **Needs:** static overlays **Longevity:** NASA — files frozen but permanent
WTH-10	**NOAA GFS 10 m wind**	Global	File download	GRIB2 via NOMADS filter_gfs.pl , converted with grib2json	Global, 1° grid, forecast	None	NOAA public data	**Yes**	**Moderate** P3	● Inst.	**How:** Pre-bake wind vector snapshot (GRIB is parsed offline) **Assessment:** NOMADS subset → decode GRIB2, or pre-bake JSON **Needs:** `grib` crate **Longevity:** NOAA NOMADS
WTH-11	**OpenAQ v3 air quality**	Global	REST / JSON API	REST api.openaq.org/v3 (/locations , /sensors/{id}/measurements); bulk copy on AWS S3 without a key	~170 countries: PM2.5/PM10/O3/NO2/SO2/CO/BC from reference + low-cost monitors	Free key	Per-provider terms	**Check**	**Moderate** P3	● Comm.	**Terms:** Open with per-provider terms — OpenAQ demands attribution and disclaims aggregate licensing. 60 req/min, 2,000/hour; don't rehost as a competing service **How:** Key client (X-API-Key); bulk bundle from the AWS S3 copy needs no key **Assessment:** Per-provider licences; honour `/v3/licenses` **Needs:** key (D) **Longevity:** NGO; per-provider licences vary — fetch /v3/licenses and honour flags at runtime
WTH-12	**DWD Open Data (de)**	Germany	File download	HTTPS tree opendata.dwd.de (ICON forecasts, MOSMIX, RADOLAN radar). Python client wetterdienst . JSON wrapper: Bright Sky	Germany radar and observations. MOSMIX point forecasts for 5,400 stations worldwide	None	CC BY 4.0	**Yes**	**Moderate** P3	● Inst.	**Terms:** CC BY 4.0. No SLA, no support ticket (opendata@dwd.de) **How:** Geo fetch over the HTTPS file tree **Longevity:** DWD CC BY 4.0
WTH-13	**dynamical.org Icechunk Zarr (AWS Open Data)**	Global	Cloud files (S3/COG/Zarr)	Anonymous S3 in us-west-2: `s3://dynamical-ecmwf-aifs-single`, `s3://dynamical-ecmwf-aifs-ens` (51 members); `aws s3 ls --no-sign-request`	Global forecasts initialised every 6 hours	None	CC BY 4.0	**Yes**	**Pre-bake** P3	● Org	**How:** Pre-bake. Rust `icechunk` + `zarrs` crates exist (TL-19) but are heavy **Needs:** offline step **Longevity:** dynamical.org via the AWS Open Data Sponsorship Program

ID	Source	Region	Format	Endpoint / access	Coverage / resolution	Auth	Licence	Comm. use	Integration	Longevity	Notes, terms and caveats
WTH-14	**ERA5 (Copernicus CDS)**	Global	REST / JSON API	Python API / CLI (cds.climate.copernicus.eu)	Global reanalysis since 1940, GRIB / netCDF	Free account	Copernicus licence	**Yes**	**Pre-bake** P3	● Inst.	**Terms:** Free, Copernicus licence **How:** Pre-bake only; CDS queue/credentials make runtime fetch pointless **Needs:** account login; offline step **Longevity:** Copernicus
WTH-15	**OSCAR ocean currents**	Global	File download	NetCDF via S3 s3://podaac-ops-cumulus-protected/OSCAR_L4_OC_* with temporary credentials from the s3credentials endpoint, or Earthdata Search	Global 0.25° daily surface currents (top 30 m), 1993–present, v2.0 (final / interim / NRT latencies)	Free account	CC BY 4.0	**Yes**	**Pre-bake** P3	● Inst.	**Terms:** CC BY 4.0. Verified 2026-10-18 **How:** Pre-bake netCDF per month; Earthdata auth blocks anonymous runtime fetch **Needs:** account login; offline step **Longevity:** NASA, CC BY 4.0
WTH-16	**WorldClim**	Global	File download	Raster download	Global climate normals (1950–2000)	None	Check terms	**Check**	**Pre-bake** P3	● Comm.	**How:** Vendored snapshot (small) **Needs:** offline step **Longevity:** research product — terms to verify before shipping
WTH-17	**NVIDIA Earth2Studio data sources**	Global	Reference (not data)	Python library whose data-source modules document how to read GFS (NOMADS and AWS), HRRR, IFS/AIFS, dynamical.org, GDAS observations and IBTrACS	Reference only	—	Library licence †	—	n/a	● Org	**How:** Use as a URL/pattern reference; not a dependency **Assessment:** Reference only **Longevity:** NVIDIA

Wind & Solar Resource (RES) — 6 sources

ID	Source	Region	Format	Endpoint / access	Coverage / resolution	Auth	Licence	Comm. use	Integration	Longevity	Notes, terms and caveats
RES-01	**NASA POWER**	Global	REST / JSON API	REST API (power.larc.nasa.gov)	Global, daily 1981–now: irradiance, wind, humidity, rain	None	Free (NASA)	**Check**	**Easy** P2	● Inst.	**Terms:** Free. Throttles rapid repeat queries **How:** Geo fetch client with backoff for throttling **Assessment:** Point JSON; throttles rapid repeats **Needs:** backoff **Longevity:** NASA
RES-02	**PVGIS 5.3 (EU JRC)**	Global	REST / JSON API	GET: https://re.jrc.ec.europa.eu/api/v5_3/PVcalc?lat=45&lon=8&peakpower=1&loss=14	Global except poles. Hourly series, TMY, PV yield	None	Free (JRC)	**Check**	**Easy** P2	● Inst.	**Terms:** Free. 30 calls/s per IP. Not callable from browser AJAX (not an issue natively) **How:** Geo fetch GET client (30 calls/s is plenty) **Assessment:** Single GET per click; 30 calls/s cap **Longevity:** EU JRC — free, best-effort, no SLA
RES-03	**NSRDB solar (PSM v4) (NLR, ex-NREL)**	Americas	REST / JSON API	REST returning CSV: developer.nlr.gov/api/nsrdb/v2/solar/nsrdb-GOES-aggregated-v4-0-0-download.json (PSM v3 endpoints deprecated). POST with a WKT point/multipoint	Americas (GOES) 4 km, 30-min, 1998–2023+, extended annually; Himawari / Meteosat endpoints for the rest of the world	Free key	CC BY 3.0 US	**Yes**	**Moderate** P3	● Inst.	**Terms:** Free, CC BY 3.0 US (not 4.0). Verified 2026-10-18 **How:** Key client, point-or-area CSV pull at runtime **Needs:** key (D) **Longevity:** US DOE lab (now NLR) — CC BY 3.0 US
RES-04	**Wind Toolkit (NLR, ex-NREL)**	United States	REST / JSON API	Download API (developer.nlr.gov/docs/wind/wind-toolkit/wtk-download/ — the nrel.gov domains were rebranded to NLR and developer.nrel.gov no longer resolves, verified 2026-10-18)	US, 2.4M+ sites, 2007–2014	Free key	CC BY	**Yes**	**Moderate** P3	● Inst.	**Terms:** CC BY. Dataset is dated **How:** Key client; the dataset is frozen (2007–2014), so one runtime import or a bundled copy **Needs:** key (D) **Longevity:** US DOE (CC BY class on data)
RES-05	**Global Solar Atlas**	Global	File download	GeoTIFF (globalsolaratlas.info/download)	Global. GHI / DNI / DIF / GTI 250 m, PVOUT and TEMP 1 km	None	CC BY 4.0	**Yes**	**Pre-bake** P3	● Org	**Terms:** CC BY 4.0 (World Bank). World PVOUT file is 3.6 GB. Pre-bake **How:** Pre-bake the 3.6 GB world raster into tiles outside the app **Needs:** offline step **Longevity:** World Bank ESMAP (CC BY 4.0) — redistribution permitted with attribution
RES-06	**Global Wind Atlas 3**	Global	File download	GIS downloads for a point, area or country (globalwindatlas.info)	Global, 250 m, 10 / 50 / 100 / 150 / 200 m heights: wind speed, power density, capacity factor	None	Free (check terms)	**Check**	**Pre-bake** P3	● Inst.	**Terms:** Free. No tile endpoint found, so pre-bake to tiles **How:** Pre-bake per country of interest; no tile endpoint exists **Needs:** offline step **Longevity:** / World Bank ESMAP + DTU — attribution keeps bundling fine

Energy Infrastructure & Emissions (INF) — 9 sources

ID	Source	Region	Format	Endpoint / access	Coverage / resolution	Auth	Licence	Comm. use	Integration	Longevity	Notes, terms and caveats
INF-01	**WRI Global Power Plant Database**	Global	File download	CSV (github.com/wri/global-power-plant-database)	About 35,000 plants, 167 countries: capacity, fuel, owner CSV verified 2026-10-19: 11.97 MB (`output_database/global_power_plant_database.csv`).	None	Open (unmaintained)	**Yes**	**Easy** P2	● Comm.	**Terms:** Open. Unmaintained since v1.3.0 (June 2021) **How:** Vendored CSV import **Assessment:** 11.97 MB CSV (verified) → compact placemark file **Needs:** bundle **Longevity:** WRI dataset frozen since 2021 — stable as a frozen snapshot
INF-02	**US Wind Turbine Database (USWTDB) V8.2 (10 Dec 2025)**	United States	REST / JSON API	API, CSV and shapefile at `energy.usgs.gov/uswtdb/data/`; USGS viewer; quarterly updates	US land-based and offshore turbines with technical specs and project info; 60,000+ turbines, position-verified from aerial imagery	None	US federal (CC BY paper)	**Yes**	**Easy** P2	● Inst.	**Terms:** US federal data (USGS, LBNL, AWEA); paper CC BY (Scientific Data 7:15) **How:** Geo fetch by bounding box via the API; point layer at close zoom **Assessment:** API by bbox; verified current release **Longevity:** USGS Energy Resources Program
INF-03	**Climate TRACE**	Global	File download	Download packages (CSV per country or per sector), geospatial files (gridded data, ship voyages), Public API (beta) at `climatetrace.org/data`	Global per-facility greenhouse-gas emissions with confidence labels	None	CC BY 4.0 (confirm)	**Yes**	**Moderate** P3	● Org	**Terms:** CC BY 4.0 (per a third-party listing of the free API; confirm on the site) **How:** Vendored CSV into a facility layer **Longevity:** Coalition (non-profit)
INF-04	**Global Energy Monitor trackers (`gem.wiki`)**	Global	File download	Downloads from the GEM website (27 open datasets)	Global Solar Power Tracker (operating ≥ 1 MW; announced and construction > 20 MW), Global Coal Plant Tracker (≥ 30 MW), Coal Mine and Terminal trackers, Latin America Energy Portal (plants, pipelines, LNG, wind, solar)	None	Per dataset †	**Check**	**Moderate** P2	● Org	**Terms:** Licence per dataset † **How:** Vendored CSV into the placemark pipeline; a maintained successor to INF-01 for solar and coal **Assessment:** Vendored CSV into placemarks **Needs:** licence † **Longevity:** NGO (Global Energy Monitor)
INF-05	**Open Infrastructure Map**	Global	Vector tiles / PMTiles	Vector tiles via TileJSON (best-effort). Overpass for small GeoJSON extracts	Power lines, substations, wind turbines, solar generators, pipelines, telecom	None	ODbL (OSM)	**Yes · SA**	**Moderate** P2	● Comm.	**Terms:** OSM licence. Tile server may be restricted in future **How:** Vector-tile decode or reuse the Overpass pattern; safest is a bundled PMTiles copy **Longevity:** community project over OSM data — ODbL applies to derived distributions
INF-06	**ENTSO-E Transparency Platform**	Europe	REST / JSON API	RESTful API `https://web-api.tp.entsoe.eu/api?securityToken=…` (XML); register on the platform and email for "Restful API access"	Europe: load, generation, transmission, balancing	Free key	EU Reg. 543/2013 †	**Check**	**Moderate** P3	● Inst.	**Terms:** Open under EU Regulation 543/2013; terms † **How:** Key client plus XML parser; live generation by country as an overlay **Needs:** key (D) **Longevity:** EU regulation
INF-07	**Renewable Energy Data Explorer (NREL)**	Selected countries	File download	GIS layer downloads	Selected non-US countries	None	Per layer	**Check**	**Moderate** P3	● Inst.	**Terms:** Check per layer **How:** Bundle per selected layer **Longevity:** US DOE
INF-08	**UMN "Primary Resources Extracted"**	Global	File download	Shapefile (U-Spatial)	Global points: mines, oil and gas, nuclear, hydro, solar and wind farms	None	Research use only	**No**	**Excluded**	● Fragile	**Terms:** Research use. Redistribution or commercial use needs permission **How:** Bundle only after explicit permission **Assessment:** Redistribution needs permission **Needs:** gate: permission **Longevity:** redistribution needs permission — don't plan around it
INF-09	**GRanD dams, global solar-farm inventory, Global Wind Atlas API †**	Global	—	Leads, not verified	Dams; utility-scale solar polygons; wind API	—	—	—	n/a	—	**How:** Not assessed **Assessment:** Leads only **Status:** Lead only, not verified

Land Cover, Farmland, Soil & Forest (LND) — 26 sources

ID	Source	Region	Format	Endpoint / access	Coverage / resolution	Auth	Licence	Comm. use	Integration	Longevity	Notes, terms and caveats
LND-01	**Global Forest Watch tiles (Hansen layers)**	Global	Tiles (XYZ/WMTS)	Keyless tile URLs (live-tested): `https://tiles.globalforestwatch.org/umd_tree_cover_loss/v1.13/mercator/{z}/{x}/{y}.png` (512 px, PBF also available; implementations mercator / gfw / dynamic ; covers alerts umd_glad_* , carbon flux, etc.). Analysis endpoints on data-api.globalforestwatch.org need a free key	Global 30 m raster tiles (z≤12)	None (tiles); free key (API)	CC BY 4.0	**Yes**	**Easy** P1	—	**Terms:** CC BY 4.0 (UMD/Hansen layers). Complements LND-12: the same data served as plain XYZ-style PNG tiles — no GDAL, fits the existing tile path today **Assessment:** Keyless 512 px XYZ PNG; easiest forest layer **Longevity:** Tile path (keyless PNG) — the easiest live forest layer; cache like imagery
LND-02	**ESA WorldCereal**	Global	GeoParquet	GeoParquet farm points (source.coop/streambatch/worldcereal). Rasters via Digital Earth Africa	10 m maize, winter and spring cereal masks, irrigation	None	Open	**Yes**	**Moderate** P3	● Org	**Terms:** Open. Cite the WorldCereal paper **How:** Geo fetch GeoParquet (source.coop range reads) **Longevity:** ESA-funded WorldCereal programme
LND-03	**Pre-baked TESSERA PCA tiles (`geotessera visualize` + `webmap`)**	Global	Tiles (XYZ/WMTS)	`geotessera download --bbox … --year 2024 -o ./tiffs`, then `geotessera visualize ./tiffs pca.tif --balance histogram|percentile|adaptive` (3-component PCA, reprojected to EPSG:3857), then `geotessera webmap pca.tif --min-zoom 8 --max-zoom 15` writes XYZ web tiles	Any downloaded region. PCA colours are arbitrary per fit, so two mosaics are not colour-comparable	None	CC0 data, MIT tool	**Yes**	**Moderate** P2	● Inst.	**How:** Pre-bake a showcase region (a country or farmland area), pack the XYZ folder into an MBTiles file and load it as a local provider (EarthCraft's cache already speaks MBTiles). Fit one PCA basis and reuse it across regions. End-to-end packing not tested † **Assessment:** Showcase-region false-colour layer; colours not comparable across regions **Needs:** offline PCA + MBTiles pack † **Longevity:** once baked the file is yours (CC0)
LND-04	**France RPG (Registre Parcellaire Graphique) at source**	France	Vector tiles / PMTiles	data.gouv.fr dataset; also mirrored in LND-05 as GeoParquet and PMTiles (latest edition 2024)	France crop parcels with majority crop group	None	Licence Ouverte (Etalab)	**Yes**	**Moderate** P3	● Inst.	**How:** Same as LND-05 (France collection) **Needs:** `pmtiles` (C) **Longevity:** IGN / French state open data
LND-05	**FTW harmonized field data catalogue (Source Cooperative `ftw/harmonized-field-data`)**	Europe (per country)	Vector tiles / PMTiles	STAC/portolan catalogue of every known non-AI-generated field-boundary dataset. Per country and year GeoParquet (`s3://ftw/harmonized-field-data/fr/year=*/*.parquet` via `https://data.source.coop`) and a PMTiles asset per collection	France RPG 2017–2024 (9.68 M fields); Switzerland (1.35 M); Andalusia (4.4 M); Castile and León (9.1 M); Basque Country (0.56 M); Lower Saxony/Bremen/Hamburg (0.88 M); NRW (0.74 M); Portugal (3.57 M); more	None	Per collection	**Check**	**Moderate** P2	● Org	**Terms:** Per-collection licence: CC-BY-4.0, DL-DE-BY-2.0, Licence Ouverte, or "other". Read each `collection.json` **How:** Tile path for PMTiles (via `pmtiles` crate) or Geo fetch of one country; start with France (Licence Ouverte) as a field-boundary layer at high zoom **Needs:** `pmtiles` (C) **Longevity:** FTW on Source Cooperative; each underlying source is national/regional open data
LND-06	**ESA WorldCover 2021 v200**	Global	Cloud files (S3/COG/Zarr)	COGs: s3://esa-worldcover/v200/2021/map (about 120 GB)	Global 10 m, 11 classes (cropland is one)	None	CC BY 4.0	**Yes**	**Pre-bake** P3	● Inst.	**Terms:** CC BY 4.0, attribution required **How:** Pre-bake COG subsets per region (async-tiff) **Needs:** offline step **Longevity:** ESA (CC BY 4.0)
LND-07	**Fields of The World (FTW) Global Field Boundary Map v1.1 (Microsoft Research + FTW)**	Global	Cloud files (S3/COG/Zarr)	Open data product in cloud-native formats: a confidence-thresholded default map, the full unfiltered dataset and aggregated layers. Exact bucket path not confirmed †	First global agricultural field-boundary dataset at 10 m, years 2024 and 2025	None	CC BY	**Yes**	**Pre-bake** P3	● Org	**Terms:** CC-BY (stated in the paper) **How:** Pre-bake regional extracts; locate the cloud-native paths on the project page before planning **Needs:** offline step **Longevity:** Microsoft Research + FTW consortium (Taylor Geospatial Engine and NASA open-source software award funding)
LND-08	**GAEZ+ 2015 gridded crop dataset (FAO / IIASA; Scientific Data 2022)**	Global	File download	Dataset described at PMC8776749 (harvested area, production, yield, monthly physical area)	Global 5-minute grid circa 2015, 26 crops and crop categories, irrigated and rainfed	None	Unverified †	**Check**	**Pre-bake** P3	● Inst.	**Terms:** Licence text † **How:** Pre-bake like LND-13; newer than MapSPAM's 2010 base **Needs:** offline step **Longevity:** FAO / IIASA
LND-09	**GLC_FCS30D (zh / en)**	Global	File download	Zenodo 10.5281/zenodo.8239305 (36 zip files, 135–194 GB)	Global 30 m land cover 1985–2022, 35 classes incl. rainfed vs irrigated cropland and orchards	None	CC BY 4.0 (paper); check dataset	**Check**	**Pre-bake** P3	● Inst.	**Terms:** Check Zenodo record. The paper is CC BY 4.0 **How:** Pre-bake from the Zenodo record **Needs:** offline step **Longevity:** Zenodo permanent DOI; paper CC BY 4.0
LND-10	**Global Synergy Cropland Map v3.0 †**	Global	File download	Listed in GEOKUR as an input to MapSPAM; not verified directly	Global cropland extent	None	Unverified †	**Check**	**Pre-bake** P3	—	**How:** Lead only; ESA WorldCover (LND-06) covers the same need at 10 m **Needs:** offline step **Status:** Lead only, not verified

ID	Source	Region	Format	Endpoint / access	Coverage / resolution	Auth	Licence	Comm. use	Integration	Longevity	Notes, terms and caveats
LND-11	**GMIA v5.0 (Global Map of Irrigation Areas) and MIRCA2000 v1.1**	Global	File download	Download (both listed as inputs in the GEOKUR pipeline description)	Global irrigated areas; monthly irrigated and rainfed crop areas around 2000	None	Unverified †	**Check**	**Pre-bake** P3	● Inst.	**Terms:** Licence text † **How:** Vendored snapshot; old (2000s) data **Needs:** offline step **Longevity:** FAO / University of Frankfurt †
LND-12	**Hansen Global Forest Change**	Global	File download	Direct 10° GeoTIFF tiles, no auth: storage.googleapis.com/earthenginepartners-hansen/GFC-2025-v1.13/download.html . Also GEE asset UMD/hansen/global_forest_change_2025_v1_13	Global ~30 m, updated annually: tree cover 2000, loss, gain, loss year, datamask (v1.13 = 2000–2025)	None	CC BY 4.0	**Yes**	**Pre-bake** P3	● Inst.	**Terms:** CC BY 4.0, verified 2026-10-18 (old appspot.com v1.3 links are dead) **How:** Geo fetch of 10° GeoTIFFs via async-tiff; or consume the keyless GFW tile PNGs (LND-01) as a far easier runtime layer **Needs:** offline step **Longevity:** CC BY 4.0; GEE-published asset also mirrored through Global Forest Watch
LND-13	**MapSPAM SPAM2010 v2.0 (IFPRI)**	Global	File download	Downloads at `mapspam.info` after a short information form. Harvested area on Harvard Dataverse `doi:10.7910/DVN/PRFF8V`	Global 5 arc-minute (~10 km), 42 crops × 4 production systems (irrigated, rainfed high-input, rainfed low-input, rainfed subsistence), circa 2010. ESSD 12, 3545 (2020)	None (form)	Free †	**Check**	**Pre-bake** P3	● Org	**Terms:** Freely available; licence text † **How:** Pre-bake a crop-share raster (wheat, maize, rice…) into PNG tiles; 10 km fits globe-scale zooms **Needs:** offline step **Longevity:** IFPRI / IIASA
LND-14	**NASA SEDAC Global Agricultural Lands**	Global	File download	Raster download	Global extent and intensity of agriculture, year 2000	None	CC BY	**Yes**	**Pre-bake** P3	● Inst.	**Terms:** CC BY. Old data **How:** Vendored snapshot **Needs:** offline step **Longevity:** NASA SEDAC (CC BY) — frozen 2000 data
LND-15	**SoilGrids 2.0 (ISRIC)**	Global	File download	GeoTIFF (files.isric.org/soilgrids/latest/data/), WCS	Global 250 m, 6 depths: pH, carbon, texture, CEC, N	None	CC BY 4.0	**Yes**	**Pre-bake** P3	● Inst.	**Terms:** CC BY 4.0. REST API currently paused **How:** Pre-bake GeoTIFF volumes; do not depend on the paused REST API **Needs:** offline step **Longevity:** ISRIC (CC BY 4.0)
LND-16	**TESSERA v1 embeddings (GeoTessera, global 2024)**	Global	File download	NPY tiles on Source Cooperative: `https://data.source.coop/tessera/tessera/npy/v1/{year}/grid_{lon}_{lat}/grid_{lon}_{lat}.npy` plus `_scales.npy` (int8 values × float32 scales = embedding); landmask TIFFs under `…/tessera/landmasks/v1/`; per-dataset `manifest.parquet`; client `pip install geotessera` (Python 3.12+)	128 channels per pixel at 10 m from Sentinel-1 + Sentinel-2, years 2017–2025 where generated (tiles are randomly sampled per region); only v1 / 2024 has full global coverage. Tiles are 0.1° × 0.1° (~11 km), named by centre, about 1200 × 1200 × 128 each	None	CC0 (MIT software)	**Yes**	**Pre-bake** P3	● Comm.	**Terms:** Embeddings and model weights CC0 (commercial use allowed); software MIT; inputs are open Sentinel data **How:** Not a runtime layer: ~184 MB per tile (int8), and a back-of-envelope global land total is ~190 TB uncompressed (1.49 × 10^{12} land pixels × 128 B; my estimate, not a published figure). Use only as input to LND-03 **Needs:** offline step **Longevity:** Cambridge-led research project; hosting moved from an AWS bucket (since retired) to Source Cooperative behind Cloudflare; maintainers have flagged egress cost
LND-17	**TESSERA v1.1 and v2 variants**	Global	File download	`dataset_version` / `dataset_variant`: v1 (`vultr`, default, `npy/v1/`); v1.1 (`cambridge`, `npy/v1.1-cam/`; `dclimate` announced); v2.0 (`2B-L~beta1`, `2B-L~beta2`, `npy/v2-2B-L~beta1/` …). `geotessera info` lists them	v1.1 is smoother and more temporally stable where clear observations are sparse (maintainers' June 2026 post). v2 embeddings are produced slowly with no guaranteed turnaround; v2 stores add matryoshka prefixes (first 4 or 16 dimensions) for cheaper reads	None	CC0 (confirm v2)	**Yes**	**Pre-bake** P3	● Comm.	**Terms:** CC0 stated for the project's embeddings; confirm the licence of each v2 beta variant † **How:** Pin one (version, variant) per bundle; directory names and manifests change between versions **Needs:** offline step **Longevity:** beta variants will change
LND-18	**TESSERA Zarr v3 store (cloud-native)**	Global	Cloud files (S3/COG/Zarr)	`https://data.source.coop/tessera/tessera/zarr/v1` (also `zarr/v2-2B-L~beta1`, `zarr/v2-2B-L~beta2`): 60 UTM-zone groups plus an RGB pyramid †. Python `GeoTesseraZarr`: `probe`, `sample_points`, `read_region`, `read_patch`, `iter_region`; layout follows the `geoemb:` convention (`zarr-cm`)	Native 10 m UTM grid per zone, never resampled; dequantised float32 on read; per-point status `valid` / `water` / `nodata` / `outside`	None	CC0	**Yes**	**Pre-bake** P3	● Comm.	**Terms:** CC0 (as LND-16) **How:** Rust has `zarrs` 0.23.14 (Zarr v3, MSRV 1.91) and `object_store`, so a click-probe reading a few chunks is conceivable, but zone routing, quantisation and sharding make it a research project. The RGB pyramid † is the only part that could feed a globe overlay **Assessment:** Research project; RGB pyramid unverified **Needs:** `zarrs` + `object_store` **Longevity:** same host as LND-16

ID	Source	Region	Format	Endpoint / access	Coverage / resolution	Auth	Licence	Comm. use	Integration	Longevity	Notes, terms and caveats
LND-19	**MapBiomas (pt)**	Brazil (+LatAm)	File download	GeoTIFF export via Earth Engine (mapbiomas-user-toolkit), downloads at brasil.mapbiomas.org	Brazil annual 30 m land use 1985–present: agriculture, pasture, forest. Also Bolivia, Ecuador and others	None / Earth Engine	Attribution (check)	Check	Pre-bake P3	● Comm.	**Terms:** Attribution. Check terms. Matches the pt-BR README audience **How:** Pre-bake from an Earth Engine export **Needs:** offline step **Longevity:** MapBiomas NGO pipeline + attribution
LND-20	**USDA Cropland Data Layer**	United States	File download	GeoTIFF (croplandcros.scinet.usda.gov)	US, 30 m, crop-specific, annual	None	Public domain	Yes	Pre-bake P3	● Inst.	**How:** Pre-bake US mosaic **Needs:** offline step **Longevity:** USDA public domain
LND-21	**Dynamic World 10 m land cover**	Global	REST / JSON API	Verified: CC BY 4.0 and available in Earth Engine (GOOGLE/DYNAMICWORLD/V1 , 9 classes, Sentinel-2 refresh every 2–5 days). Direct access refuted: the long-cited GCS bucket dynamic-world.earthengine does not exist; non-GEE path = Earth Engine Python API with a GCP project (free only for research/education/nonprofit)	Global 10 m, 2015-06–now	Free account	CC BY 4.0 (no usable path)	Yes	Excluded	—	**Terms:** CC BY 4.0 + Copernicus Sentinel notice. Not usable for EarthCraft as-is — only worth revisiting if Google opens the raster export **How:** No usable runtime path — revisit only if Google opens raster export **Needs:** blocked **Longevity:** blocked
LND-22	**GeoTessera client design (reference)**	Global	Reference (not data)	Python library, registry and cache design: one `manifest.parquet` per dataset (block-indexed 5° × 5°); only manifests are cached (tens to ~200 MB); downloads checked against `Content-Length` and, when the `ETag` is a content MD5, an MD5 over the stream; exponential-backoff retries; skip-if-exists resume; optional chunk cache (`CacheStore`)	Pattern reference	—	MIT	—	n/a	● Comm.	**How:** Two ideas fit EarthCraft: a small availability manifest per sparse provider (avoids requesting tiles that do not exist, as with Trek or Mapterhorn regions) and checksum-verified downloads for any bundled data **Longevity:** research group
LND-23	**Tessera coverage map and globe viewer**	Global	File download	`ucam-eo/tessera-coverage-map` (a GitHub Action refreshes an index Parquet of all manifests); `geotessera coverage` writes a PNG, `coverage.json` and an interactive `globe.html`	Which 0.1° tiles exist per year and per (version, variant); manifest columns `lon`, `lat`, `year`, `file_size`	None	MIT / CC0	Yes	n/a	● Comm.	**How:** Reference only: 0.1° tile outlines from a manifest are trivial to draw as a coverage overlay **Longevity:** project-run
LND-24	**TESSERA model and inference (`ucam-eo/tessera`)**	Global	Reference (not data)	Model weights and inference code; v2 code live. Inputs: Sentinel-1 RTC and Sentinel-2 L2A from Planetary Computer or AWS. Papers: arXiv 2506.20380 (TESSERA), arXiv 2607.03949 (v2)	Global, annual, 10 m, pixel-wise (spectral-temporal features only)	None	MIT software, CC0 weights	—	n/a	● Comm.	**Terms:** Software MIT; weights and embeddings CC0 **How:** Not for EarthCraft: running inference is out of scope for a viewer **Longevity:** research group
LND-25	**WaPOR (FAO) and Gridded Livestock of the World (GLW4) †**	Global	—	Leads. Searches returned nothing usable this session	Water productivity (WaPOR); livestock density (GLW4)	None	—	—	n/a	—	**How:** Not assessed **Assessment:** Leads only **Status:** Lead only, not verified
LND-26	**AI4Boundaries (JRC)**	Europe	File download	Dataset of 7,831 samples: 10 m Sentinel-2 monthly composites plus 1 m orthophotos with field-boundary labels (ESSD 15, 317–329)	Labels from open parcel data of Austria, Catalonia, France, Luxembourg, the Netherlands, Slovenia and Sweden	None	Open (JRC) †	Check	n/a	● Org	**Terms:** Open (JRC); licence text † **How:** Research benchmark, not a map layer. Useful as a list of which countries open their parcels **Assessment:** Research benchmark, not a layer **Longevity:** EU JRC

Buildings & Population (BLD) — 5 sources

ID	Source	Region	Format	Endpoint / access	Coverage / resolution	Auth	Licence	Comm. use	Integration	Longevity	Notes, terms and caveats
BLD-01	**VIDA Google + Microsoft footprints**	Global	Vector tiles / PMTiles	PMTiles / GeoParquet / FlatGeobuf (source .coop/vida/google-microsoft-open-buildings)	Global, about 2.58 billion footprints	None	Source licences (Google, Microsoft)	Check	Moderate P2	● Comm.	**Terms:** Source licences apply (Google, Microsoft) **How:** PMTiles over HTTP range-requests, or pre-extract a metro area for interactive use **Needs:** `pmtiles` (C) **Longevity:** source.coop mirror of Google/Microsoft releases; per-source licences vary — verify the release you bundle

ID	Source	Region	Format	Endpoint / access	Coverage / resolution	Auth	Licence	Comm. use	Integration	Longevity	Notes, terms and caveats
BLD-02	**GHS-POP COG mirror (unofficial)**	Global	Cloud files (S3/COG/Zarr)	COGs at data.source.coop/nlebovits/ghsl/	GHS-POP R2023A, 100 m, 1975–2030	None	Open (JRC mirror)	**Yes**	**Pre-bake** P3	● Fragile	**Terms:** Mirror of JRC data **How:** Pre-bake; convenience mirror only **Needs:** offline step **Longevity:** unofficial mirror — cite JRC, keep BLD-03 as the primary
BLD-03	**GHSL population (EU JRC)**	Global	File download	GeoTIFF (human-settlement.emergency.copernicus.eu)	Global grids 1975–2100, 5-year steps. World Mollweide, so reproject	None	Open (JRC)	**Yes**	**Pre-bake** P3	● Inst.	**How:** Pre-bake per region **Needs:** offline step **Longevity:** EU JRC
BLD-04	**Overture Maps buildings**	Global	GeoParquet	GeoParquet on S3 / Azure (use the latest release under s3://overturemaps-us-west-2/release/)	Global. Merges OSM, Esri, Google, Microsoft	None	Per-source licences	**Check**	**Pre-bake** P3	● Org	**How:** Pre-extract at build time (5.6 GB GeoParquet is not a runtime source) **Needs:** offline step **Longevity:** Overture Foundation (ODbL) — industry-backed and stable
BLD-05	**PLATEAU 3D city models (ja)**	Japan	File download	geospatial.jp/ckan/dataset/plateau . CityGML, 3D Tiles, MVT. Converter: PLATEAU GIS Converter (also outputs Minecraft MCA)	Japan cities (about 200 targeted for fiscal 2023, 500 for 2027). Buildings with height, floors, use, year built	None	Free incl. commercial (CC BY / ODbL)	**Yes**	**Pre-bake** P3	● Inst.	**Terms:** Free including commercial use. Buildings also under CC BY 4.0 / ODbL / ODC-BY. Site policy applies **How:** Bundle per city (3D Tiles or GeoJSON/GeoParquet extract) **Needs:** offline step **Longevity:** MLIT statutory open; sub-licences CC BY 4.0 / ODbL — strong

Boundaries, Places & Reference (REF) — 23 sources

ID	Source	Region	Format	Endpoint / access	Coverage / resolution	Auth	Licence	Comm. use	Integration	Longevity	Notes, terms and caveats
REF-01	**GeoNames gazetteer**	Global	File download	https://download.geonames.org/export/dump/ (daily dumps)	12M+ place names, feature classes, admin links	None	CC BY 4.0	**Yes**	**Easy** P3	● Comm.	**How:** Great fallback/dedup layer behind Natural Earth populated places; the geonames "cities15000" subset is a nice small bundle **Assessment:** CC BY 4.0 **Needs:** bundle `cities15000` **Longevity:** single-maintainer project but CC BY, so bundle freely
REF-02	**Natural Earth / GSHHG**	Global	File download	Shapefile download	Coastlines, borders, lakes	None	Check terms	**Check**	**Easy** P3	● Inst.	**How:** Vendored GeoJSON in build **Needs:** bundle **Longevity:** public domain lineage, updated for decades
REF-03	**Natural Earth 1:10m ladder (the immediate win)**	Global	File download	GeoJSON on raw.githubusercontent.com/nvkelso/natural-earth-vector/master/geojson/... — verified sizes: countries 13.3 MB, map units 13.5 MB, admin_1 states 40.7 MB, urban areas 27.9 MB, populated places 4.9 MB	~10× more vertices than the 1:50m file EarthCraft currently loads The 4.9 MB populated-places figure is the `ne_10m_populated_places_simple` file (verified 2026-10-19); the full `ne_10m_populated_places` is 19.4 MB.	None	Public domain	**Yes**	**Easy** P1	● Inst.	**Terms:** Public domain — vendoring fully allowed **How:** Drop-in: swap ne_50m_admin_0_countries for the 10m URL. Also add map_units (its type field carries "Disputed"/"Indeterminate"/"Lease") **Assessment:** 13.3 MB countries (verified) **Needs:** URL swap **Longevity:** Public domain since 2009; vendor files into each release, keep GitHub raw as the live-update channel
REF-04	**ne_10m_urban_areas**	Global	File download	GeoJSON, 27.9 MB (verified)	Global city-extent polygons	None	Public domain	**Yes**	**Easy** P3	● Inst.	**How:** One-shot fetch or bundle; the only free global "urban footprint" polygon set besides Overture **Assessment:** 27.9 MB GeoJSON (verified)
REF-05	**timezone-boundary-builder**	Global	File download	GitHub releases: evansiroky/timezone-boundary-builder — simplified/full shapefile + GeoJSON, tzid per polygon	Global tz boundaries matching IANA identifiers	None	ODbL	**Yes · SA**	**Easy** P2	● Org	**Terms:** Verbatim: "The outputted data is licensed under the Open Data Commons Open Database License (ODbL)" **How:** Bundle — one-time, replaces per-app "what time is it there" guessing; share-alike on any redistributed modifications The `tzf-rs` crate (2.1.3, MIT, MSRV 1.88) embeds these polygons, so no bundling step is needed. **Assessment:** Embedded polygons; ODbL attribution **Needs:** `tzf-rs` crate **Longevity:** active GitHub project, yearly releases

ID	Source	Region	Format	Endpoint / access	Coverage / resolution	Auth	Licence	Comm. use	Integration	Longevity	Notes, terms and caveats
REF-06	**France IGN ADMIN EXPRESS / COG**	France	File download	Keyless Atom API `https://data.geopf.fr/telechargement/download/ADMIN-EXPRESS-COG/…` + WFS/WMS data.geopf.fr	Commune → canton → département → région; updated monthly	None	Licence Ouverte 2.0	**Yes**	**Easy** P1	● Inst.	**Terms:** Licence Ouverte / Open Licence 2.0 — attribution, commercial and redistribution OK **How:** Per-country plugin (sgf → GeoJSON fetch; the already-wired data.geopf.fr host also serves altimetry IMG-16) **Assessment:** Keyless Atom API **Longevity:** French state open licence
REF-07	**Germany BKG VG250 / VG2500**	Germany	File download	Open data www.bkg.bund.de + WFS sgx.geodatenzentrum.de	Bund → Länder → Regierungsbezirke → Kreise → Gemeinden; yearly	None	GeoNutzV	**Yes**	**Easy** P1	● Inst.	**Terms:** GeoNutzV "nutzungsfrei" (de-facto CC0-equivalent) — attribution not even mandatory **How:** Per-country plugin **Assessment:** GeoNutzV **Longevity:** strongest licence in the group
REF-08	**Mexico INEGI Marco Geoestadístico**	Mexico	File download	Zip download from www.inegi.org.mx/geo/contenidos/geoestadistica/...	States / municipios / AGEB / manzana (Brazil-grade detail)	None	Libre Uso MX	**Yes**	**Easy** P1	● Inst.	**Terms:** "Libre Uso MX" — free incl. commercial, attribution required **How:** Per-country plugin (shp → GeoJSON at build) **Assessment:** Libre Uso MX **Longevity:** national statistical institute
REF-09	**NZ LINZ / Stats NZ boundaries**	New Zealand	File download	data.linz.govt.nz + Stats NZ digital boundaries	Regions, territorial authorities, wards, SA1/SA2	None	CC BY 4.0	**Yes**	**Easy** P3	● Inst.	**How:** Per-country plugin, small country so easy **Assessment:** CC BY 4.0
REF-10	**Spain CNIG / IGN municipalities**	Spain	File download	centrodedescargas.cnig.es (keyless)	Provincias / municipios / CCAA	None	CC BY 4.0	**Yes**	**Easy** P1	● Inst.	**Terms:** CC BY 4.0 (Orden FOM/2807/2015), redistribution OK **How:** Per-country plugin; pairs with TER-14's WCS for Spain in general **Assessment:** CC BY 4.0
REF-11	**Statistics Canada boundary files**	Canada	File download	www12.statcan.gc.ca/.../boundary-limites/ (SHP/GML/-gdb) + WMS / ESRI-REST at geo.statcan.gc.ca	Provinces, census divisions, census subdivisions (≈ municipality), CMA/CA metros	None	OGL – Canada	**Yes**	**Easy** P1	● Inst.	**Terms:** Open Government Licence – Canada (attribution; commercial OK) **How:** Same special-case pattern as REF-12 **Assessment:** OGL-Canada **Longevity:** statutory open-data framework
REF-12	**US Census cartographic boundaries**	United States	File download	www2.census.gov/geo/tiger/GENZ2023/shp/cb_2023_us_* — county_20m 0.90 MB, county_500k 11.6 MB, cbsa_20m 0.35 MB (counties, CBSA/CSA metros, ZCTAs, tracts)	US detail down to tracts; new vintage each year	None	Public domain	**Yes**	**Easy** P1	● Inst.	**Terms:** Public domain (US govt) **How:** First-class "US special case" like Brazil's IBGE path — tiny fetches, heavier TIGER/Line files available too **Assessment:** Tiny public-domain files (verified sizes) **Longevity:** permanent US-gov open data
REF-13	**GEBCO Undersea Feature Names gazetteer**	Global	File download	Download from GEBCO data-productspage (page URL moved, re-locate on gebco.net) †	~30k named seamounts, trenches, abyssal plains — ocean-floor placemarks	None	Public domain	**Yes**	**Moderate** P3	● Inst.	**Terms:** Public domain (same GEBCO policy as TER-22) **How:** Bundle as undersea placemarks — the "Atlantis-finder" layer for ocean zoom-ins **Assessment:** Locate download on gebco.net **Needs:** URL moved † **Longevity:** GEBCO/IHO
REF-14	**geoBoundaries gbOpen (William & Mary geoLab)**	Global	File download	API `https://www.geoboundaries.org/api/current/gbOpen/{ISO3}/{ADM0-5}/` returns per-country GeoJSON/TopoJSON/shp URLs on GitHub raw; simplifiedGeometryGeoJSON per boundary	~1M boundaries, 200+ entities, ADM0–ADM5; full + simplified per country; CGAZ composites at ADM0/1/2	None	CC BY 4.0	**Yes**	**Moderate** P1	● Org	**Terms:** CC BY 4.0 verbatim ("only requirement... is acknowledgement"); redistribution and bundling allowed **How:** Targeted per-country fetch (a country ADM2 file is ~2–15 MB) fits the existing one-shot GeoJSON path Files are in Git LFS: `raw.githubusercontent.com` returns a ~132-byte pointer (verified 2026-10-19); use `github.com/wmgeolab/geoBoundaries/raw/main/…` (302 to `media.githubusercontent.com`). **Assessment:** `raw.githubusercontent.com` returns an LFS pointer; use `github.com/…/raw/…` **Needs:** LFS-aware URL **Longevity:** academic lab, funded since 2016; releases already public can't be revoked — vendor what you ship

ID	Source	Region	Format	Endpoint / access	Coverage / resolution	Auth	Licence	Comm. use	Integration	Longevity	Notes, terms and caveats
REF-15	**geoBoundaries CGAZ composites**	Global	File download	github.com/wmgeolab/geoBoundaries/raw/main/releaseData/CGAZ/geoBoundariesCGAZ_ADM0.geojson (ADM1/ADM2 same pattern)	Global composites, ADM0 401 MB / ADM1 361 MB / ADM2 550 MB full-resolution GeoJSON	None	CC BY 4.0	**Yes**	**Pre-bake** P3	● Org	**How:** Not a first-load fetch; pre-simplify offline (or bundle a simplified regional subset). Disputed areas are clipped to US State Dept lines Same LFS caveat as REF-14: `raw.githubusercontent.com` returns a ~134-byte pointer (verified 2026-10-19). **Assessment:** Hundreds of MB in Git LFS **Needs:** LFS + pre-simplify **Longevity:** same lab, but the full-size composites will shift every release
REF-16	**GeoFabrik OSM extracts**	Global	File download	download.geofabrik.de daily .osm.pbf / .gpkg per country (0.7– 33 GB)	All admin_level 2–10 + everything else	None	ODbL	**Yes · SA**	**Pre-bake** P3	● Org	**Terms:** ODbL 1.0 **How:** Too broad for runtime; use for offline preprocessing only **Needs:** offline step **Longevity:** 15+-year-old German service, funded; ODbL keeps the data base open
REF-17	**HydroSHEDS**	Global	File download	Vector / raster download	Global rivers and basins, from SRTM	None	Check terms	**Check**	**Pre-bake** P3	● Comm.	**How:** Vendored snapshot **Needs:** offline step **Longevity:** USGS/WWF lineage — terms to verify
REF-18	**Overture Divisions**	Global	GeoParquet	Anonymous GeoParquet over HTTP: overturemapswestus2.blob.core.windows.net/release/2026-09-23.1/theme=divisions/type=* (also s3://overturemaps-us-west-2/release/…)	~5.8M divisions: country → region → county → locality → borough → neighborhood; localized names, population, perspectives (disputed_by / accepted_by) for the politics UI	None	ODbL (+ CC BY sources)	**Yes · SA**	**Pre-bake** P2	● Org	**Terms:** ODbL 1.0 (OSM-derived); per-feature sources[].license also carries geoBoundaries CC BY 4.0; share-alike attribution required **How:** Pre-bake per-country slices at build time (theme is ~5.6 GB zstd; bbox-scanning in-app is awkward). Best source for upgrading the Overpass metro/locality layer into bundled polygons **Needs:** offline step **Longevity:** Microsoft/Meta/Amazon/TomTom-backed foundation — monthly releases since 2023; shares OSM's perpetual ODbL guarantee
REF-19	**GADM v4.1**	Global	File download	Static files geodata.ucdavis.edu/gadm4.1/…	Global ADM0–3 (+ADM4/5 in some countries), 3.5M+ units	None	Non-commercial	**No**	**Excluded**	—	**Terms:** "Freely available for academic and other non-commercial use. Redistribution or commercial use is not allowed without prior permission" — cannot be bundled in an OSS build **How:** Excluded for shipping. Fetch-only personal reference at most; geoBoundaries REF-14 is the open equivalent **Assessment:** Non-commercial, no redistribution **Needs:** licence **Longevity:** (excluded by licence)
REF-20	**osm-boundaries.com**	Global	Viewer only	Web UI, download needs OAuth-OSM login; monthly import	All admin levels, configurable export	OAuth login	ODbL (OAuth-gated)	**Check**	**Excluded**	—	**Terms:** ODbL passthrough; commercial operator (Ground Zero Communications AB) **How:** Not usable as an anonymous fetch path — keep Overpass (already wired) for live OSM boundaries instead **Assessment:** No anonymous fetch path **Needs:** OAuth-gated **Longevity:** ✗ (no server-based API)
REF-21	**UN SALB**	Global	File download	Per-country on salb.un.org (uneven coverage), UN ArcGIS FeatureServers, no registration	ADM1/ADM2 to a 1:1m target	None	Non-commercial	**No**	**Excluded**	—	**Terms:** Non-commercial only, revocable, no geometry changes; UN copyright + contributor credit. gbAuthoritative inherits the restriction **How:** Excluded for shipping **Assessment:** Non-commercial, revocable **Needs:** licence **Longevity:** (excluded by licence)
REF-22	**Rosreestr Публичная кадастровая карта / НСПД (ru)**	Russia	Viewer only	Web viewer; unofficial community WMS endpoints exist	Russia: cadastral parcels, ortho/basemap in the viewer	None	No open licence	**No**	**Excluded**	● Fragile	**Terms:** No open licence; commercial use of the cadastral map is not permitted (OSM community note). Unofficial WMS = legally murky †. Treat as display-only and effectively restricted **How:** Do not ship **Assessment:** No open licence **Needs:** restricted **Longevity:** effectively closed licensing

ID	Source	Region	Format	Endpoint / access	Coverage / resolution	Auth	Licence	Comm. use	Integration	Longevity	Notes, terms and caveats
REF-23	**TKGM Ada/Parsel Sorgu / TKKG (tr)**	Turkey	REST / JSON API	Parcel query API + WMS/WMTS behind registration (parselsorgu.tkgm.gov.tr)	Turkey: cadastral parcels, orthophoto in the TKKG viewer	Free account	Restricted †	**No**	**Excluded**	● Fragile	**Terms:** Government service, restricted use, no open bulk † **How:** Registration-gated parcel API; no open bulk **Assessment:** Restricted use **Needs:** registration **Longevity:** restricted use

Live Tracking & Transport (TRK) — 8 sources

ID	Source	Region	Format	Endpoint / access	Coverage / resolution	Auth	Licence	Comm. use	Integration	Longevity	Notes, terms and caveats
TRK-01	**OpenSky Network**	Global	REST / JSON API	https://opensky-network.org/api/states/all?lamin=..&lomin=..&lamax=..&lomax=..	Live aircraft, bbox queries	Optional key	ToU (research-oriented)	**Check**	**Easy** P3	● Comm.	**Terms:** Anonymous access is rate-limited, credits apply **How:** Geo fetch with credit budget; degrade gracefully to adsb.lol (already wired in the binary) **Assessment:** Already merged with ADSB.lol **Needs:** already in binary **Longevity:** non-profit; the anonymous tier has been shrinking — non-default layer
TRK-02	**OurAirports**	Global	File download	CSV: https://davidmegginson.github.io/ourairports-data/airports.csv (+ runways.csv , navaids.csv , airport-frequencies.csv)	~85,000+ airports worldwide with runway geometry; rebuilt daily	None	Unlicense (public domain)	**Yes**	**Easy** P2	● Comm.	**Terms:** Public domain (Unlicense) **How:** One-shot CSV straight into the existing placemark pipeline **Assessment:** CSV into placemarks **Longevity:** single-maintainer but Unlicense: bundle a snapshot and the data is yours forever
TRK-03	**AISstream.io**	Global	WebSocket stream	WebSocket wss://stream.aisstream.io/v0/stream . Send subscription (key + bbox) within 3 s	Live global vessel positions	Free key	Free, fair use (beta)	**Check**	**Moderate** P3	● Comm.	**Terms:** Free, fair use, beta / unstable. Key must not ship in client builds **How:** WS client + key env; keep the HTTP fallback (OpenSeaFeed / digitraffic are already in the binary) **Assessment:** HTTP fallbacks already wired; AISstream is beta **Needs:** WebSocket client + key (D) **Longevity:** beta service — treat the fallbacks as the primary plan
TRK-04	**CelesTrak GP**	Global	Feed (JSON/CSV/text)	https://celestrak.org/NORAD/elements/gp.php?GROUP=active&FORMAT=json	About 16,000 active satellites. Needs SGP4 propagation	None	Free	**Yes**	**Moderate** P2	● Comm.	**Terms:** Free. Cache, refresh every few hours **How:** Geo fetch of GP daily + an SGP4 crate Prefer the `sgp4` crate (2.4.0, pure Rust, MIT) over `satellite` (0.1.2, stale since 2024); see TL-11. **Assessment:** OMM decode exists; replace two-body with SGP4 **Needs:** `sgp4` crate **Longevity:** run by T.S. Kelso since the 1980s; dependable but community-run
TRK-05	**Earth Traffic (iTIC/ELEMNT platform)**	Global	Vector tiles / PMTiles	`https://traffic.elemnt.earth/api/v1/tiles/h3-traffic/{z}/{x}/{y}.mvt?at=`<ISO hour>	Global H3 congestion hexes from 3.4B GPS probes, hourly archive 2017–now; live ingestion "forthcoming"	None	CC BY 4.0	**Yes**	**Moderate** P3	● Comm.	**Terms:** CC BY 4.0, attribution **How:** MVT decode is new code (no vector parser for H3 yet); great for replay/time-slider, not for live jam colouring **Longevity:** one small foundation, demo service — vendor the CC BY archive if relied on
TRK-06	**openAIP airspace**	Global	REST / JSON API	Free API key; weekly airports + airspace volumes + navaids dumps	Global aviation data (airspace polygons would draw the "no-fly zones" over the globe)	Free key	CC BY-NC 4.0 †	**No**	**Moderate** P3	● Fragile	**Terms:** CC BY-NC 4.0 † — non-commercial only **How:** Key client — opt-in plugin only, like GFW MAR-02; never a default layer **Needs:** key (D); gate: non-commercial **Longevity:** NC licence + community model
TRK-07	**NDW (Netherlands road data)**	Netherlands	Feed (JSON/CSV/text)	https://opendata.ndw.nu/ DATEX II XML	NL only; 24k point-speed sites, ~1 min	None	Open government data	**Check**	**Pre-bake** P3	● Inst.	**Terms:** Open gov **How:** Custom DATEX II parser + location tables; useful per-country pattern only **Needs:** offline step **Longevity:** statutory national traffic authority

ID	Source	Region	Format	Endpoint / access	Coverage / resolution	Auth	Licence	Comm. use	Integration	Longevity	Notes, terms and caveats
TRK-08	**ADS-B Exchange (live aircraft)**	Global	REST / JSON API	Paid now: Community API via RapidAPI $10/mo per 10k req (non-commercial), Enterprise tiers for real use; volume keys never ship in a client	Global, ~500+ community-fed receivers, 500 ms updates	Paid	Paid, commercial licence	**No**	**Excluded**	● Comm.	**Terms:** Monthly sub, commercial licence only. Free paths: self-hosted receiver (RTL-SDR + dump1090/readsb), or community aggregators adsb.lol (free, ODbL API) and airplanes.live (free; access granted by email with a project description) **How:** adsb.lol HTTP path (already in the binary); ODbL attribution **Assessment:** Use adsb.lol (already in binary) **Needs:** paid **Longevity:** community network — ODbL guarantees the data stays open even if the host dies

Hazards & Space Weather (HAZ) — 12 sources

ID	Source	Region	Format	Endpoint / access	Coverage / resolution	Auth	Licence	Comm. use	Integration	Longevity	Notes, terms and caveats
HAZ-01	**NOAA SWPC aurora (OVATION)**	Global	Feed (JSON/CSV/text)	https://services.swpc.noaa.gov/json/ovation_aurora_latest.json	Global probability grid, 30– 90 min forecast	None	NOAA public data	**Yes**	**Easy** P1	● Inst.	**Terms:** NOAA public **How:** Geo fetch every ~30 min into a filled-polygon overlay **Assessment:** One JSON → filled polygons **Longevity:** NOAA
HAZ-02	**NOAA SWPC space weather**	Global	Feed (JSON/CSV/text)	JSON under services.swpc.noaa.gov/json/ (Kp index, X-ray flares, solar wind)	Global	None	NOAA public data	**Yes**	**Easy** P1	● Inst.	**Terms:** NOAA public **How:** Same pattern as HAZ-01 **Longevity:** NOAA
HAZ-03	**Smithsonian GVP volcano list**	Global	File download	Download at volcano.si.edu/database/list_volcano_holocene_excel.cfm (XML-Excel / CSV ~1,400 Holocene volcanoes; VEI lives in the separate eruption list). Gotchas: Cloudflare 403s plain HTTP clients (browser-like fetch required), and search/download availability is uncertain during the 2026 site rebuild	Global	None	Non-commercial (Smithsonian)	**No**	**Easy** P2	● Comm.	**Terms:** Smithsonian terms: free for personal / educational / non-commercial with attribution. Better claim-free alternative (live-tested, ~1,600 entries): NOAA NCEI https://www.ngdc.noaa.gov/hazel/hazard-service/api/v1/volcanolocs (JSON, public domain, VEI per eruption under /volcanoes) **How:** Vendored CSV; or live Geo fetch from NOAA NCEI instead (public domain, no terms friction) **Assessment:** Prefer NOAA NCEI JSON over Smithsonian **Needs:** use NCEI endpoint **Longevity:** Smithsonian terms are non-commercial; the NCEI alternative is
HAZ-04	**USGS earthquakes**	Global	Feed (JSON/CSV/text)	GeoJSON summary feeds: earthquake.usgs.gov/earthquakes/feed/v1.0/summary/{significant,4.5,2.5,1.0,all}_{hour,day,week,month}.geojson ; FDSN event search at /fdsnws/event/1/query	Global quakes, live (hour/day feeds update every minute)	None	Public domain	**Yes**	**Easy** P1	● Inst.	**Terms:** Verified 2026-10-18: public domain, no key. USGS recommends the feeds for real-time display and asks clients not to hammer /query ; the old "fetch details 1/s" advice stays unconfirmed **How:** Geo fetch (already how the app consumes it) **Assessment:** GeoJSON summary feeds, no key **Longevity:** US federal public domain
HAZ-05	**NWS active alerts**	United States	REST / JSON API	api.weather.gov/alerts/active	US	None	Public domain	**Yes**	**Easy** P1	● Inst.	**Terms:** Public domain. Verified 2026-10-18: a User-Agent header identifying the app with contact info is required (official format (myweatherapp.com, contact@myweatherapp.com)); NWS says it may be replaced by an API key in future. Rate limits exist but are unpublished — on a limit, retry after ~5 s **How:** Geo fetch; send a User-Agent like EarthCraft/<version> (+https://github.com/therichronicles/EarthCraft) **Assessment:** GeoJSON alerts; send an identifying UA **Needs:** User-Agent header **Longevity:** US federal public domain

ID	Source	Region	Format	Endpoint / access	Coverage / resolution	Auth	Licence	Comm. use	Integration	Longevity	Notes, terms and caveats
HAZ-06	**FreeLightning (parsed GLM)**	Global	Feed (JSON/CSV/text)	https://www.freelightning.com/glm/… plain-text placefile, 40 s	GLM flashes reparsed	None	Credit required (hobby host)	Check	Moderate P3	● Fragile	**Terms:** "Must credit FreeLightning.com" **How:** Trivial text parse — good starter lightning layer **Assessment:** Trivial parse; hobbyist host **Needs:** text parser **Longevity:** hobbyist-run single maintainer; keep HAZ-11 as the durable fallback
HAZ-07	**GloFAS river discharge (Copernicus CEMS)**	Global	File download	GRIB/netCDF via ewds.climate.copernicus.eu/datasets/cems-glofas-forecast ; WMS-T at global-flood.emergency.copernicus.eu	Global 0.05° (v4), 30-day forecasts, reanalysis back to 1979	Free account	Free & open (CEMS)	Yes	Moderate P3	● Inst.	**Terms:** Free & open (CEMS); registration; explicitly not a time-critical operational service **How:** Not runtime material (GRIB + queue); use the WMS-T flood maps as a tile layer and pre-bake analytics **Assessment:** WMS-T flood maps; account for analytics **Needs:** WMS adapter (A) **Longevity:** Copernicus institutional
HAZ-08	**IBTrACS tropical cyclone best tracks (NOAA NCEI)**	Global	File download	CSV/netCDF at NCEI	Global historical tracks combining best-track data from 10+ forecast centres	None	NOAA public †	Yes	Moderate P3	● Inst.	**How:** Vendored CSV; animate history **Longevity:** NOAA NCEI
HAZ-09	**NASA FIRMS**	Global	REST / JSON API	API / WMS / WFS. Key at firms.modaps.eosdis.nasa.gov/api/map_key/	Global fire detections, 375 m VIIRS / 1 km MODIS, within 3 h	Free key	Free (NASA)	Yes	Moderate P3	● Inst.	**Terms:** Free. 5,000 requests per 10 min **How:** Key client + prefetched fire tiles **Assessment:** Key client; 5,000 requests / 10 min **Needs:** key (D) **Longevity:** NASA — 5,000 req/10 min free tier
HAZ-10	**NHC GIS products (US National Hurricane Center)**	Atlantic, E/C Pacific	Feed (JSON/CSV/text)	nhc.noaa.gov/gis/: shapefile/KMZ per basin (Atlantic, Eastern and Central Pacific) for cone, track, warnings, wind radii, surge, outlooks; a JSON status file lists active storms with zip/KMZ links; an active KML feed for all cyclones	Atlantic and eastern/central Pacific	None	US federal	Yes	Moderate P2	● Inst.	**Terms:** US federal; NHC notes the GIS data is provided as a convenience **How:** Geo fetch of the JSON status plus KMZ (the app already reads KML via GeoRust kml); cone and track overlay **Longevity:** NOAA
HAZ-11	**NOAA GOES GLM lightning**	Americas	Cloud files (S3/COG/Zarr)	Anonymous S3: s3://noaa-goes{18,19}/GLM-L2-LCFA/YYYY/DDD/HH/ (~20-s netCDF flash-group-event files)	Americas + Central America discs, ~8 km	None	Public domain	Yes	Pre-bake P3	● Inst.	**How:** S3 netCDF polling is real engineering; practical variant is a small pre-parsed flash JSON snapshot per 10 minutes **Needs:** offline step **Longevity:** NOAA — permanent open archive
HAZ-12	**Blitzortung.org**	Global	Feed (JSON/CSV/text)	data.blitzortung.org (protected endpoints), min-latency global regions	Global community receiver network	Members only	Prohibited	No	Excluded	—	**Terms:** Redistribution and commercial use prohibited **How:** Excluded — no app-tier feed without joining the network **Assessment:** No app-tier feed **Needs:** participants only

Marine & Maritime (MAR) — 5 sources

ID	Source	Region	Format	Endpoint / access	Coverage / resolution	Auth	Licence	Comm. use	Integration	Longevity	Notes, terms and caveats
MAR-01	**IMF PortWatch**	Global	REST / JSON API	ArcGIS FeatureServer queries on services9.arcgis.com: Daily_Trade_Data (filter portid) and Daily_Chokepoints_Data; exact URL in the PortWatch FAQ	Ports and chokepoints: daily vessel calls and trade estimates	None	IMF terms †	Check	Easy P3	● Inst.	**How:** Geo fetch per port id. Trade values are estimates: smooth with a 7–14-day mean (per the FAQ) **Assessment:** ArcGIS query per port
MAR-02	**Global Fishing Watch (4WINGS)**	Global	REST / JSON API	REST v3 at gateway.api.globalfishingwatch.org/v3 : gridded apparent fishing effort as PNG/MVT tiles + CSV/JSON/TIFF reports. Datasets: public-global-fishing-effort:latest , -sar-presence , -presence	Global ocean 0.01° grid, daily, 2012–present	Free key	CC BY-NC 4.0	No	Moderate P3	● Org	**Terms:** CC BY-NC 4.0 — non-commercial only, "Powered by Global Fishing Watch" attribution mandatory; 50,000 req/day, 1.5M/mo **How:** Key client; PNG/MVT effort + CC BY-NC means non-commercial builds only **Assessment:** CC BY-NC: non-commercial builds only **Needs:** key (D); gate: non-commercial **Longevity:** well-funded non-profit, but the -NC licence is a hard constraint on shipping

ID	Source	Region	Format	Endpoint / access	Coverage / resolution	Auth	Licence	Comm. use	Integration	Longevity	Notes, terms and caveats
MAR-03	**Marine Regions: EEZ v12 + High Seas v2 (VLIZ, IOC-UNESCO)**	Global	Vector tiles / PMTiles	Shapefile/GeoJSON via marineregions.org; WFS/WMS/WMTS on the VLIZ GeoServer; GeoParquet `https://data.source.coop/cboettig/high-seas/iho/eez.parquet` and PMTiles `…/eez.pmtiles`	Global maritime boundaries and EEZs	None	CC BY 4.0	Yes	Moderate P2	Org	**Terms:** CC BY 4.0. VLIZ asks users not to host the files for download elsewhere and to credit marineregions.org **How:** Tile path for the PMTiles (vector); draw as a border overlay like the boundary layers (REF-*) **Needs:** `pmtiles` (C); WMS adapter (A) **Longevity:** VLIZ, funded via LifeWatch Belgium
MAR-04	**TeleGeography submarine cable map**	Global	File download	Map images are CC BY-SA 4.0. The data behind them is licensed annually for commercial use; community GitHub copies and scrapers have unclear rights	Global cables and landing points	—	Not open (commercial data)	No	Excluded	Fragile	**Terms:** Not open **How:** Do not ship. OSM telecom lines via INF-05 are the open alternative **Assessment:** Commercial data **Needs:** not open **Longevity:** not open
MAR-05	**WDPA, EMODnet vessel density, GDACS, Copernicus Marine †**	Global	—	Leads, not verified	Protected areas; vessel density; disaster alerts; ocean models	—	—	—	n/a	—	**How:** Not assessed **Assessment:** Leads only **Status:** Lead only, not verified

Heritage, Archaeology & Nature (HER) — 25 sources

ID	Source	Region	Format	Endpoint / access	Coverage / resolution	Auth	Licence	Comm. use	Integration	Longevity	Notes, terms and caveats
HER-01	**GBIF species occurrences**	Global	REST / JSON API	Keyless REST verified: api.gbif.org/v1/species/match?name=… (live) + /v1/occurrence/search?taxonKey=…&hasCoordinate=true ; also bulk occurrence snapshots on AWS S3 (registry.opendata.aws/gbif)	~2.5 B geo-tagged occurrence records, all taxa, global	None	CC BY 4.0 / CC0 per dataset	Yes	Easy P2	Org	**Terms:** CC BY 4.0 / CC0 per contributing dataset; attribution "GBIF" **How:** Geo fetch — species-searchplacemark layer ("where do golden eagles live"); bulk pre-bake from the S3 snapshot **Assessment:** Keyless REST **Longevity:** GBIF secretariat (30+ countries), permanent DOIs
HER-02	**GeoNames heritage feature codes**	Global	File download	Already-downloaded dumps (REF-01): filter featureClass=S codes — verified on the codes page: RUIN ruins, PYR/PYRS pyramids, RLG "ancient site of significant religious importance", WALLA "ancient wall", TMB tombs, SHRN shrines, HSTS historical sites, FT forts/earthworks, WRCK wrecks	Global points; density varies by country	None	CC BY 4.0	Yes	Easy P2	Comm.	**How:** No new fetch — a filter over the REF-01 pipeline (placemark layer "archaeology"); the codes list above is the exact filter set **Needs:** filter over existing dumps **Longevity:** GeoNames CC BY; bundle a snapshot and it is yours
HER-03	**Paleobiology Database (fossils)**	Global	REST / JSON API	Keyless JSON verified: paleobiodb.org/data1.2/colls/list.json?lngmin=…&latmin=…&show=full (live — returns lat/lon + geological age per fossil collection); also colls/geo.json GeoJSON	~200k fossil collections worldwide with age (Maastrichtian, Pleistocene…)	None	CC BY †	Check	Easy P2	Org	**Terms:** CC BY per PBDB terms † (attribution; cite collection refs) **How:** Geo fetch — "fossil sites" placemark layer with epoch filter; pairs with the Other Worlds rows (OTH-*) for deep-time mode **Assessment:** Keyless JSON; use `colls/list.json` **Longevity:** 400+ institutions; two decades
HER-04	**Wikidata "archaeological site" (Q839954) SPARQL**	Global	REST / JSON API	query.wikidata.org SPARQL with coordinate export; also PeriodO (period definitions, CC0) for time-sliders	Global; completeness wildly uneven by country	None	CC0	Yes	Easy P3	Org	**Terms:** CC0 (Wikidata) — attribution-free **How:** Geo fetch / SPARQL one-shot into the placemark pipeline; good cross-check layer for dedup **Assessment:** CC0 **Needs:** SPARQL one-shot **Longevity:** Wikimedia infrastructure, CC0
HER-05	**EMODnet Human Activities — Cultural Heritage**	Europe	WMS	WMS verified in GetCapabilities: layers heritageshipwrecks , heritageshipwrecksv2 , wwshipwrecks ; WFS + CSV download on the portal; also Submerged Prehistoric Archaeology and Landscapes points	European seas: wreck positions (heritage + WWI/WWII), submerged prehistoric sites	None	CC BY 4.0 †	Check	Easy P2	Inst.	**Terms:** EMODnet open-data (CC BY 4.0 † — confirm on the download page) **How:** Tile path (WMS) or WFS one-shot into the vector overlay — zero new infrastructure **Assessment:** Shipwreck layers **Needs:** WMS adapter (A) **Longevity:** EU/CINEA-funded service line

ID	Source	Region	Format	Endpoint / access	Coverage / resolution	Auth	Licence	Comm. use	Integration	Longevity	Notes, terms and caveats
HER-06	**Pleiades (ancient Mediterranean places)**	Mediterranean	REST / JSON API	JSON per place: pleiades.stoa.org/places/{id}/json (live-tested); bulk "Pleiades Datasets 4.1" downloads on the site	41,480 citable ancient places (v4.1, May 2025): Greek/Roman/Levantine sites, forts, tombs	None	CC BY 3.0	**Yes**	**Easy** P2	● Org	**Terms:** Footer verified: "Creative Commons Attribution 3.0 License (cc-by)" — attribution + cite DOI per place **How:** Geo fetch or bundle the v4.1 CSV/GeoJSON dumps into the placemark pipeline (one-time asset, small) **Assessment:** 41k places, CC BY 3.0 **Needs:** bundle CSV **Longevity:** academic consortium (NYU/ISAW), two decades running
HER-07	**Stonehenge LiDAR Archaeology Landscape (Zenodo)**	United Kingdom	File download	10.5281/zenodo.10334054 , direct download	Stonehenge, Avenue, Cursus, barrow field	None	CC BY 4.0	**Yes**	**Easy** P3	● Org	**Terms:** CC BY 4.0, open **How:** Bundle — small pre-encoded show-case site (Neolithic/Medieval analogue for tells) **Longevity:** Zenodo DOI
HER-08	**Movebank animal tracking**	Global	REST / JSON API	REST www.movebank.org/movebank/service/direct-read (Earthdata login)	Live GPS tracks of tagged animals (birds, whales), 3B+ locations	Free account	Per study	**Check**	**Moderate** P3	● Org	**Terms:** Per-study licences (CC BY / CC0 / restricted) † — check each study's licence flag **How:** Geo fetch — animated migration overlay for flagship open-licence studies only **Needs:** account login **Longevity:** Max Planck/AGB consortium
HER-09	**UNESCO World Heritage list**	Global	File download	Official list XML whc.unesco.org/en/list/xml — exists but Cloudflare-gates plain HTTP clients (verified) — browser-UAfetch or mirror needed †	1,200+ WH sites incl. all major archaeology properties	None	UNESCO terms †	**Check**	**Moderate** P3	● Inst.	**Terms:** UNESCO terms apply — redistribution text unverified † **How:** Geo fetch with browser-like headers; kiosk-quality fallback: OSM heritage=2 tag mining via Overpass (already wired) **Longevity:** UNESCO institution; mirror risk low
HER-10	**Acre geoglyph site vectors (de Souza 2018, Nat Commun)**	Brazil (Acre)	File download	nature.com/articles/s41467-018-03510-7 supplementary (OA)	~47,887 km² southern rim of Amazonia; geoglyph/earthwork polygons	None	CC BY 4.0	**Yes**	**Moderate** P3	● Org	**Terms:** Article + supplement CC BY 4.0; raw LiDAR never published **How:** Bundle site vectors as placemark overlay on the ANADEM (TER-11) hillshade **Longevity:** Springer OA, stable DOI
HER-11	**Geoglyph DTM-anomaly toolkit (Zenodo)**	Peru	File download	10.5281/zenodo.5087620 — code + sample data for "Fast computation of digital terrain model anomalies... for geoglyph detection"	Nazca/Peru sample; method not full coverage	None	CC BY 4.0	**Yes**	**Moderate** P3	● Org	**Terms:** CC BY 4.0, open **How:** Offline guidance only — the algorithm EarthCraft could run over TER-11 ANADEM tiles **Longevity:** Zenodo DOI
HER-12	**Kilwa Kisiwani 3D (CyArk, Zenodo)**	Tanzania	File download	10.5281/zenodo.10377387 — .glb meshes	Swahili port-town monuments; terrestrial LiDAR + photogrammetry	None	CC BY-NC-SA 2.0	**No**	**Moderate** P3	● Org	**Terms:** CC BY-NC-SA 2.0 — non-commercial + share-alike **How:** Bundle glTF as site overlay — only if the NC licence is acceptable **Assessment:** Non-commercial licence **Needs:** gate: NC-SA **Longevity:** Zenodo DOI stable
HER-13	**Loreto sites gazetteer (Coomes 2021, Sci Data)**	Peru	File download	figshare collection 10.6084/m9.figshare.c.5262530 (Excel + API verified)	415 geolocated archaeological sites, Loreto, Peruvian Amazon	None	CC BY 4.0	**Yes**	**Moderate** P3	● Org	**Terms:** CC BY 4.0 (verified in figshare) **How:** Bundle vector placemarks — the target list for any future LiDAR there **Longevity:** figshare DOI + Springer OA
HER-14	**National heritage registers: UK / IE**	UK & Ireland	REST / JSON API	Historic England NHLE data services (API, free key) — Cloudflare-blocks plain clients † · Canmore (Scotland, OGL †) · Coflein (Wales) · Ireland Historic Environment Viewer / RMP shapefiles (301-redirect verified, exact open-licence text †)	UK + Ireland: listed buildings, scheduled monuments, SMR/RMP sites	Free key	OGL / open reuse (verify)	**Check**	**Moderate** P3	● Inst.	**Terms:** OGL / open-reuse family (attribution) — per-source text to verify before shipping **How:** Per-country plugin in the REF-06, REF-07, REF-10, REF-08 pattern (key from config if required) **Needs:** key (D) **Longevity:** statutory heritage agencies
HER-15	**NPS National Register of Historic Places**	United States	File download	Bulk data page nps.gov/subjects/nationalregister/database.htm (page live; bulk CSV link not grep-visible — JS) †	~95,000 US historic places incl. archaeological districts	None	Public domain	**Yes**	**Moderate** P3	● Inst.	**Terms:** Public domain (US federal) **How:** One-shot CSV into the placemark pipeline **Assessment:** Bulk CSV link not confirmed **Needs:** JS-hidden link † **Longevity:** US federal

ID	Source	Region	Format	Endpoint / access	Coverage / resolution	Auth	Licence	Comm. use	Integration	Longevity	Notes, terms and caveats
HER-16	**OpenTopography archaeology LiDAR collections**	Central America	File download	Direct DOI → public .laz tiles on opentopography.s3.sdsc.edu (Caracol verified keyless)	Caracol/Chiquibul Belize (10.5069/G9833Q42: 64 km², 1.5 B pts, ~24 pts/m²) · Middle Usumacinta, Mexico (10.5069/G95B00NF) · Quirigua UAV, Guatemala (10.5069/G9416V95)	None	CC BY 4.0 (page: not provided)	Check	Moderate P3	● Org	**Terms:** CC BY 4.0 in DCAT; Caracol page itself says "Use License: Not Provided" † — cite PI + DOI, keep the ambiguity flag **How:** Pre-bake — ground-classified DEM tiles per landmark Maya site, bundled; .laz too heavy for runtime **Longevity:** / single-project datasets on durable NSF storage; DOIs stable
HER-17	**SE-Acre mound villages (Iriarte 2020, JCAA)**	Brazil (Acre)	File download	journal.caa-international.org/articles/10.5334/jcaa.45 — full OA text	65.6 km² transects (Oct-2018 RIEGL flights); Table 1 = UTM coords of 57 mound villages	None	CC BY 4.0	Yes	Moderate P3	● Org	**Terms:** Article CC BY 4.0 ; DTM not published (ERC PAST/FUTURES) **How:** Extract Table 1 → placemark layer; terrain stays research-held **Longevity:** Ubiquity/CAA OA journal
HER-18	**Randolph Glacier Inventory (RGI 7.0) / GLIMS**	Global	File download	Site live (glims.org/RGI/); RGI 7.0 via NSIDC; GLIMS GLIMS glacier database WMS	~215,000 glacier outlines, global, versioned snapshots	None	Research / education †	Check	Pre-bake P3	● Inst.	**Terms:** Freely available for research and education † (exact statement to re-read on the RGI page) **How:** Pre-bake — glacier polygons as terrain "icing" mask or white-cap overlay **Needs:** offline step **Longevity:** NSIDC + GLIMS consortium
HER-19	**Figshare: ML-ready Maya remote sensing (Chactún)**	Mexico	File download	API + HTTPS verified: api.figshare.com/v2/articles/22202395 — Chactun_ML_ready_lidar.zip ≈ 1.08 GB ALS raster stack + masks/CHM, ~2.1 GB total	Chactún area, Campeche, Maya lowlands; ALS-derived DTM/CHM tiles	None	CC BY 4.0	Yes	Pre-bake P3	● Org	**Terms:** CC BY 4.0 (verified in Figshare API licence field) **How:** Pre-bake — decode ALS rasters to hillshade/relief tiles offline, bundle as the first fly-over Maya landmark **Needs:** offline step **Longevity:** Figshare DOI permanent; files stable
HER-20	**NOAA ENC chart data — WRECKS/OBSTRN**	United States	File download	S-57 ENC downloads (keyless) + chart-vector pipeline; wreck (WRCK), obstruction (OBSTRN) are standard S-57 object classes	US waters and coasts, chart-accurate	None	Public domain	Yes	Pre-bake P3	● Inst.	**Terms:** Public domain (US federal) **How:** Pre-bake — S-57 → GeoJSON offline (no runtime S-57 decoder), ship as wreck placemarks **Needs:** offline step **Longevity:** NOAA permanent
HER-21	**UNESCO Underwater Cultural Heritage database**	Global	Viewer only	Web viewer only (unesco.org/en/underwater-cultural-heritage); no open API/bulk found (fetches empty) †	Global designated UCH sites	None	Unclear (viewer only)	No	Excluded	● Fragile	**Terms:** Unclear; viewer-only **How:** Not usable now — re-check if UNESCO opens bulk data **Assessment:** Viewer only **Needs:** no API **Longevity:** no API — excluded for now
HER-22	**Dryad PACUNAM 2,144 km² Guatemala LiDAR**	Guatemala	File download	10.5061/dryad.k51j708 — only 25.04 MB of derived shapes/stats	Northern Guatemala (the famous 2018 Science survey) shapes only — no DTM/point cloud	None	CC0 † (derived data only)	Yes	Excluded	● Inst.	**Terms:** CC0 † for the derived data; raw terrain closed / on request **How:** Not a terrain source — cite HER-16-style pre-bakes instead **Assessment:** Derived shapes only **Needs:** terrain closed **Longevity:** metadata; raw stays closed
HER-23	**Great Zimbabwe (Zamani Project)**	Zimbabwe	Viewer only	zamaniproject.org viewer	Great Enclosure + Hill Complex terrestrial laser-scan models	None	Academic courtesy only	No	Excluded	● Comm.	**Terms:** Academic courtesy reuse — clarify per request **How:** Link 3-D models; no bulk data **Assessment:** No bulk data **Needs:** on request **Longevity:** university-hosted, request-based
HER-24	**Peru / Colombia / Ecuador national LiDAR**	South America	—	None found	Ministry of Culture (Machu Picchu, Nasca): restricted heritage data; IDEAM Colombia: claim unverifiable (datos.gov.co Socrata search = 0 hits); Ecuador: no ministerial portal	—	Not open	No	Excluded	● Fragile	**Terms:** Not open / not verifiable **How:** No action — re-check if portals appear **Assessment:** Closed or non-existent **Needs:** not open **Longevity:** closed or non-existent
HER-25	**Archaeology Data Service (ADS)**	United Kingdom	Catalog / portal	Catalogue + DOI archives, mostly UK (Environment-Agency-grade LiDAR)	UK-dominant; no non-UK LiDAR surfaced (search is an SPA — not exhaustive) †	None	Per collection	Check	n/a	● Inst.	**Terms:** "…licence conditions clearly identified within each data collection"; metadata CC BY 4.0 **How:** Not a pipeline — copy its per-collection licence-tagging pattern; UK bundle-quilt option only **Assessment:** Pattern to copy, not a pipeline **Longevity:** CoreTrustSeal curated

Other Worlds (OTH) — 3 sources

ID	Source	Region	Format	Endpoint / access	Coverage / resolution	Auth	Licence	Comm. use	Integration	Longevity	Notes, terms and caveats
OTH-01	**NASA Mars Trek**	Global	Tiles (XYZ/WMTS)	RESTful WMTS (api.nasa.gov/mars-wmts/catalog)	Mars: MOLA hillshade, Viking, THEMIS, 21 products	None	NASA open data	**Yes**	**Moderate** P3	● Inst.	**Terms:** NASA open. Missing tile means no coverage **How:** Same as OTH-02 **Assessment:** As OTH-02 **Needs:** per-body projection **Longevity:** NASA
OTH-02	**NASA Moon Trek**	Global	Tiles (XYZ/WMTS)	RESTful WMTS. Product list: trek.nasa.gov/tiles/apidoc/trekAPI.html?body=moon	Moon mosaics, DEMs, equirectangular + polar	None	NASA open data	**Yes**	**Moderate** P3	● Inst.	**Terms:** NASA open. Read each product's capabilities for matrix set and extension **How:** Tile path + non-WGS84 projection handling per body **Assessment:** Needs body-aware tile math (Moon is an embedded image today) **Needs:** per-body projection **Longevity:** NASA
OTH-03	**NASA Vesta Trek**	Global	Tiles (XYZ/WMTS)	RESTful WMTS	Vesta, 6 Dawn products	None	NASA open data	**Yes**	**Moderate** P3	● Inst.	**Terms:** NASA open **How:** Same as OTH-02 **Assessment:** As OTH-02 **Needs:** per-body projection **Longevity:** NASA

Catalogs & Discovery (CAT) — 4 sources

ID	Source	Region	Format	Endpoint / access	Coverage / resolution	Auth	Licence	Comm. use	Integration	Longevity	Notes, terms and caveats
CAT-01	**Earth Search (Element84)**	Global	Catalog / portal	STAC https://earth-search.aws.element84.com/v1	Sentinel-2 COGs (11M+ scenes), Landsat, NAIP	None	Free (scene licences apply)	**Check**	**n/a**	● Inst.	**Terms:** Free, scene licences apply **How:** Discovery portal (STAC) for scene pulls; items are pre-bake input, not runtime layers **Assessment:** Discovery catalogue, not a layer **Longevity:** (Sentinel-2 source) / (Element 84 mirror) — AWS Open Data Registry
CAT-02	**Esri Open Data Hub**	Global	Catalog / portal	Web portal	250,000+ datasets from 5,000+ organisations	Varies	Per dataset	**Check**	**n/a**	—	**Terms:** Per-dataset **How:** Discovery portal only **Longevity:** mixed — evaluate per dataset
CAT-03	**Microsoft Planetary Computer**	Global	Catalog / portal	STAC https://planetarycomputer.microsoft.com/api/stac/v1	20+ PB, anonymous access to most datasets	None	Per dataset	**Check**	**n/a**	● Org	**Terms:** Per-dataset **How:** Discovery + per-dataset access **Assessment:** Discovery catalogue, not a layer **Longevity:** Microsoft Planetary Computer programme
CAT-04	**Google Earth Engine (Data Catalog and REST tile API)**	Global	REST / JSON API	Data Catalog: https://developers.google.com/earth-engine/datasets. REST: maps.create (map ID valid a few hours) then tiles.get with an OAuth token. Client libraries: Python and JavaScript (`earthengine-api`). No Rust client.	Global archive (Landsat, Sentinel, MODIS, climate and more); resolution and dates per dataset	Free account (Cloud project)	Per dataset + Earth Engine terms	**Check**	**Moderate** P3	● Comm.	**Terms:** Noncommercial use is free with a monthly compute quota since 2026-04-27 (Community, Contributor and Partner tiers); noncommercial status must be re-verified every year; unregistered or commercially registered projects can be charged. Quota belongs to one Cloud project and cannot be transferred, so a shared project inside a distributed app would run out quickly. The older Additional API Terms (page last updated 2024) limit the API to development, research and education and bar sustained commercial use. **How:** Opt-in bring-your-own-Google-Cloud-project layer, like the key-based traffic layer: OAuth loopback, expression JSON built in Rust from fixed templates only, tiles cached under a provider identity per (collection, parameters). The JavaScript client suits the planned web frontend. **Needs:** REST client + OAuth + user project (D) **Longevity:** Google service: depends on one company's goodwill and quota policy.

Appendix A. Findings

Topic	Finding	Related
Real 3D terrain	The README gap is closed by AWS Terrain Tiles (TER-06) and Mapterhorn (TER-09). Mapterhorn already bundles national LiDAR for many countries (NL, ES, CH, JP, all German states, Austria, Denmark, Finland, Italy regions, UK, US, NZ and more), so per-country terrain rows matter only where it has no data. AWS Terrain Tiles are an older compilation in Europe.	TER-06, TER-09
Borders and disputed areas	Natural Earth map_units is the always-renderable default (type / disp fields); geoBoundaries per-country gives "as the country represents itself" views; Overture perspectives is the only first-class multi-narrative dataset. OSM has one consensus set plus unofficial disputed=* tags (what the Overpass layers see today). A 10m URL swap roughly triples country-border detail.	REF-03, REF-14, REF-18
Traffic	No free, key-free, global, live traffic feed exists (OSM serves none; Google/Apple tiles are ToS-forbidden). TomTom's free tier (200,000 flow tiles/month, no card) stays the practical live option. Earth Traffic gives hourly CC BY history.	TRK-05, TRK-07
Archaeological LiDAR	Only OpenTopography plus two small bundles are shipping-grade: Caracol/Usumacinta ground DEMs (HER-16) and Chactún (HER-19) give two fly-over Maya sites as pre-bakes. ANADEM (TER-11) is the only open bare-earth layer that could reveal Amazonian earthworks and Sahel tells at 30 m. OpenTopography keys may not be embedded in a distributed app. Open municipal and insular stacks exist: GeoSampa and INEGI (Americas), Cape Town, Rwanda, Réunion/Mayotte (Africa); Kruger is the open point-cloud exemplar. Site-vector bundles (HER-10, HER-17, HER-13) are the immediate wins.	TER-11, HER-16, HER-19, TER-15, TER-18, HER-10, TER-13, TER-28, TER-19, TER-17
Heritage layers	Start with GeoNames heritage feature codes (already in the dumps), then Pleiades and EMODnet shipwrecks (clean licences). National registers are per-country plugins. EU waters are open via EMODnet; US wrecks via NOAA ENC pre-bakes; elsewhere no open global wreck list exists (UNESCO is viewer-only).	HER-02, HER-06, HER-05, HER-20, HER-15, HER-14, HER-21
Nature and time-travel	GBIF, Paleobiology Database, time zones and seafloor names are cheap, licence-clean and distinctive. Historical USGS/NLS/NL maps give a time-travel mode once tile layers accept a year parameter. openAIP stays opt-in (non-commercial).	HER-01, HER-03, REF-05, IMG-25, IMG-21, REF-13, TRK-06
Farmland	WorldCover (LND-06) gives 10 m cropland, FTW/RPG (LND-05) gives actual parcels where governments publish them, and MapSPAM (LND-13) answers "what grows here" at country scale.	LND-06, LND-05, LND-13
Energy trackers	USWTDB (INF-02) and Global Energy Monitor (INF-04) are the quick wins for "where are the wind and solar farms"; the WRI database (INF-01) stays useful as a 2021 snapshot for other fuels.	INF-04, INF-02, INF-01
Foundation-model embeddings	TESSERA's CC0 licence and global 10 m scale are impressive, but the data is a feature vector, not a picture (about 190 TB uncompressed for global land, my estimate). Record it; the only EarthCraft-sized use is a pre-baked PCA showcase layer (LND-03).	LND-16, LND-03
Non-English coverage	Searches ran in 20 languages (Japanese, French, German, Spanish, Portuguese, Chinese, Dutch, Korean, Italian, Polish, Danish, Swedish, Norwegian, Finnish, Icelandic, Russian, Hindi, Turkish and others). Arabic-language searches found no open national source and only pointed back to global sources.	TER-05, IMG-09, IMG-13
Time jump and Earth Engine	Earth Engine does not replace Wayback: Wayback is sub-metre in discrete releases, while the Earth Timelapse mosaics (released 2026-10-20 under CC BY 4.0) give one 30 m / 19 m image per year for 1984–2022. They extend the time jump about 30 years further back. Access needs a user-supplied Earth Engine project. The CC BY licence would allow a pre-baked mirror, but native detail is several TB (my estimate) and Earth Engine export terms were not checked. Keyless alternatives for change over time: NASA GIBS (IMG-04), EOX yearly mosaics (IMG-01), Global Forest Watch loss-year tiles (LND-01), historical topographic maps (IMG-25, IMG-21).	IMG-28, CAT-04, IMG-02, IMG-04, IMG-01

Appendix B. Suggested implementation order

Step	What
1	Boundary detail upgrade (REF-03 first, then REF-14 / REF-12, REF-11, REF-06, REF-07, REF-10, REF-08). One URL change (NE 50m → 10m) roughly triples country-border detail; then geoBoundaries ADM2 as the generic detailed-local layer, plus Census/France/Germany/Spain/Mexico plugins in the Brazil pattern. Overture per-country slices (REF-18) can replace the fragile Overpass metro layer.
2	Terrarium terrain (TER-06). Closes the "real 3D terrain" gap with the existing tile and cache code.
3	GIBS (IMG-04) and weather overlays (IMG-03 night lights, WTH-04 / WTH-05, WTH-06). Reuse the imagery path, add a layer picker and timeline.
4	Offline tiling script (GDAL to PNG / PMTiles) for RES-06, RES-05, LND-06, LND-15, TER-08, TER-07. Keep it outside the app to preserve the no-system-deps promise.
5	Point and vector layers: INF-05, INF-01, TRK-01, TRK-03, TRK-04, HAZ-09, HAZ-01, HAZ-02, HAZ-05, WTH-01, WTH-02, HAZ-04, WTH-11, LND-01, WTH-07, HAZ-03, TRK-08, TRK-02 (airports), rendered like existing placemarks.
6	PMTiles reader for TER-09, IMG-14, BLD-01 (and REF-18 slices).

Step	What
7	Other worlds (OTH-*) after a body and projection abstraction exists.
8	Regional terrain and imagery. Start with TER-05 (Japan, plain XYZ), IMG-09 / IMG-10 (Netherlands) and IMG-13 (Switzerland). They need no key and reuse the tile path. Add a per-source decoder for terrain, because encodings differ (Terrarium vs GSI).
9	Cheap wins last: IMG-03 (night lights — plain GIBS layer), lightning via HAZ-06 with HAZ-11 as fallback.
10	Archaeology pre-bakes (HER-16, HER-19, HER-07, plus TER-13 Cape Town / TER-19 Rwanda / TER-17 Réunion-Mayotte) with the same offline GDAL step, and the free site-vector overlays (HER-10, HER-17, HER-13) on top; prototype the ANADEM hillshade (TER-11) for Amazonia/Sahel before committing to any runtime key handling.
11	Life and history layers: HER-02 (GeoNames filter — already downloaded), then HER-06/HER-05 + HER-01/HER-03 (placemark fetches), REF-05 (bundle time zones), REF-13 (seafloor names); time-travel imagery (IMG-25/IMG-21) after a tile-layer year-parameter exists.
12	Shared enabling work first. Build four adapters once and many rows unlock together: (A) WMS-as-XYZ, (B) elevation decoders (Terrarium, GSI, WebP) + hillshade, (C) PMTiles over HTTP via the `pmtiles` crate, (D) key management from env/config. Then the remaining rows marked Easy are each a small change.
13	TESSERA last. Record only: a pre-baked PCA showcase layer for one region (LND-03) is the only EarthCraft-sized use; skip the Zarr/NPY paths.
14	New data layers by value: FTW harmonized fields for France (LND-05), USWTDB (INF-02) and GEM trackers (INF-04), EEZ and NHC storms (MAR-03, HAZ-10), `sgp4` upgrade for satellites (TL-11), ECMWF/AIFS wind via the `grib` crate prototype (WTH-08).
15	Time-jump extension: add Earth Timelapse (IMG-28) as an opt-in 1984–2022 year layer after the Wayback picker, reusing the shared REST, OAuth and key work (D). If licence and hosting allow, consider a pre-baked z0–6 overview (about 1 GB, my estimate).

Shared enabling work: build once, unlock many

Tag	Work	Sources it unlocks
A	WMS-as-XYZ adapter (tile bbox → `GetMap`, EPSG:3857)	TER-04, TER-01, WTH-06, HAZ-07, HER-05
B	Elevation decoders (Terrarium, GSI, WebP) + hillshade pass	TER-06, TER-09, TER-05, TER-03 (then true relief in the ray-caster)
C	PMTiles over HTTP (`pmtiles` crate; `image-webp`; `mvt-reader` for vector tiles)	TER-09, INF-05, IMG-14, BLD-01, LND-05, MAR-03
D	Key management from env/user config (never shipped)	WTH-04, TRK-03, HAZ-09, MAR-02, WTH-11, IMG-08, INF-06, IMG-28, CAT-04
E	`grib` crate prototype (GRIB2 subsets)	WTH-10, WTH-08
F	COG window reads (`async-tiff`) for region extracts	TER-21, LND-06, LND-20, BLD-02, TER-25, TER-29
G	`sgp4` crate for satellites	TRK-04

Appendix C. What EarthCraft already does

Capability	Status in the app	Related sources
Imagery with disk cache	Esri satellite, OSM streets, OpenTopoMap, Wayback releases. Decoded tile cache (Moka, 256 MiB) + MBTiles (128 MiB), keyed per provider/release/format	IMG-02, TER-06
Live aircraft	OpenSky and ADSB.lol, merged, 17-minute/90-second refresh	TRK-01, TRK-08
Vessels	AIS vessels with HTTP fallbacks	TRK-03
Satellites	CelesTrak OMM with an approximate two-body model (not SGP4)	TRK-04, TL-11
Weather	Cloud and radar fetching in the data crate	WTH-04, WTH-05, WTH-06
Traffic	Traffic rasters (key-based)	TRK-05, TRK-07
Roads, municipal data	OSM via Overpass	INF-05, REF-18

Capability	Status in the app	Related sources
Borders and names	Natural Earth 1:50m countries, admin_1, populated places, datasets/geo-countries, IBGE for Brazil	REF-03, REF-14, REF-12, REF-06
Other bodies	Moon (embedded image) and custom .earthplanet bodies in Explorer	OTH-02, OTH-01, OTH-03
Files	KML/KMZ import and export (GeoRust kml)	heritage and placemark layers
Rendering	CPU ray-casting (Rayon + wide), optional wgpu compute (experimental)	all terrain rows

Repo rules that shape every integration judgment (from AGENTS.md): tests must not depend on public map services; new dependencies need a demonstrated use; Rust 1.95 minimum; coordinates stay f64; public OSM stays on-demand (no prefetch); tile imagery persists through MBTiles with separate provider identities; GPU rendering stays experimental.

Appendix D. Tools, reference projects and Rust crates

ID	Project / crate	What it provides	Notes
TL-01	cambecc/earth + cambecc/grib2json	Particle-style wind and ocean-current visualisation from GFS and OSCAR JSON	MIT. Prototype for the wind layer
TL-02	TomHeaven/netcdf2json	OSCAR netCDF to JSON conversion script	Offline step
TL-03	VrushankPatel/godseye	Browser 3D globe wiring OpenSky, CelesTrak, SWPC, METAR, NDBC, AISstream	Good feed checklist. Includes military-site overlays. Take feeds only
TL-04	tapiab/heimdall	Rust raster viewer with STAC browser, COG streaming, Sentinel-2 cloudless basemap	Uses GDAL, which conflicts with EarthCraft's no-system-deps goal. Read, don't copy
TL-05	async-tiff (Rust crate)	Pure-Rust async COG / tiled TIFF reader via object_store	No GDAL. Fits TER-21, LND-06, LND-20, BLD-02
TL-06	rain_viewer (Rust crate)	RainViewer client Crate last released 2021-11; a small serde client is preferable (see TL-21).	Fits WTH-05
TL-07	PMTiles tools (go-pmtiles)	pmtiles extract for regional cutouts	For TER-09, IMG-14, BLD-01
TL-08	TiTiler	Server-side COG to XYZ tiles	Option for □ rasters if a tile proxy is acceptable
TL-09	Arnis (`louis-e/arnis`)	Rust (99.8%), Apache-2.0. Generates Minecraft Java 1.17+ and Bedrock worlds from OpenStreetMap (Overpass) with elevation from AWS Terrain Tiles, cached locally. The closest sibling to EarthCraft's data path	Proves Terrarium tiles + Overpass is a workable Rust stack. Read its elevation module before writing TER-06
TL-10	Terra++ / Terra121 (`BuildTheEarth/terraplusplus`)	MIT Java mod: elevation and biomes from public datasets, OSM roads, oceans from elevation with depth data, `/tpll lat lon`	Reference for source selection (BuildTheEarth)
TL-11	`sgp4` crate	Pure Rust SGP4, 2.4.0 (2026-02-23), MIT, 402k downloads	Upgrade path for the approximate two-body satellite model in `earthcraft-core::orbit`. The older `satellite` crate (0.1.2, 2024-09) is stale
TL-12	`pmtiles` crate	0.25.0 (2026-10-15), MIT/Apache-2.0, MSRV 1.88, sync and async backends including HTTP	One reader unlocks TER-09, IMG-14, BLD-01, LND-05, MAR-03
TL-13	`mvt-reader` 2.5.0 (MIT), `galileo-mvt` 0.2.1	Decode Mapbox vector tiles (pure Rust, also WASM)	Unlocks vector sources: INF-05, TRK-05, PMTiles vector layers
TL-14	`grib` crate	0.20.0 (updated 2026-10-18), MIT/Apache-2.0, pure-Rust GRIB parser and writer	Could turn WTH-10 and WTH-08 from pre-bake into a runtime fetch of small subsets
TL-15	`async-tiff` 0.4.0 + `object_store` 0.14.2 + `tiff` 0.11.3	Pure-Rust COG reads over HTTP/S3, MSRV 1.88, no GDAL	Region extracts for TER-21, LND-06, LND-20, BLD-02, TER-25, TER-29 without leaving Rust
TL-16	`image-webp` 0.2.4	Pure-Rust WebP decode (MIT/Apache-2.0)	Mapterhorn tiles are WebP
TL-17	`proj4rs` 0.2.1	Pure-Rust PROJ4 adaptation (MIT/Apache-2.0)	Polar stereographic (EPSG:3413/3031), Mollweide (GHSL), per-body projections for the Other Worlds rows (OTH-*)
TL-18	`h3o` 0.11.0, `tzf-rs` 2.1.3, `flatgeobuf` 6.0.1, `geojson` 1.0.0, `shapefile` 0.9.0, `osmpbf` 0.3.8, `geoarrow` / `geoparquet` 0.9.0	Format and index crates. `tzf-rs` embeds time-zone polygons (MSRV 1.88). `geoarrow` and `geoparquet` declare MSRV 1.95	Fit TRK-05 (H3), REF-05, BLD-*, REF-*, LND-05

ID	Project / crate	What it provides	Notes
TL-19	zarrs 0.23.14 + icechunk 2.3.0	Rust Zarr and Icechunk (MSRV 1.91 / 1.95.0)	Heavy; only for WTH-13 if a runtime path is ever wanted
TL-20	netcdf 0.12.1	Bindings to libnetcdf — a system dependency	Conflicts with the no-system-deps goal: keep OSCAR (WTH-15) as a pre-bake
TL-21	Not worth adding: rain_viewer (last release 2021-11), satellite (2024-09), srtm (2018), demtile (0.2.0, 1k downloads)	.hgt is a trivial big-endian 16-bit grid; the GSI decode (TER-05) is ~15 lines; the RainViewer index is a small serde client	Write the few lines instead of taking a stale or single-purpose dependency
TL-22	earthengine-api (google/earthengine-api)	Python and JavaScript client libraries for the Earth Engine API (Apache-2.0; about 3.4k stars, 902 commits). Issues are tracked in Google's Issue Tracker, not on GitHub.	No Rust client: call the REST API directly. Needs a registered Google Cloud project. Related rows: IMG-28, CAT-04.

Crate audit (crates.io API, 2026-10-19)

Crate	Version	Updated	Licence	MSRV	Note
pmtiles	0.25.0	2026-10-15	MIT OR Apache-2.0	1.88	OK for MSRV 1.95
mvt-reader	2.5.0	2026-09-06	MIT	—	OK for MSRV 1.95
satellite	0.1.2	2024-09-14	MIT	—	stale (2024)
sgp4	2.4.0	2026-02-23	MIT	—	OK for MSRV 1.95
h3o	0.11.0	2026-08-29	BSD-3-Clause	—	OK for MSRV 1.95
proj4rs	0.2.1	2026-10-09	MIT OR Apache-2.0	—	OK for MSRV 1.95
grib	0.20.0	2026-10-18	MIT OR Apache-2.0	—	OK for MSRV 1.95
flatgeobuf	6.0.1	2025-12-28	BSD-2-Clause	1.85.0	OK for MSRV 1.95
geoarrow	0.9.0	2026-09-11	MIT OR Apache-2.0	1.95	OK for MSRV 1.95
geojson	1.0.0	2026-03-16	MIT/Apache-2.0	—	OK for MSRV 1.95
osmpbf	0.3.8	2025-10-24	MIT OR Apache-2.0	1.88	OK for MSRV 1.95
shapefile	0.9.0	2026-07-05	MIT	—	OK for MSRV 1.95
tzf-rs	2.1.3	2026-10-11	MIT	1.88	OK for MSRV 1.95
async-tiff	0.4.0	2026-09-18	MIT OR Apache-2.0	1.88	OK for MSRV 1.95
rain_viewer	0.3.1	2021-11-08	MIT	—	stale (2021)
demtile	0.2.0	2026-08-26	MIT	—	tiny; reimplement
geographiclib-rs	0.2.7	2026-02-17	MIT	1.70.0	OK for MSRV 1.95
object_store	0.14.2	2026-09-15	MIT/Apache-2.0	1.85	OK for MSRV 1.95
tiff	0.11.3	2026-02-10	MIT	1.85.0	OK for MSRV 1.95
image-webp	0.2.4	2025-08-27	MIT OR Apache-2.0	1.80.1	OK for MSRV 1.95
srtm	0.1.1	2018-07-21	BSD-3-Clause	—	stale (2018)
zarrs	0.23.14	2026-08-14	MIT OR Apache-2.0	1.91	OK for MSRV 1.95
icechunk	2.3.0	2026-10-06	Apache-2.0	1.95.0	OK for MSRV 1.95
netcdf	0.12.1	2026-07-26	MIT OR Apache-2.0	1.77.0	needs libnetcdf (system dependency)

Crate	Version	Updated	Licence	MSRV	Note
geoparquet	0.9.0	2026-09-11	MIT OR Apache-2.0	1.95	OK for MSRV 1.95

Appendix E. Papers and citations

ID	Reference	Covers
PP-01	Poggio et al. 2021, SoilGrids 2.0, SOIL 7, 217–240	LND-15
PP-02	Hansen et al. 2013, High-Resolution Global Maps of 21st-Century Forest Cover Change, Science 342, 850–853	LND-12
PP-03	Zanaga et al. 2022, ESA WorldCover 10 m 2021 v200	LND-06
PP-04	World Bank, Global Solar Atlas 2.0 Technical Report	RES-05
PP-05	WRI, A Global Database of Power Plants	INF-01
PP-06	GEBCO Compilation Group 2026, The GEBCO_2026 Grid, doi:10.5285/4f68d5c7-45eb-f999-e063-7086abc036fa (released April 2026; supersedes the 2025 grid)	TER-22
PP-07	ESA WorldCereal system paper (see WorldCereal project site)	LND-02
PP-08	Zhang et al. 2024, GLC_FCS30D: the first global 30 m land-cover dynamics monitoring product..., Earth System Science Data 16, 1353	LND-09
PP-09	Souza et al. 2020, Reconstructing three decades of land use and land cover changes in Brazilian biomes with Landsat archive and Earth Engine, Remote Sensing 12(17), doi:10.3390/rs12172735	LND-19
PP-10	Reetz et al. 2022, DWD Geoportal – A central hub for Open Data, API and communication, EMS2022-468, doi:10.5194/ems2022-468	WTH-06, WTH-12
PP-11	JAXA EORC, ALOS World 3D-30m (AW3D30) Version 4.0 Product Description, April 2023	TER-23
PP-12	Protomaps blog, Mapterhorn – Terrain for Web Mapping (2025); Wipfli, Mapterhorn Terrain and Imagery, State of the Map 2026	TER-09

Appendix F. Risks and licence rules

Source	Risk	Mitigation
Esri World Imagery / Wayback (IMG-02 — EarthCraft's default today)	Non-commercial-only ToU, revocable at will, unauthed use ToS-grey per Esri staff, some reference layers marked "(Retiring)"	Make it opt-in; default to GIBS + EOX; document the free-Developer-account path (2M tiles/mo, still non-commercial)
RainViewer (WTH-05)	Personal/educational terms, no SLA, radar owners can withdraw data at any time	Best-effort overlay only; fallback to NWS/DWD composites where available
OpenSky (TRK-01)	Anonymous = 400 credits/day, OAuth2-only since 2024, ToU non-commercial-research oriented	Route daily use through adsb.lol (ODbL, already in the binary); keep OpenSky as a higher-fidelity opt-in
EOX cloudless (IMG-01)	Free by goodwill for a decade, but not a licensed right; licence terms still unsettled (IMG-01 row)	Keep as imagery option with GIBS fallback; verify licence before shipping
Mapterhorn (TER-09)	NLnet grant money is project-scoped; tiles served from individual-hosted infrastructure	Don't make it sole-source: TER-06 AWS Terrain (Registry of Open Data, very durable) + GEBCO/EMODnet cover the failure modes
FreeLightning (HAZ-06), Earth Traffic (TRK-05)	Single maintainer / demo platform	Treat as accelerators with the direct NOAA S3 / vendored-archive fallback
OpenAQ (WTH-11)	v1/v2 retired Jan 2025 → v3 needs a key; per-source licences vary	Fetch /v3/licenses, respect flags, cache aggressively
AISstream (TRK-03)	Beta/unstable	OpenSeaFeed + digitraffic fallbacks are already wired
EWDS / GloFAS (HAZ-07)	Explicitly "not for time-critical operation"	WMS-T tiles for visuals; GRIB analytics offline
tile.openstreetmap.org in-app	OSM's tile usage policy caps heavy third-party usage of the main tile host	Prefer Protomaps PMTiles (IMG-14) for offline-grade basemap; keep OSM raster as light fallback
NREL→NLR rebrand (RES-04, RES-03)	Domains changed once; may change again after rename	Keep the DOI/AWS mirrors (registry.opendata.aws) as anchors, not developer.nlr.gov URLs

Source	Risk	Mitigation
OpenTopography API (TER-11)	Free key: 50 calls/day non-academic; keys may not be embedded in an application; commercial integration requires a paid Enterprise key	Never ship a key — user-key from config or a pre-bake of ANADEM/archaeology tiles (HER-16); treat the API as opt-in, not default
OT archaeology collection licences (HER-16)	DCAT says CC BY 4.0 for Caracol Belize, but the dataset page itself reads "Use License: Not Provided"	Keep the † flag; cite DOI + PI (Weishampel 2019) and re-check the dataset page before shipping
geoBoundaries raw files (REF-14, REF-15)	Files live in Git LFS: `raw.githubusercontent.com` returns a ~130-byte pointer, not data (verified 2026-10-19). GitHub LFS bandwidth quotas can throttle the media host	Use `github.com/<repo>/raw/main/…` (302 to `media.githubusercontent.com`), vendor what you ship, or mirror
TeleGeography cable map (MAR-04)	Map images are CC BY-SA; the data behind them is sold under annual licence; community copies have unclear rights	Do not bundle or scrape; use OSM telecom lines via INF-05
FABDEM (TER-32), Viewfinder Panoramas (TER-33)	Non-open licences (commercial use requires purchase / research and private use only)	Excluded from shipping; use TER-06, TER-09, Copernicus GLO-30
Marine Regions (MAR-03)	CC BY 4.0, but VLIZ asks users not to host the files for download elsewhere	Link to marineregions.org; prefer streaming the PMTiles over bundling
OpenZenith (TER-03)	Single hobby host (`openzenith.cyopsys.com`)	Vendor the z0–8 pyramid (MIT) or fall back to TER-06
TESSERA hosting (LND-16, LND-17, LND-18)	Data moved from an AWS bucket (since retired) to Source Cooperative behind Cloudflare; maintainers cite egress cost; v2 variants are beta	Do not depend on live reads; if used, pre-bake from a pinned (version, variant) and keep the CC0 output
Source Cooperative as a host (LND-05, MAR-03, BLD-01, LND-02, BLD-02, 26.x)	Many rows in this document live on one community-run host	Prefer a bundled or mirrored copy for anything the app needs at first run
TESSERA hosting (LND-16, LND-17, LND-18)	Data moved from an AWS bucket (since retired) to Source Cooperative behind Cloudflare; maintainers cite egress cost; v2 variants are beta	Do not depend on live reads; if used, pre-bake from a pinned (version, variant) and keep the CC0 output
Source Cooperative as a host (LND-05, MAR-03, BLD-01, LND-02, BLD-02, 26.x)	Many rows in this document live on one community-run host	Prefer a bundled or mirrored copy for anything the app needs at first run
Earth Engine and Earth Timelapse (CAT-04, IMG-28)	Per-project quota, yearly noncommercial re-verification, commercial use is paid; the 2024 API terms page may conflict with current practice; the dataset is labelled v0, is not managed by Earth Engine and ends in 2022.	Opt-in bring-your-own-project plugin; never ship credentials; keep Wayback and GIBS/EOX as defaults; if a mirror is needed, check export terms and keep the CC BY attribution.

Rules computed from the master table

Rule	Sources
Non-commercial, restricted or paid (Comm. use = No)	HAZ-03, HAZ-12, HER-12, HER-21, HER-23, HER-24, IMG-02, IMG-15, IMG-27, INF-08, MAR-02, MAR-04, REF-19, REF-21, REF-22, REF-23, TER-32, TER-33, TRK-06, TRK-08, WTH-03, WTH-05
Share-alike (ODbL, CC BY-SA): derived databases must carry the same terms	IMG-14, INF-05, REF-05, REF-16, REF-18
Needs a key or account (never ship keys; load from env or user config)	CAT-04, HAZ-07, HAZ-09, HAZ-12, HER-08, HER-14, IMG-08, IMG-20, IMG-22, IMG-23, IMG-24, IMG-26, IMG-28, INF-06, LND-01, LND-21, MAR-02, REF-20, REF-23, RES-03, RES-04, TER-02, TER-11, TER-12, TER-23, TER-28, TRK-01, TRK-03, TRK-06, TRK-08, WTH-04, WTH-11, WTH-14, WTH-15
Licence unverified or mixed (Comm. use = Check): read the terms first	BLD-01, BLD-04, CAT-01, CAT-02, CAT-03, CAT-04, HAZ-06, HER-03, HER-05, HER-08, HER-09, HER-14, HER-16, HER-18, HER-25, IMG-01, IMG-06, IMG-11, IMG-12, IMG-16, IMG-17, IMG-18, IMG-20, IMG-21, IMG-22, IMG-23, IMG-26, IMG-28, INF-04, INF-06, INF-07, LND-05, LND-08, LND-09, LND-10, LND-11, LND-13, LND-19, LND-26, MAR-01, REF-02, REF-17, REF-20, RES-01, RES-02, RES-06, TER-04, TER-05, TER-10, TER-11, TER-12, TER-15, TER-16, TER-23, TER-27, TER-29, TER-30, TRK-01, TRK-03, TRK-07, WTH-04, WTH-11, WTH-16
Unmaintained, paused or beta	INF-01 (WRI power plants, frozen 2021), LND-15 (SoilGrids REST API paused), TRK-02 (AISstream beta)
Attribution	Always show the provider's required attribution string in the UI footer.
Renamed host	NREL is now NLR: use developer.nlr.gov (affects the Wind Toolkit and NSRDB rows).

Appendix G. Dead ends and negative results

Topic	Finding	Related
Dynamic World direct access	The long-cited GCS bucket does not exist; the only path is Earth Engine with a GCP project (free only for research/education/non-profit).	LND-21
ADS-B Exchange free tier	No longer exists. Free paths: self-hosted receiver, adsb.lol (ODbL) and airplanes.live.	TRK-08
Key-free tile hosts	No key-free XYZ host exists for Dynamic World, VIDA buildings or NASA DEM: keep them as pre-bake.	BLD-01, LND-21
Shipping, ocean, geology	No free shipping-density feed; no new ocean/chlorophyll layer beyond GIBS and OSCAR; no on-the-fly geology raster (use ETOPO and EMAG2).	IMG-04, WTH-15, TER-08, TER-07
Closed archaeological data	Raw PACUNAM DTM tiles are not open; Santa Rosa Xtampak LiDAR DEM (Zenodo 10.5281/zenodo.7983013) is restricted; El Shincal (Zenodo 10.5281/zenodo.7382162) is an open CC BY paper but not a DEM; no open Sahel-tell LiDAR exists on Zenodo, Dryad or Figshare.	HER-22
Amazonia campaigns	Pärssinen et al. 2026 (Nature 657:420–425): raw LiDAR restricted under Brazilian law. Prümers 2022 (Bolivian Mojos): DTM only on request via the DAI. Rostain 2024 (Upano Valley): supplementary vectors only. IBGE has no airborne LiDAR archive.	HER-10, HER-17, HER-24
Africa archaeological LiDAR	North Africa (Egypt, Sudan, Libya, Tunisia, Morocco): none open (Giza is photogrammetry/GIS only). Sahel/West Africa: DAI–MOWAA Benin survey uses drone photogrammetry with no release. Kenya: commercial contractor only. Tanzania: no airborne archaeology LiDAR. Ethiopia (Aksum): PDF chapter only. Madagascar: nothing.	TER-13, TER-28, TER-19, TER-17
Broken endpoints and paths	Grottocenter cave API returned nothing on the documented v1 endpoint; the GEBCO data-products URL restructure broke the gazetteer link (data still public); PBDB colls/geo.json returns 404 (use colls/list.json).	HER-03, REF-13
Submarine cables, FABDEM, Viewfinder	TeleGeography cable data is commercial; FABDEM needs a purchase for commercial use; Viewfinder Panoramas is research/private use only.	MAR-04, TER-33, TER-32
Reddit	Reddit threads could not be surfaced by search. Community finds came from GitHub, Hugging Face, the OSM community forum and blogs.	TER-03, TER-30

Appendix H. Link audit

Check	Result
25 GitHub repositories referenced (arnis, terraplusplus, heimdall, godseye, cambecc/earth, grib2json, netcdf2json, mapterhorn, global-power-plant-database, natural-earth-vector, geoBoundaries, timezone-boundary-builder, opentopodata, open-elevation, HeightMap, rwanda-dem, earth2studio, esa-worldcover-datasets, wayback-core, TeslaInstituteSpaceWeather, open-traffic-collection, gibs-api-docs, async-tiff, go-pmtiles, titiler)	All HTTP 200
Natural Earth 10m GeoJSON file sizes	Countries 13.3 MB, map units 13.5 MB, admin_1 40.7 MB, urban areas 27.9 MB, populated places (simple) 4.93 MB (full file 19.4 MB), 50m countries 3.08 MB
WRI power plant CSV	11.97 MB
geoBoundaries files	Finding: raw.githubusercontent.com serves a 132–134 byte Git LFS pointer; github.com/…/raw/… redirects (302) to media.githubusercontent.com
timezone-boundary-builder	Latest release tag 2026e
38 Rust crates (crates.io API, 2026-10-19)	See crate table. All declared MSRVs are ≤ 1.95
All other hosts (tile servers, WMS/WMTS, S3, ArcGIS, ministries)	Not reachable from the research sandbox (DNS blocked). Not live-tested: run the smoke tests

Appendix I. Smoke tests to run on your machine

Not part of the test suite (repo rules forbid tests that depend on public map services). Run by hand or in a scheduled workflow.

```
curl -sI https://s3.amazonaws.com/elevation-tiles-prod/terrarium/3/4/2.png | head -1            # TER-06
curl -s 'https://gibs.earthdata.nasa.gov/wmts/epsg3857/best/wmts.cgi?SERVICE=WMTS&REQUEST=GetCapabilities' | head -c 300   # IMG-04
curl -sI https://download.mapterhorn.com/attribution.json | head -1                            # TER-09
curl -s 'https://wms.gebco.net/mapserv?REQUEST=GetCapabilities&SERVICE=WMS&version=1.3.0' | head -c 300   # TER-01
curl -s 'https://ows.emodnet-bathymetry.eu/wms?SERVICE=WMS&VERSION=1.3.0&REQUEST=GetCapabilities' | head -c 300  # TER-04
curl -s 'https://service.pdok.nl/hwh/luchtfotorgb/wmts/v1_0?request=GetCapabilities&service=wmts' | head -c 300   # IMG-09
curl -s https://api.rainviewer.com/public/weather-maps.json | head -c 300                      # WTH-05
curl -s https://services.swpc.noaa.gov/json/ovation_aurora_latest.json | head -c 200           # HAZ-01
curl -sI https://earthquake.usgs.gov/earthquakes/feed/v1.0/summary/all_hour.geojson | head -1    # HAZ-04
curl -sIL https://github.com/wmgeolab/geoBoundaries/raw/main/releaseData/CGAZ/geoBoundariesCGAZ_ADM0.geojson | grep -i -E '^(HTTP|location)'   # REF-15
```

Appendix J. Source links

ID	Source	Links
TER-01	GEBCO WMS — zero-effort ocean-floor tiles	https://wms.gebco.net/mapserv?REQUEST=GetCapabilities&SERVICE=WMS&version=1.3.0
TER-02	NASA GEDI L3 + ICESat-2 ATL08 (spaceborne, combined)	https://www.earthdata.nasa.gov/engage/open-data-services-software · https://nsidc.org/data/atl08
TER-03	OpenZenith DEM (Hugging Face `aliasfox/openzenith-dem`)	https://huggingface.co/datasets/aliasfox/openzenith-dem/blob/main/README.md
TER-04	EMODnet Bathymetry	https://ows.emodnet-bathymetry.eu/wms?SERVICE=WMS&VERSION=1.3.0&REQUEST=GetCapabilities
TER-05	GSI Japan elevation tiles 地理 院標高タイル (ja)	https://maps.gsi.go.jp/development/elevation.html · https://maps.gsi.go.jp/development/demtile.html · https://docs.rs/crate/demtile/latest · https://zenn.dev/shi_works/articles/6f50506d5bc3b0
TER-06	AWS Terrain Tiles (Terrarium)	https://registry.opendata.aws/terrain-tiles
TER-07	EMAG2 v3 magnetic anomalies	https://www.ngdc.noaa.gov/mgg/global/ · https://www.ngdc.noaa.gov/geomag/emag2_download.html
TER-08	ETOPO 2022 / ETOPO1 relief	https://www.ngdc.noaa.gov/mgg/global/ · https://www.ngdc.noaa.gov/geomag/emag2_download.html
TER-09	Mapterhorn	https://mapterhorn.com/ · https://protomaps.com/blog/mapterhorn-terrain/ · https://github.com/mapterhorn/mapterhorn · https://nlnet.nl/project/Mapterhorn/ · https://download.mapterhorn.com/attribution.json
TER-10	Open Topo Data and Open-Elevation (point APIs)	https://github.com/ajnisbet/opentopodata · https://github.com/Jorl17/open-elevation · https://open-elevation.com
TER-11	OpenTopography global DEM API — incl. ANADEM	https://opentopography.org/developers · https://portal.opentopography.org/apidocs/ · https://portal.opentopography.org/stac/raster_catalog.json · https://portal.opentopography.org/csw/catalog.json · https://doi.org/10.5069/G9736P4G · https://doi.org/10.5069/G9028PQB · https://doi.org/10.5069/G9V12301
TER-12	CD:NGI South Africa + eThekwini footprints	https://ngigeoportal.dlrrd.gov.za/cdngiportal/ · https://gis-ethekwini.opendata.arcgis.com/
TER-13	City of Cape Town LiDAR DEM/DTM	https://odp-cctegis.opendata.arcgis.com/search?tags=dem · https://cityimg.capetown.gov.za/erdas-iws/esri/GeoSpatial%20Datasets/rest/services/Elevation_CCT_GLM2019_5m_DEM/MapServer · https://gis.westerncape.gov.za/server2/rest/services/SpatialDataWarehouse/Elevation/MapServer/layers
TER-14	CNIG MDT05 / MDS05 (es)	https://inspire-geoportal.ec.europa.eu/srv/api/records/spaignMDT05 · https://inspire-geoportal.ec.europa.eu/srv/api/records/spaignwcs_mdt · https://datos.gob.es/en/catalogo/e00125901-spaignmdt
TER-15	GeoSampa — city of São Paulo LiDAR	https://geosampa.prefeitura.sp.gov.br/PaginasPublicas/_SBC.aspx · https://igc.sp.gov.br
TER-16	IGC-SP — São Paulo state LiDAR 2023/24	https://geosampa.prefeitura.sp.gov.br/PaginasPublicas/_SBC.aspx · https://igc.sp.gov.br
TER-17	IGN RGE ALTI + LiDAR HD — Réunion & Mayotte	https://cartes.gouv.fr/rechercher-une-donnee/dataset/IGNF_RGE-ALTI · https://www.data.gouv.fr/datasets/mnt-lidar-hd

ID	Source	Links
TER-18	INEGI MDE / historical LiDAR tiles (Mexico)	https://www.inegi.org.mx/programas/mde/ · https://www.inegi.org.mx/app/biblioteca/ficha.html?upc=702825796341 · https://www.inegi.org.mx/app/geo2/elevacionesmex/
TER-19	Rwanda 10 m DEM (via Mapterhorn)	https://download.mapterhorn.com/sources/rw.tar · https://github.com/WASAC/rwanda-dem
TER-20	USGS 3DEP	https://epqs.nationalmap.gov/v1/docs · https://portal.opentopography.org/apidocs/#/Public/getUsgsDem
TER-21	Copernicus DEM (GLO-30 / GLO-90)	https://registry.opendata.aws/copernicus-dem
TER-22	GEBCO grid (GEBCO_2026, released April 2026)	https://www.gebco.net/data-products/gridded-bathymetry-data · https://www.gebco.net/news/release-gebco2026-grid · https://data.ceda.ac.uk/bodc/gebco/global/gebco_2026
TER-23	JAXA ALOS World 3D (AW3D30) (ja / en)	https://www.eorc.jaxa.jp/ALOS/en/dataset/aw3d30/data/aw3d30v4.0_product_e_1.0.pdf · https://data.amerigeoss.org/dataset/alos-world-3d-30m
TER-24	AHN Actueel Hoogtebestand Nederland (nl)	https://community.openstreetmap.org/t/ahn-gebruiken-als-achtergrondlaag-voor-bewerken/73056 · https://download.mapterhorn.com/attribution.json
TER-25	ArcticDEM (Polar Geospatial Center)	https://registry.opendata.aws/pgc-arcticdem · https://aws.amazon.com/marketplace/pp/prodview-ut7h7gaxvkyrq · https://opentopography.org/node/3533
TER-26	geoportal.gov.pl NMT / LiDAR point clouds (pl)	https://www.geoportal.gov.pl/pl/dane/dane-pomiarowe-lidar-lidar/
TER-27	IGN LiDAR HD (fr)	https://www.data.gouv.fr/datasets/mns-lidar-hd/ · https://geoservices.ign.fr/sites/default/files/2025-09/Offre_Produit_LiDAR_2025-08.pdf
TER-28	Kruger NP river LiDAR 2012 (CEDA)	https://catalogue.ceda.ac.uk/uuid/a2e82c7f92dc4f389a7fb7e4e6629c9e/
TER-29	REMA (Reference Elevation Model of Antarctica)	https://registry.opendata.aws/pgc-arcticdem · https://aws.amazon.com/marketplace/pp/prodview-ut7h7gaxvkyrq · https://opentopography.org/node/3533
TER-30	Sonny's LiDAR Digital Terrain Models of Europe	https://sonny.4lima.de/ · https://community.openstreetmap.org/t/sonny-s-lidar-digital-terrain-models-of-europe/98043
TER-31	swissALTI3D / swissSURFACE3D (de / fr)	https://catalog.data.gov/harvest_record/add61b59-6572-44b4-9b15-17a193319ae3/raw · https://object.cloud.sdsc.edu/v1/AUTH_opentopography/www/metadata/CH_50cmDEM_metadata.pdf
TER-32	FABDEM	https://viewfinderpanoramas.org · https://science.ecosyste.ms/projects/10121
TER-33	Viewfinder Panoramas (J. de Ferranti)	https://viewfinderpanoramas.org · https://science.ecosyste.ms/projects/10121
TER-34	GMRT 4.0 (100 m multibeam) and SRTM15+ V2.1 (15″) †	https://github.com/jaluebbe/HeightMap · https://www.opendem.info/link_dem.html · https://science.ecosyste.ms/projects/10121
TER-35	HeightMap (`jaluebbe/HeightMap`)	https://github.com/jaluebbe/HeightMap · https://www.opendem.info/link_dem.html · https://science.ecosyste.ms/projects/10121
TER-36	opendem.info link directory	https://github.com/jaluebbe/HeightMap · https://www.opendem.info/link_dem.html · https://science.ecosyste.ms/projects/10121
IMG-01	EOX Sentinel-2 cloudless	https://eox.at/2017/03/sentinel-2-cloudless-original-tiles-available/ · https://osapiens.maps.eox.at/
IMG-02	Esri World Imagery + Wayback (EarthCraft's current default imagery)	https://www.esri.com/en-us/legal/terms/web-site-service · https://github.com/Esri/wayback-core
IMG-03	GIBS Earth at Night (Black Marble city lights)	https://gibs.earthdata.nasa.gov/wmts/epsg3857/best/wmts.cgi?SERVICE=WMTS&REQUEST=GetCapabilities
IMG-04	NASA GIBS	https://wiki.earthdata.nasa.gov/x/24soCw
IMG-05	Dataforsyningen (Klimadatastyrelsen, da)	https://dataforsyningen.dk/ · https://datafordeler.dk/dataoversigt/geodanmark-ortofoto/ortofoto-foraar-web-mercator-wmts/
IMG-06	geoportal.gov.pl Ortofotomapa WMTS (pl)	https://mapy.geoportal.gov.pl/wss/service/PZGIK/ORTO/WMTS/StandardResolution · https://mapy.geoportal.gov.pl/wss/service/PZGIK/ORTO/WMTS/HighResolution · https://www.geoportal.gov.pl/pl/dane/ortofotomapa-orto/
IMG-07	Kartverket topo4 WMTS + NiB + hoydedata (no)	https://www.kartverket.no/en/api-and-data/terms-of-use · https://norgeibilder.no
IMG-08	Maanmittauslaitos avoimet aineistot (fi)	https://www.maanmittauslaitos.fi/en/maps-and-spatial-data/datasets-and-interfaces/product-descriptions/elevation-model-2-m · https://www.maanmittauslaitos.fi/ortokuvien-ja-korkeusmallien-kyselypalvelu

ID	Source	Links
IMG-09	PDOK aerial photos Luchtfoto (nl)	https://service.pdok.nl/hwh/luchtfotorgb/wmts/v1_0?request=GetCapabilities&service=wmts
IMG-10	PDOK BRT Achtergrondkaart (nl)	https://data.overheid.nl/en/dataset/48457-brt-achtergrondkaart · https://www.pdok.nl/introductie/-/article/basisregistratie-topografie-achtergrondkaarten-brt-a-
IMG-11	Portale Geocartografico Trentino (it)	http://www.territorio.provincia.tn.it/portal/server.pt/community/lidar/847/lidar/23954
IMG-12	Regione Piemonte Geoportale (it)	https://www.geoportale.piemonte.it/geonetwork/srv/search?format=OGC:WMTS
IMG-13	swisstopo geo.admin.ch (de / fr)	https://atlas.co/data-portals/swisstopo-geo-admin-ch/ · https://opendata.swiss/en/group/envi?page=9 · https://www.baselland.ch/politik-und-behorden/direktionen/volkswirtschafts-und-gesundheitsdirektion/amt-fur-geoinformation/newsletter/freier-zugang-zu-geodaten-und-geodiensten-von-swisstopo
IMG-14	Protomaps OSM basemap	https://docs.protomaps.com/basemaps/downloads
IMG-15	Geoportale Nazionale / PCN (it)	http://www.pcn.minambiente.it/viewer/ · https://www.alspergis.altervista.org/data/ortofoto_pcn.html
IMG-16	IGN Géoplateforme (fr)	https://geoservices.ign.fr/sites/default/files/2024-06/Tuto_Geoplateforme.pdf
IMG-17	İBB Şehir Haritası + data.ibb.gov.tr (tr)	https://sehirharitasiapi.ibb.gov.tr/ · https://data.ibb.gov.tr
IMG-18	Landmælingar Íslands gagnatorg (is)	https://gagnatorg.lmi.is · https://gee-community-catalog.org/projects/iceland_dem/
IMG-19	Lantmäteriet öppna geodata (sv)	https://www.lantmateriet.se/sv/geodata/vara-produkter/oppnadata/
IMG-20	NGII background and satellite maps 영상지도·배경지도 (ko)	https://www.data.go.kr/catalog/15059358/openapi.json
IMG-21	NLS Scotland historic maps + NL Topotijdreis	https://maps.nls.uk · https://service.pdok.nl/hwh/topotijdreis/wmts/v1_0
IMG-22	Tianditu 天地图 (zh)	https://lbs.tianditu.gov.cn/server/MapService.html · https://iclient.supermap.io/docs/leaflet/TiandituTileLayer.html · https://cloud.tencent.com/developer/inventory/23674/article/1709969
IMG-23	VWorld 브이월드 (ko)	https://www.data.go.kr/data/15058704/openapi.do · https://www.data.go.kr/data/3073144/openapi.do
IMG-24	VIIRS night lights (NASA Black Marble)	https://ladsweb.modaps.eosdis.nasa.gov/missions-and-measurements/products/VNP46A2/ · https://developers.google.com/earth-engine/datasets/catalog/NASA_VIIRS_002_VNP46A2
IMG-25	USGS historical topographic maps (topoView)	https://www.usgs.gov/programs/national-geospatial-program/topoview · https://basemap.nationalmap.gov/arcgis/rest/services/USGSTopo/MapServer
IMG-26	Bhuvan / NRSC Open EO Data Archive (NOEDA) (hi)	https://bhuvan-app3.nrsc.gov.in/data/ · https://bhuvan.nrsc.gov.in/
IMG-27	map.is / Loftmyndir ehf (is)	https://www.map.is/
IMG-28	Earth Timelapse annual mosaics (Google, via Earth Engine)	https://medium.com/google-earth/four-decades-of-planetary-change-now-open-for-everyone-to-build-on-9b32d6cb8c86 · https://developers.google.com/earth-engine/datasets/catalog/projects_ee-earth-timelapse_assets_annual · https://developers.google.com/earth-engine/timelapse/about · https://code.earthengine.google.com/?scriptPath=Examples:Datasets/ee-earth-timelapse/projects_ee-earth-timelapse_assets_annual · https://developers.google.com/earth-engine/reference/rest/v1/projects.maps.tiles/get
WTH-01	NOAA AviationWeather	https://github.com/VrushankPatel/godseye
WTH-02	NOAA NDBC buoys	https://github.com/VrushankPatel/godseye
WTH-03	Open-Meteo	https://zenodo.org/records/8112599 · https://freeapihub.com/apis/open-meteo-historical
WTH-04	OpenWeatherMap Maps 1.0	https://openweathermap.org/api/weathermaps
WTH-05	RainViewer radar	https://www.rainviewer.com/api/weather-maps-api.md · https://rainviewer.com/api.html · https://docs.rs/crate/rain_viewer/latest
WTH-06	DWD radar and weather WMS (de)	https://opendata.dwd.de/README.txt · https://www.ki-syndikat.de/tools/dwd-opendata/ · https://wetterdienst.readthedocs.io/en/latest/_sources/data/provider/dwd/index.md · https://meetingorganizer.copernicus.org/EMS2022/EMS2022-468.html · https://gitlab.com/brun-soft/das-etb/-/work_items/210
WTH-07	NOAA CO-OPS tides and currents	https://api.tidesandcurrents.noaa.gov/api/prod/datagetter · https://tidesandcurrents.noaa.gov/web_services_info.html
WTH-08	ECMWF Open Data (IFS + AIFS)	https://www.ecmwf.int/en/forecasts/datasets/open-data · https://www.ecmwf.int/node/37688

ID	Source	Links
WTH-09	NASA Earth Observations (NEO)	https://gisgeography.com/best-free-gis-data-sources-raster-vector/
WTH-10	NOAA GFS 10 m wind	https://github.com/cambecc/earth · https://github.com/IMAGINARY/earth
WTH-11	OpenAQ v3 air quality	https://docs.openaq.org/using-the-api/api-key · https://docs.openaq.org/about/terms · https://docs.openaq.org/using-the-api/rate-limits
WTH-12	DWD Open Data (de)	https://opendata.dwd.de/README.txt · https://www.ki-syndikat.de/tools/dwd-opendata/ · https://wetterdienst.readthedocs.io/en/latest/_sources/data/provider/dwd/index.md · https://meetingorganizer.copernicus.org/EMS2022/EMS2022-468.html · https://gitlab.com/brun-soft/das-etb/-/work_items/210
WTH-13	dynamical.org Icechunk Zarr (AWS Open Data)	https://registry.opendata.aws/dynamical-ecmwf-aifs-single/ · https://registry.opendata.aws/dynamical-ecmwf-aifs-ens/
WTH-14	ERA5 (Copernicus CDS)	https://casrai.org/guides/copernicus-climate-data-store-c3s
WTH-15	OSCAR ocean currents	https://github.com/TomHeaven/netcdf2json · https://registry.opendata.aws/nasa-oscar/ · https://www.earthdata.nasa.gov/data/catalog/pocloud-oscar-l4-oc-interim-v2.0-2.0 · https://www.esr.org/data-products/oscar/
WTH-16	WorldClim	https://en.wikipedia.org/wiki/List_of_GIS_data_sources · https://guides.library.duke.edu/gisdata/gisdata_international
WTH-17	NVIDIA Earth2Studio data sources	https://github.com/NVIDIA/earth2studio/releases
RES-01	NASA POWER	https://power.larc.nasa.gov/
RES-02	PVGIS 5.3 (EU JRC)	https://joint-research-centre.ec.europa.eu/photovoltaic-geographical-information-system-pvgis/getting-started-pvgis/api-non-interactive-service_en
RES-03	NSRDB solar (PSM v4) (NLR, ex-NREL)	https://developer.nlr.gov/docs/solar/nsrdb · https://developer.nlr.gov/docs/solar/nsrdb/psm3-2-2-download · https://registry.opendata.aws/nrel-pds-nsrdb
RES-04	Wind Toolkit (NLR, ex-NREL)	https://developer.nlr.gov/docs/wind/wind-toolkit/wtk-download/
RES-05	Global Solar Atlas	https://datacatalog.worldbank.org/search/dataset/0042316 · https://energydata.info/application/
RES-06	Global Wind Atlas 3	https://globalwindatlas.info/about/KeyFeatures
INF-01	WRI Global Power Plant Database	https://old-datasets.wri.org/dataset/globalpowerplantdatabase · https://github.com/wri/global-power-plant-database
INF-02	US Wind Turbine Database (USWTDB) V8.2 (10 Dec 2025)	https://energy.usgs.gov/uswtdb/data/ · https://doi.org/10.1038/s41597-020-0353-6 · https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6957502/
INF-03	Climate TRACE	https://climatetrace.org/data · https://apify.com/jungle_synthesizer/climate-trace-facility-emissions-scraper.md
INF-04	Global Energy Monitor trackers (`gem.wiki`)	https://gem.wiki
INF-05	Open Infrastructure Map	https://openinframap.org/about.html
INF-06	ENTSO-E Transparency Platform	https://transparency.entsoe.eu/ · https://pypi.org/project/entsopy · https://gist.github.com/dirkjanfaber/4a99336d48dde52181514d6e4a06b9eb
INF-07	Renewable Energy Data Explorer (NREL)	https://guides.library.duke.edu/gisdata/gisdata_international
INF-08	UMN "Primary Resources Extracted"	https://geodata.libraries.mit.edu/record/gisogm:edu.umn:3d26e110c136
LND-01	Global Forest Watch tiles (Hansen layers)	https://tiles.globalforestwatch.org/openapi.json · https://data-api.globalforestwatch.org/ · https://data.globalforestwatch.org/
LND-02	ESA WorldCereal	https://source.coop/streambatch/worldcereal · https://docs.digitalearthafrica.org/en/latest/data_specs/ESA_World_Cereal_specs.html
LND-03	Pre-baked TESSERA PCA tiles (`geotessera visualize` + `webmap`)	https://github.com/ucam-eo/tessera-coverage-map · https://github.com/ucam-eo/tessera-interactive-map · https://github.com/ucam-eo/geotessera-examples
LND-04	France RPG (Registre Parcellaire Graphique) at source	https://www.data.gouv.fr/en/datasets/registre-parcellaire-graphique-rpg-contours-des-parcelles-et-ilots-culturaux-et-leur-groupe-de-cultures-majoritaire/ · https://etalab.gouv.fr/licence-ouverte-open-licence
LND-05	FTW harmonized field data catalogue (Source Cooperative `ftw/harmonized-field-data`)	https://source.coop/ftw/harmonized-field-data · https://data.source.coop/ftw/harmonized-field-data/fr/collection.json
LND-06	ESA WorldCover 2021 v200	https://worldcover2021.esa.int/download · https://github.com/ESA-WorldCover/esa-worldcover-datasets

ID	Source	Links
LND-07	Fields of The World (FTW) Global Field Boundary Map v1.1 (Microsoft Research + FTW)	https://www.microsoft.com/en-us/research/wp-content/uploads/2026/04/FTW_Global_Map_v1.1.pdf · https://fieldsofthe.world · https://arxiv.org/abs/2409.16252
LND-08	GAEZ+ 2015 gridded crop dataset (FAO / IIASA; Scientific Data 2022)	https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8776749/
LND-09	GLC_FCS30D (zh / en)	https://zenodo.org/records/8239305 · https://essd.copernicus.org/articles/16/1353/2024/ · https://gee-community-catalog.org/projects/glc_fcs/
LND-10	Global Synergy Cropland Map v3.0 †	https://geokur-dmp.geo.tu-dresden.de/dataset/output_dataset
LND-11	GMIA v5.0 (Global Map of Irrigation Areas) and MIRCA2000 v1.1	https://geokur-dmp.geo.tu-dresden.de/dataset/output_dataset
LND-12	Hansen Global Forest Change	https://developers.google.com/earth-engine/datasets/catalog/UMD_hansen_global_forest_change_2025_v1_13 · https://glad.earthengine.app/view/global-forest-change · https://storage.googleapis.com/earthenginepartners-hansen/GFC-2025-v1.13/download.html
LND-13	MapSPAM SPAM2010 v2.0 (IFPRI)	https://www.mapspam.info/ · https://doi.org/10.7910/DVN/PRFF8V · https://pure.iiasa.ac.at/16963 · https://geokur-dmp.geo.tu-dresden.de/dataset/mapspam-harvested-area · https://agrodep.org/node/6070
LND-14	NASA SEDAC Global Agricultural Lands	https://en.wikipedia.org/wiki/List_of_GIS_data_sources · https://guides.library.duke.edu/gisdata/gisdata_international
LND-15	SoilGrids 2.0 (ISRIC)	https://www.isric.org/explore/soilgrids
LND-16	TESSERA v1 embeddings (GeoTessera, global 2024)	https://github.com/ucam-eo/geotessera · https://geotessera.readthedocs.io · https://pypi.org/project/geotessera/0.10.0/ · https://github.com/ucam-eo/geotessera/releases/tag/v0.10.0 · https://data.source.coop/tessera/tessera/zarr/v1
LND-17	TESSERA v1.1 and v2 variants	https://github.com/ucam-eo/geotessera · https://geotessera.readthedocs.io · https://pypi.org/project/geotessera/0.10.0/ · https://github.com/ucam-eo/geotessera/releases/tag/v0.10.0 · https://data.source.coop/tessera/tessera/zarr/v1 · https://rogue-scholar.org/records/h0wq3-hpy81 · https://anil.recoil.org/notes/tessera-v11-out
LND-18	TESSERA Zarr v3 store (cloud-native)	https://github.com/ucam-eo/geotessera · https://geotessera.readthedocs.io · https://pypi.org/project/geotessera/0.10.0/ · https://github.com/ucam-eo/geotessera/releases/tag/v0.10.0 · https://data.source.coop/tessera/tessera/zarr/v1
LND-19	MapBiomas (pt)	https://developers.google.com/earth-engine/datasets/tags/mapbiomas-public · https://brasil.mapbiomas.org/wp-content/uploads/sites/4/2023/08/Nota_Tecnica_MapBiomas_Pastagem.pdf · https://api.openstreetmap.org/user/pedro_tharg/diary/406596
LND-20	USDA Cropland Data Layer	https://www.nass.usda.gov/Research_and_Science/Cropland/metadata/metadata_CDL24_FGDC-STD-001-1998.htm
LND-21	Dynamic World 10 m land cover	https://developers.google.com/earth-engine/datasets/catalog/GOOGLE_DYNAMICWORLD_V1 · https://dynamicworld.app/about/
LND-22	GeoTessera client design (reference)	https://github.com/ucam-eo/geotessera · https://geotessera.readthedocs.io
LND-23	Tessera coverage map and globe viewer	https://github.com/ucam-eo/tessera-coverage-map · https://github.com/ucam-eo/tessera-interactive-map · https://github.com/ucam-eo/geotessera-examples
LND-24	TESSERA model and inference (`ucam-eo/tessera`)	https://github.com/ucam-eo/tessera · https://arxiv.org/abs/2506.20380 · https://arxiv.org/abs/2607.03949
LND-26	AI4Boundaries (JRC)	https://essd.copernicus.org/preprints/essd-2022-298/essd-2022-298-ATC1.pdf · https://agris.fao.org/search/fr/records/68879c937fd4d06c32a9879c
BLD-01	VIDA Google + Microsoft footprints	https://source.coop/vida/google-microsoft-open-buildings/pmtiles
BLD-02	GHS-POP COG mirror (unofficial)	https://data.source.coop/nlebovits/ghsl/README.md
BLD-03	GHSL population (EU JRC)	https://human-settlement.emergency.copernicus.eu/ghs_wup_pop_r2025a.php
BLD-04	Overture Maps buildings	https://docs.overturemaps.org/guides/buildings
BLD-05	PLATEAU 3D city models (ja)	https://www.mlit.go.jp/plateau/learning/tpc30/ · https://www.moguravr.com/?p=315618 · https://wiki.openstreetmap.org/wiki/JA:MLIT_PLATEAU · https://zenn.dev/rescuenow/articles/ff5dca69861b94
REF-01	GeoNames gazetteer	https://download.geonames.org/export/dump/
REF-02	Natural Earth / GSHHG	https://www.helenmakesmaps.com/post/guide-to-free-gis-data-sources

ID	Source	Links
REF-03	Natural Earth 1:10m ladder (the immediate win)	https://github.com/nvkelso/natural-earth-vector
REF-04	ne_10m_urban_areas	https://github.com/nvkelso/natural-earth-vector
REF-05	timezone-boundary-builder	https://github.com/evansiroky/timezone-boundary-builder
REF-06	France IGN ADMIN EXPRESS / COG	https://data.geopf.fr/telechargement/download/ADMIN-EXPRESS-COG/
REF-07	Germany BKG VG250 / VG2500	https://www.bkg.bund.de
REF-08	Mexico INEGI Marco Geoestadístico	https://www.inegi.org.mx/geo/contenidos/geoestadistica/
REF-09	NZ LINZ / Stats NZ boundaries	https://data.linz.govt.nz/ · https://www.stats.govt.nz/products/digital-boundary-files
REF-10	Spain CNIG / IGN municipalities	https://centrodedescargas.cnig.es
REF-11	Statistics Canada boundary files	https://www12.statcan.gc.ca/census-recensement/2021/geo/sip-pis/boundary-limites/index2021-eng.cfm · https://geo.statcan.gc.ca/geo_wa/rest/services/2021/Cartographic_boundary_files/MapServer
REF-12	US Census cartographic boundaries	https://www2.census.gov/geo/tiger/GENZ2023/shp/
REF-13	GEBCO Undersea Feature Names gazetteer	https://www.gebco.net/data-products/
REF-14	geoBoundaries gbOpen (William & Mary geoLab)	https://www.geoboundaries.org/api/current/gbOpen/ALL/ADM0/ · https://github.com/wmgeolab/geoBoundaries/tree/main/releaseData/CGAZ
REF-15	geoBoundaries CGAZ composites	https://www.geoboundaries.org/api/current/gbOpen/ALL/ADM0/ · https://github.com/wmgeolab/geoBoundaries/tree/main/releaseData/CGAZ
REF-16	GeoFabrik OSM extracts	https://download.geofabrik.de · https://osm-boundaries.com
REF-17	HydroSHEDS	https://en.wikipedia.org/wiki/List_of_GIS_data_sources · https://guides.library.duke.edu/gisdata/gisdata_international
REF-18	Overture Divisions	https://docs.overturemaps.org/schema/reference/divisions/
REF-19	GADM v4.1	https://gadm.org/license.html · https://salb.un.org/terms-of-use
REF-20	osm-boundaries.com	https://download.geofabrik.de · https://osm-boundaries.com
REF-21	UN SALB	https://gadm.org/license.html · https://salb.un.org/terms-of-use
REF-22	Rosreestr Публичная кадастровая карта / НСПД (ru)	https://rosreestr.gov.ru/press/news/470917/ · https://community.openstreetmap.org/t/wms/50567?page=14
REF-23	TKGM Ada/Parsel Sorgu / TKKG (tr)	https://parselsorgu.tkgm.gov.tr
TRK-01	OpenSky Network	https://opensky-network.org/api/states/all
TRK-02	OurAirports	https://davidmegginson.github.io/ourairports-data/
TRK-03	AISstream.io	https://aisstream.io/documentation · https://github.com/aisstream
TRK-04	CelesTrak GP	https://celestrak.org/NORAD/elements/
TRK-05	Earth Traffic (iTIC/ELEMNT platform)	https://traffic.elemnt.earth/ · https://opendata.ndw.nu/ · https://github.com/graphhopper/open-traffic-collection · https://developer.tomtom.com/store/
TRK-06	openAIP airspace	https://www.openaip.net
TRK-07	NDW (Netherlands road data)	https://traffic.elemnt.earth/ · https://opendata.ndw.nu/ · https://github.com/graphhopper/open-traffic-collection · https://developer.tomtom.com/store/
TRK-08	ADS-B Exchange (live aircraft)	https://www.adsbexchange.com/community/developer-hub/ · https://api.adsb.lol/ · https://airplanes.live/
HAZ-01	NOAA SWPC aurora (OVATION)	https://www.spaceweather.gov/node/846 · https://github.com/NikolaTeslaInstitute/TeslaInstituteSpaceWeather
HAZ-02	NOAA SWPC space weather	https://www.spaceweather.gov/node/846 · https://github.com/NikolaTeslaInstitute/TeslaInstituteSpaceWeather

ID	Source	Links
HAZ-03	Smithsonian GVP volcano list	https://volcano.si.edu/gvp_termsofuse.cfm · https://volcano.si.edu/database/list_volcano_holocene_excel.cfm · https://www.ngdc.noaa.gov/hazel/hazard-service/api/v1/volcanolocs
HAZ-04	USGS earthquakes	https://earthquake.usgs.gov/earthquakes/feed/v1.0/geojson.php · https://earthquake.usgs.gov/fdsnws/event/1/ · https://earthquake.usgs.gov/earthquakes/feed/policy.php
HAZ-05	NWS active alerts	https://github.com/VrushankPatel/godseye · https://www.weather.gov/documentation/services-web-api · https://api.weather.gov/
HAZ-06	FreeLightning (parsed GLM)	https://www.blitzortung.org/en/coverage_maps.php · https://www.freelightning.com/
HAZ-07	GloFAS river discharge (Copernicus CEMS)	https://ewds.climate.copernicus.eu/datasets/cems-glofas-forecast · https://global-flood.emergency.copernicus.eu
HAZ-08	IBTrACS tropical cyclone best tracks (NOAA NCEI)	https://data.commerce.gov/node/141861
HAZ-09	NASA FIRMS	https://appliedsciences.nasa.gov/sites/default/files/2025-04/FIRMS_Part3_DMM_JO_MFC.pdf · https://atlas.co/data-sources/firms
HAZ-10	NHC GIS products (US National Hurricane Center)	https://www.nhc.noaa.gov/gis/ · https://www.nhc.noaa.gov/productexamples/NHC_Tropical_Cyclone_Status_JSON_File_Reference.pdf
HAZ-11	NOAA GOES GLM lightning	https://www.blitzortung.org/en/coverage_maps.php · https://www.freelightning.com/
HAZ-12	Blitzortung.org	https://www.blitzortung.org/en/coverage_maps.php · https://www.freelightning.com/
MAR-01	IMF PortWatch	https://portwatch.imf.org/pages/faqs
MAR-02	Global Fishing Watch (4WINGS)	https://globalfishingwatch.org/our-apis/documentation · https://globalfishingwatch.org/our-apis/tokens
MAR-03	Marine Regions: EEZ v12 + High Seas v2 (VLIZ, IOC-UNESCO)	https://atlas.co/data-sources/marine-regions · https://data.source.coop/cboettig/high-seas/iho/stac-collection.json · https://dc.niwa.co.nz/niwa_dc/srv/api/records/e7363de1-fddc-42eb-91e0-bd027e9fe233 · https://metadatacatalogue.lifewatch.eu/srv/api/records/5ec6b072-9e83-433b-b293-e0f75a714e34
MAR-04	TeleGeography submarine cable map	https://www2.telegeography.com/license-telegeography-map · https://www2.telegeography.com/hubfs/LP-Assets/Product-One-Pagers/geocoded-dataset-license.pdf · https://falliblevagrant.codeberg.page/finding-datasets-to-map-the-undersea-cable-network.html
HER-01	GBIF species occurrences	https://api.gbif.org/v1/species/match?name=Aquila%20chrysaetos · https://www.gbif.org/developer/summary · https://registry.opendata.aws/gbif
HER-02	GeoNames heritage feature codes	https://www.geonames.org/export/codes.html
HER-03	Paleobiology Database (fossils)	https://paleobiodb.org/data1.2/colls/list.json?lngmin=10&lngmax=15&latmin=50&latmax=55&limit=3&show=full · https://paleobiodb.org
HER-04	Wikidata "archaeological site" (Q839954) SPARQL	https://query.wikidata.org · https://perio.do
HER-05	EMODnet Human Activities — Cultural Heritage	https://ows.emodnet-humanactivities.eu/wms?SERVICE=WMS&REQUEST=GetCapabilities · https://www.emodnet-humanactivities.eu/view-data
HER-06	Pleiades (ancient Mediterranean places)	https://pleiades.stoa.org/places/423025/json · https://pleiades.stoa.org/datasets/4.1 · https://pleiades.stoa.org/about/license
HER-07	Stonehenge LiDAR Archaeology Landscape (Zenodo)	https://doi.org/10.5281/zenodo.10334054 · https://doi.org/10.5281/zenodo.5087620
HER-08	Movebank animal tracking	https://www.movebank.org
HER-09	UNESCO World Heritage list	https://whc.unesco.org/en/list/xml · https://www.unesco.org/en/underwater-cultural-heritage
HER-10	Acre geoglyph site vectors (de Souza 2018, Nat Commun)	https://www.nature.com/articles/s41467-018-03510-7 · https://journal.caa-international.org/articles/10.5334/jcaa.45
HER-11	Geoglyph DTM-anomaly toolkit (Zenodo)	https://doi.org/10.5281/zenodo.10334054 · https://doi.org/10.5281/zenodo.5087620
HER-12	Kilwa Kisiwani 3D (CyArk, Zenodo)	https://zenodo.org/records/10377387 · https://www.zamaniproject.org/site-zimbabwe-Great-Zimbabwe.html
HER-13	Loreto sites gazetteer (Coomes 2021, Sci Data)	https://doi.org/10.6084/m9.figshare.c.5262530
HER-14	National heritage registers: UK / IE	https://historicengland.org.uk/listing/the-list/ · https://canmore.org.uk · https://coflein.gov.uk · https://maps.archaeology.ie

ID	Source	Links
HER-15	NPS National Register of Historic Places	https://www.nps.gov/subjects/nationalregister/database.htm
HER-16	OpenTopography archaeology LiDAR collections	https://opentopography.org/developers · https://portal.opentopography.org/apidocs/ · https://portal.opentopography.org/stac/raster_catalog.json · https://portal.opentopography.org/csw/catalog.json · https://doi.org/10.5069/G9833Q42 · https://doi.org/10.5069/G95B00NF · https://doi.org/10.5069/G9416V95
HER-17	SE-Acre mound villages (Iriarte 2020, JCAA)	https://www.nature.com/articles/s41467-018-03510-7 · https://journal.caa-international.org/articles/10.5334/jcaa.45
HER-18	Randolph Glacier Inventory (RGI 7.0) / GLIMS	https://www.glims.org/RGI/ · https://nsidc.org/data/nsidc-0770 · https://www.glims.org
HER-19	Figshare: ML-ready Maya remote sensing (Chactún)	https://api.figshare.com/v2/articles/22202395 · `https://figshare.com/articles/dataset/…/22202395`
HER-20	NOAA ENC chart data — WRECKS/OBSTRN	https://www.nauticalcharts.noaa.gov/
HER-21	UNESCO Underwater Cultural Heritage database	https://whc.unesco.org/en/list/xml · https://www.unesco.org/en/underwater-cultural-heritage
HER-22	Dryad PACUNAM 2,144 km² Guatemala LiDAR	https://datadryad.org/dataset/doi:10.5061/dryad.k51j708 · https://doi.org/10.5281/zenodo.7983013
HER-23	Great Zimbabwe (Zamani Project)	https://zenodo.org/records/10377387 · https://www.zamaniproject.org/site-zimbabwe-Great-Zimbabwe.html
HER-24	Peru / Colombia / Ecuador national LiDAR	https://ore.exeter.ac.uk/29787914 · https://archimer.ifremer.fr/doc/00872/98412
HER-25	Archaeology Data Service (ADS)	https://archaeologydataservice.ac.uk/terms-and-conditions/
OTH-01	NASA Mars Trek	https://api.nasa.gov/mars-wmts/catalog · https://digital.gov/2016/04/01/your-trek-to-vesta-mars-starts-now
OTH-02	NASA Moon Trek	https://api.nasa.gov/mars-wmts/catalog · https://digital.gov/2016/04/01/your-trek-to-vesta-mars-starts-now
OTH-03	NASA Vesta Trek	https://api.nasa.gov/mars-wmts/catalog · https://digital.gov/2016/04/01/your-trek-to-vesta-mars-starts-now
CAT-01	Earth Search (Element84)	https://earth-search.aws.element84.com/v1 · https://element84.com/cloud-architecture/aws/sentinel-2-cloud-optimized-geotiffs-now-available-on-aws-registry-of-open-data/
CAT-02	Esri Open Data Hub	https://gisgeography.com/best-free-gis-data-sources-raster-vector/
CAT-03	Microsoft Planetary Computer	https://cng-python.gishub.org/book/case-studies/planetary-computer
CAT-04	Google Earth Engine (Data Catalog and REST tile API)	https://developers.google.com/earth-engine/datasets · https://developers.google.com/earth-engine/guides/noncommercial_tiers · https://developers.google.com/earth-engine/reference/rest/v1alpha/projects.maps/create · https://developers.google.com/earth-engine/reference/rest/v1/projects.maps.tiles/get · https://developers.google.com/earth-engine/reference/Additional.API.Terms · https://github.com/google/earthengine-api
TL-01	cambecc/earth + cambecc/grib2json	https://github.com/cambecc/earth · https://github.com/IMAGINARY/earth
TL-02	TomHeaven/netcdf2json	https://github.com/TomHeaven/netcdf2json · https://registry.opendata.aws/nasa-oscar/ · https://www.earthdata.nasa.gov/data/catalog/pocloud-oscar-l4-oc-interim-v2.0-2.0 · https://www.esr.org/data-products/oscar/
TL-03	VrushankPatel/godseye	https://github.com/VrushankPatel/godseye
TL-04	tapiab/heimdall	https://github.com/tapiab/heimdall
TL-05	async-tiff (Rust crate)	https://docs.rs/async-tiff/ · https://developmentseed.org/async-geotiff/latest/blog/2026/02/03/introducing-async-geotiff/
TL-09	Arnis (`louis-e/arnis`)	https://github.com/louis-e/arnis · https://wiki.openstreetmap.org/wiki/Arnis
TL-10	Terra++ / Terra121 (`BuildTheEarth/terraplusplus`)	https://github.com/BuildTheEarth/terraplusplus
TL-11	`sgp4` crate	https://crates.io/crates/sgp4 · https://crates.io/crates/pmtiles · https://crates.io/crates/mvt-reader · https://crates.io/crates/grib · https://crates.io/crates/async-tiff · https://crates.io/crates/image-webp · https://crates.io/crates/proj4rs · https://crates.io/crates/tzf-rs · https://crates.io/crates/icechunk · https://crates.io/crates/zarrs · https://lib.rs/science/geo
TL-12	`pmtiles` crate	https://crates.io/crates/sgp4 · https://crates.io/crates/pmtiles · https://crates.io/crates/mvt-reader · https://crates.io/crates/grib · https://crates.io/crates/async-tiff · https://crates.io/crates/image-webp · https://crates.io/crates/proj4rs · https://crates.io/crates/tzf-rs · https://crates.io/crates/icechunk · https://crates.io/crates/zarrs · https://lib.rs/science/geo

ID	Source	Links
TL-13	`mvt-reader` 2.5.0 (MIT), `galileo-mvt` 0.2.1	https://crates.io/crates/sgp4 · https://crates.io/crates/pmtiles · https://crates.io/crates/mvt-reader · https://crates.io/crates/grib · https://crates.io/crates/async-tiff · https://crates.io/crates/image-webp · https://crates.io/crates/proj4rs · https://crates.io/crates/tzf-rs · https://crates.io/crates/icechunk · https://crates.io/crates/zarrs · https://lib.rs/science/geo
TL-14	`grib` crate	https://crates.io/crates/sgp4 · https://crates.io/crates/pmtiles · https://crates.io/crates/mvt-reader · https://crates.io/crates/grib · https://crates.io/crates/async-tiff · https://crates.io/crates/image-webp · https://crates.io/crates/proj4rs · https://crates.io/crates/tzf-rs · https://crates.io/crates/icechunk · https://crates.io/crates/zarrs · https://lib.rs/science/geo
TL-15	`async-tiff` 0.4.0 + `object_store` 0.14.2 + `tiff` 0.11.3	https://crates.io/crates/sgp4 · https://crates.io/crates/pmtiles · https://crates.io/crates/mvt-reader · https://crates.io/crates/grib · https://crates.io/crates/async-tiff · https://crates.io/crates/image-webp · https://crates.io/crates/proj4rs · https://crates.io/crates/tzf-rs · https://crates.io/crates/icechunk · https://crates.io/crates/zarrs · https://lib.rs/science/geo
TL-16	`image-webp` 0.2.4	https://crates.io/crates/sgp4 · https://crates.io/crates/pmtiles · https://crates.io/crates/mvt-reader · https://crates.io/crates/grib · https://crates.io/crates/async-tiff · https://crates.io/crates/image-webp · https://crates.io/crates/proj4rs · https://crates.io/crates/tzf-rs · https://crates.io/crates/icechunk · https://crates.io/crates/zarrs · https://lib.rs/science/geo
TL-17	`proj4rs` 0.2.1	https://crates.io/crates/sgp4 · https://crates.io/crates/pmtiles · https://crates.io/crates/mvt-reader · https://crates.io/crates/grib · https://crates.io/crates/async-tiff · https://crates.io/crates/image-webp · https://crates.io/crates/proj4rs · https://crates.io/crates/tzf-rs · https://crates.io/crates/icechunk · https://crates.io/crates/zarrs · https://lib.rs/science/geo
TL-18	`h3o` 0.11.0, `tzf-rs` 2.1.3, `flatgeobuf` 6.0.1, `geojson` 1.0.0, `shapefile` 0.9.0, `osmpbf` 0.3.8, `geoarrow` / `geoparquet` 0.9.0	https://crates.io/crates/sgp4 · https://crates.io/crates/pmtiles · https://crates.io/crates/mvt-reader · https://crates.io/crates/grib · https://crates.io/crates/async-tiff · https://crates.io/crates/image-webp · https://crates.io/crates/proj4rs · https://crates.io/crates/tzf-rs · https://crates.io/crates/icechunk · https://crates.io/crates/zarrs · https://lib.rs/science/geo
TL-19	`zarrs` 0.23.14 + `icechunk` 2.3.0	https://crates.io/crates/sgp4 · https://crates.io/crates/pmtiles · https://crates.io/crates/mvt-reader · https://crates.io/crates/grib · https://crates.io/crates/async-tiff · https://crates.io/crates/image-webp · https://crates.io/crates/proj4rs · https://crates.io/crates/tzf-rs · https://crates.io/crates/icechunk · https://crates.io/crates/zarrs · https://lib.rs/science/geo
TL-20	`netcdf` 0.12.1	https://crates.io/crates/sgp4 · https://crates.io/crates/pmtiles · https://crates.io/crates/mvt-reader · https://crates.io/crates/grib · https://crates.io/crates/async-tiff · https://crates.io/crates/image-webp · https://crates.io/crates/proj4rs · https://crates.io/crates/tzf-rs · https://crates.io/crates/icechunk · https://crates.io/crates/zarrs · https://lib.rs/science/geo
TL-21	Not worth adding: `rain_viewer` (last release 2021-11), `satellite` (2024-09), `srtm` (2018), `demtile` (0.2.0, 1k downloads)	https://crates.io/crates/sgp4 · https://crates.io/crates/pmtiles · https://crates.io/crates/mvt-reader · https://crates.io/crates/grib · https://crates.io/crates/async-tiff · https://crates.io/crates/image-webp · https://crates.io/crates/proj4rs · https://crates.io/crates/tzf-rs · https://crates.io/crates/icechunk · https://crates.io/crates/zarrs · https://lib.rs/science/geo
TL-22	earthengine-api (google/earthengine-api)	https://github.com/google/earthengine-api